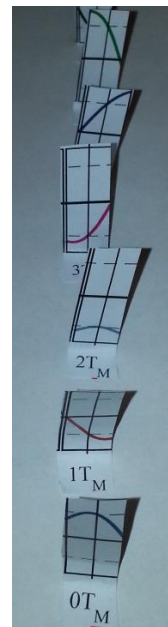
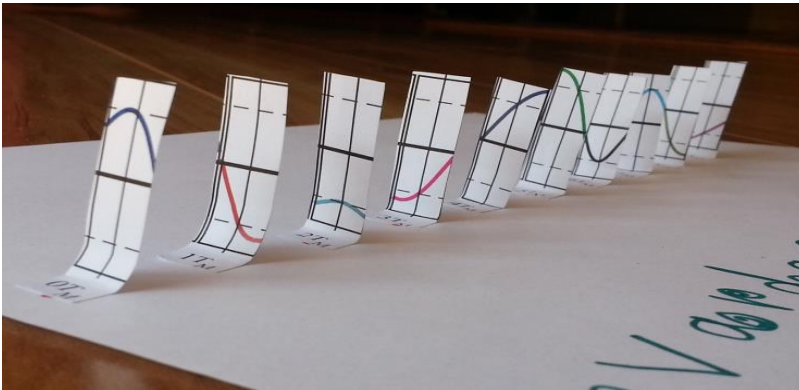


Chapter 3

Digital Communication with Linear Modulations



“The fundamental problem of communication is that of reproducing at one point either exactly or approximately a message selected at another point.”

Claude Shannon - A Mathematical Theory of Communication [7]

Our main purpose is to transfer digital information - which is a sequence of 0's and 1's - from one system to another through a communication channel. Let us return for a moment to the concept behind simple digital logic where logic 0 can be assigned to one voltage level while logic 1 to another. Provided that the static discipline is followed, all our system electronics has to do is to be able to *differentiate* between these two levels.

Of course, digital circuits also have real voltages and real currents which are never exact. The infinite voltage levels are just broken into two discrete regions. Hence, they remain close to that state when the real world noise is introduced, which makes the digital circuits very robust and noise tolerant. This breaking down of real numbers on a continuous scale into discrete regions is the basis of the whole digital revolution.

This particular philosophy is very close to human nature too. Think about it: we like to classify everyday stuff into clearly defined boxes. For example, day and night, black and white, good and bad. The reality, however, is continuous in the sense that the brightness of light varies throughout the 24 hours, there are multiple (perhaps 50) shades of gray, and people are much more complex than being simply good and bad. Nevertheless, this labeling works for our immediate purposes.

Digital communication follows the same principle but allows for multiple discrete regions. Let us explore why and how. We explain these concepts from more of a simple viewpoint without rigorous proofs. The text by Proakis [3] is highly recommended for a mathematics savvy reader.

3.1 A Simple Communication System

In every digital communication system, whether it is a home DSL or cellular data plan, one fundamental parameter we always hear about is the data rate. Data rate is simply the number of bits transmitted and received during one second, also called the *bit rate* denoted by R_b bits/second (b/s). Stated in another way, the source produces one bit during every $1/R_b = T_b$ seconds.

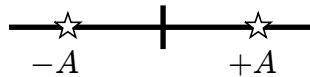


Figure 3.1: Two voltage levels representing 0 and 1

Since these bits consist of 0's and 1's, we start with assigning them two distinguishable voltage levels $-A$ and $+A$, an example of which is shown in Figure 3.1. Such a representation is called a *constellation*. Our ultimate purpose is to transmit a sequence of these voltage levels according to the bit stream at the Tx side. Then, we design the system such that these voltage levels are received undistorted in every manner at the Rx side so that the Rx can perform the inverse mapping from voltage to bits and find out the original bit stream.

Table 3.1: Bit-symbol mappings, received signal after noise addition and receiver decisions

m	0	1	2	3
b	1	0	1	0
$a[m]$	+2	-2	+2	-2
$r[m]$	+2.2	-0.4	+1.1	-2.4
$\hat{a}[m]$	+2	-2	+2	-2
$\hat{b}$	1	0	1	0

Based on this reasoning, a very simple digital communication system can be built as follows: At the Tx side, logic 0 and 1 of a bit sequence $\underline{b}$ are mapped to levels $-A$ and $+A$ of a *symbol* sequence $a[m]$, respectively, at discrete time intervals $m = 0, 1, 2, \dots$. This process is called bit-symbol mapping and the time index m counts each such *symbol time*, T_M . A symbol is basically a mapping from the binary logic levels to signal levels into the real world. The constellation in Figure 3.1 illustrates the relationship between bits in $\underline{b}$ and symbols $a[m]$ as

$$a[m] = \begin{cases} -A, & b = 0 \\ +A, & b = 1 \end{cases} \quad (3.1)$$

The block diagram of such a system is shown in Figure 3.2 and for a bit sequence 1010, the generated symbols for $A = 2$ are shown in Table 3.1.

The received signal $r[m]$ is corrupted by the addition of random noise $w[m]$ on the Rx side and given as

$$r[m] = a[m] + w[m]$$

As a consequence of noise addition, the received symbols are scattered around the actual symbol values as illustrated in the decision block of Figure 3.2. Common sense dictates that symbol decisions $\hat{a}[m]$ should be taken according to the following rule:

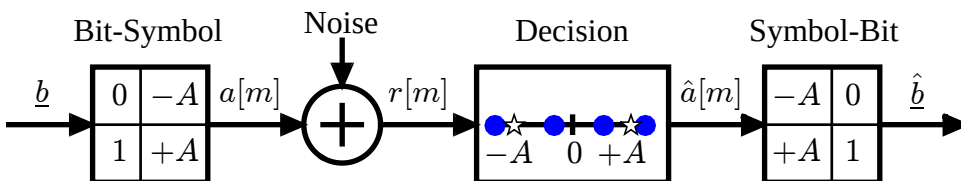


Figure 3.2: A binary digital communication system

Note 3.1 Minimum distance rule

If $r[m]$ is closer to $-A$, it is decided that symbol $-A$ was sent at time m , and if it closer to $+A$, it is taken as symbol $+A$. To construct a general intuitive rule, the point in their middle is $(-A + A)/2 = 0$, and hence

$$\hat{a}[m] = \begin{cases} -A, & r[m] < 0 \\ +A, & r[m] > 0 \end{cases} \quad (3.2)$$

Such a decision rule is applied to compute $\hat{a}[m]$ in Table 3.1 for $A = 2$ where we luckily encounter no errors. The above relation is actually a manifestation of the minimum distance rule. What is not evident here, and we will find out later in this chapter, is that we are actually performing a correlation with the two possible options and choosing the more likely one. Since we quoted Sherlock Holmes to introduce correlation in Section 1.6, we recall another of his quote in this context:

“... we balance probabilities and choose the most likely.”

Sherlock Holmes - The Hound of the Baskervilles

Finally, the subsequent symbol-bit mapping generates bit 0 for symbol $-A$ and bit 1 for symbol $+A$ to form the estimated bit sequence $\hat{\underline{b}}$. Notice that the digital communication system constructed above is sending and receiving one bit in every symbol time T_M .

3.2 Packing More Bits in One Symbol

Digital electronics are constrained to work on only two levels by electronic switches which in the simplest case are either on or off. For many reasons, practical digital communication systems require quite complicated signal processing workload both at the Tx and Rx ends that can be performed only by a device more intelligent than an electronic switch, such as an Application Specific Integrated Circuit (ASIC), Field Programmable Gate Array (FPGA), Digital Signal Processor (DSP) or a General Purpose Processor (GPP).

Departing from mapping a 0 to one symbol and 1 to another, note that if our Rx can differentiate between two signal levels like a switch, it can certainly differentiate between more than two signal levels as well. Being able to do so can result in sending more data during the same amount of time. For example, Figure 3.3 shows four different symbols, $-3A$, $-A$, $+A$, and $+3A$. To develop a fair assignment of bits to symbols, consider a shift from a binary number system to a decimal number system. Clearly 0, 1, 2 and 3 can represent those four symbols with the help of the mapping shown in Figure 3.3.

In this kind of a system, two bits can be collected during every unit of time (the symbol time T_M), converted into their decimal equivalent and a symbol out of $-3A$, $-A$, $+A$, and $+3A$ can be accordingly selected for transmission. Note that $+3A$ and $-3A$ are chosen instead of, say, $+2A$ and $-2A$ to keep an equal distance between all the symbols.

Armed with this idea, the digital communication system of Figure 3.2 can be modified to accommodate 4 symbols as shown in Figure 3.4. A serial-to-parallel (S/P) converter at the Tx collects multiple bits at a time. These bits, converted into their decimal representation, form an address to a unique symbol in a bit-symbol mapper called a *Look-Up Table (LUT)*.

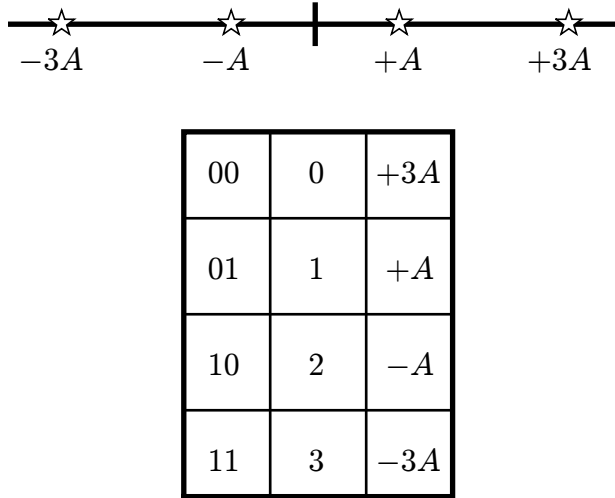


Figure 3.3: Bit to symbol mapping for 4 symbols

After the addition of noise in the channel, the Rx performs its decisions according to the minimum distance rule, maps the resulting symbols back into bits and uses a parallel-to-serial (P/S) converter to generate back the target bit stream. Finding bit sequence from the estimate of symbol sequence is a matter of one-to-one mapping. Once the symbol decisions are formed, the resultant bits can automatically be known through an LUT.

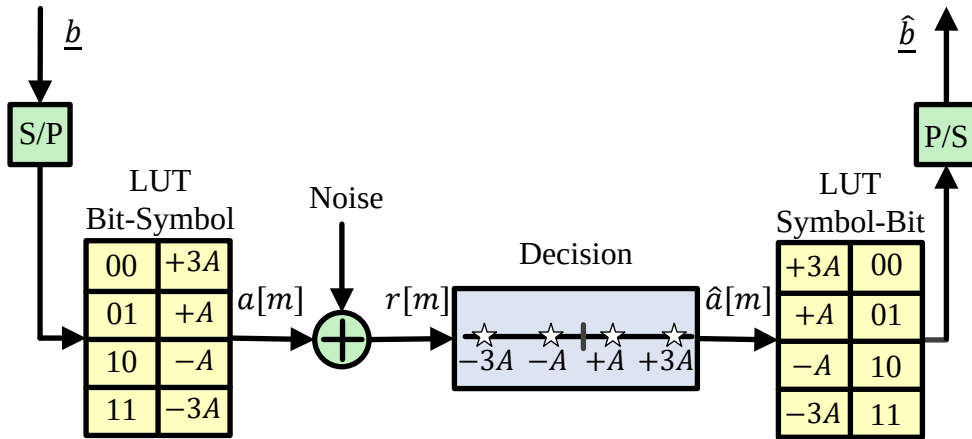


Figure 3.4: A 4 symbol communication system

Extending the above procedure, a certain number of bits can be combined and represented by one symbol to transmit information during every T_M seconds. Combining two bits generates $2^2 = 4$ symbols. Extending the same reasoning, $2^b = M$ symbols are required to transmit b bits of information. Hence, *an M -symbol scheme transmits $\log_2 M$ bits of information every T_M seconds.* As an example, we need 8 different symbols to send 3 such

bits.

In summary, a digital communication system sends a discrete set of numbers and tries to map the received values to (hopefully) the same discrete set of numbers. At this stage, the question arises: can these numbers - which are just symbols in the form of levels - be sent over **real** channels? The answer is no, and the next section explains what happens instead.

3.3 Modulation: From Numbers to Signals

The purpose of digital communications is to send digital data across a channel, some examples of which are

- wireless
- telephone lines
- coaxial cable
- optical fiber
- Ethernet
- USB
- chips on a printed circuit board.

Considering the examples shown in Figure 3.5, clearly neither a bit sequence nor a symbol sequence can be transmitted on their own through these channels – as they are nothing more than a set of numbers. This leads us to the idea of converting the numbers into a signal waveform as an appropriate tool that can travel down the channel and carry the required information – just like a train running on its track and carrying the load. For this purpose, the symbol sequence needs to be converted into an analog waveform that suits the characteristics of that particular channel. Let us start with a simple example.

Assume a rectangular pulse shape[†] $p(nT_S)$ shown in Figure 3.6 whose length is equal to one *symbol interval* in continuous time domain.

$$T_M = L \cdot T_S$$

During each symbol interval T_M , some characteristic of $p(nT_S)$ can be altered to make it a function of the symbol sequence $a[m]$. This process is called *modulation* which is why the subscript M is chosen in T_M . The amplitude of the pulse is one such candidate that can be varied according to the symbol values.

[†]Technically, it is called a Non Return to Zero (NRZ) pulse shape but we will continue using the term rectangular for ease of visualization.

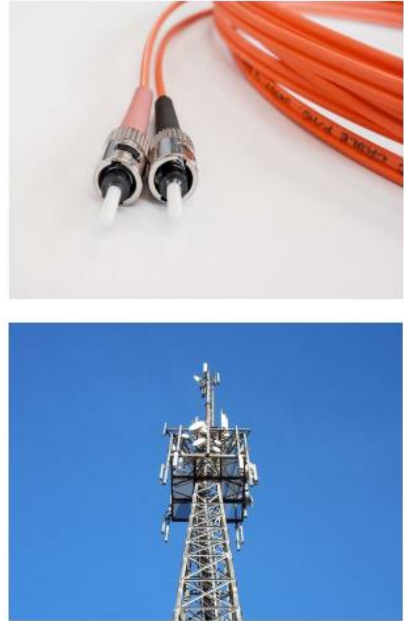


Figure 3.5: Some real world channels

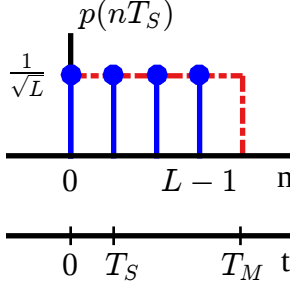


Figure 3.6: A rectangular pulse shape

Note that the pulse energy without any normalization constant is given by $\sum_{n=0}^{L-1} 1^2 = L$. To make the system independent of the pulse amplitude itself, it can be normalized with $1/\sqrt{L}$ as shown in Figure 3.6, thus turning it into a unit energy pulse.

$$E_p = \sum_{n=0}^{L-1} \left(\frac{1}{\sqrt{L}} \right)^2 = 1 \quad (3.3)$$

Let us now relate the bit rate R_b with the symbol rate $1/T_M$ and sample rate F_S . Typically, data bits 0 and 1 are generated with equal probability at a rate of $R_b = 1/T_b$ b/s. In our simple scenario, a 0 is represented by symbol $-A$ and a 1 by symbol $+A$ and the symbol rate is denoted as $R_M = 1/T_M$ (here, T_b and T_M are equal due to a single bit for every symbol duration). Subsequently, a pulse shape at a sample rate of $F_S = 1/T_S$ is generated with L number of samples.

$$L = \frac{T_M}{T_S} = \frac{F_S}{R_M} \quad (3.4)$$

This parameter L is known as *samples/symbol*, or sample rate normalized to symbol rate, or oversampling factor. It is standard practice to select an integer L so that the sample rate is an integer multiple of the symbol rate.

Now the amplitude of the pulse $p(nT_S)$ can be multiplied with $-A$ for transmitting a 0 and with $+A$ for transmitting a 1. As a consequence, two scaled versions of that pulse shape $s_0(nT_S)$ and $s_1(nT_S)$ are created whose amplitudes bear the symbol levels $-A$ and $+A$ representing 0 and 1, respectively. This amplitude scaling of the pulse $p(nT_S)$ according to the symbol value is called *Pulse Amplitude Modulation (PAM)* and is illustrated in Figure 3.7.

This can be seen as a natural result of holding the amplitude constant at a symbol level for the duration of a symbol T_M . Mathematically,

$$\begin{aligned} s_0(nT_S) &= -A \cdot p(nT_S) \\ s_1(nT_S) &= +A \cdot p(nT_S) \end{aligned} \quad (3.5)$$

During the transmission, the signal is corrupted by additive White Gaussian noise (AWGN) with power spectral density $N_0/2$ Watt/Hz. The received signal can be expressed as

$$r(nT_S) = \begin{cases} s_0(nT_S) + w(nT_S), & b = 0, \\ s_1(nT_S) + w(nT_S), & b = 1, \end{cases} \quad 0 \leq n \leq L-1$$

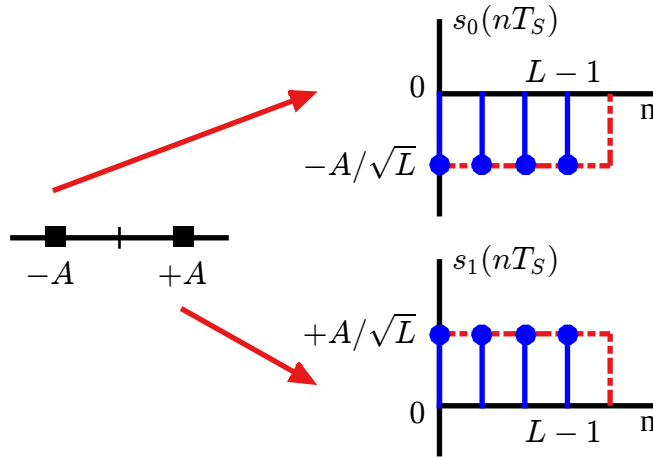


Figure 3.7: From numbers to signals: Scaling the amplitude of a rectangular pulse shape with symbols $-A$ and $+A$

The task of the receiver is to decide the most probable symbol in the interval $0 \leq n \leq L - 1$ and hence determine whether a 0 or a 1 was transmitted.

Note 3.2 Getting numbers back from signals

A natural question at this stage is the following: Basically, digital communication is all about numbers but how will we get those numbers back from such signals for detection purpose? *In the answer to this question lies most of the theory and practice of digital communication systems.*

To accomplish this task, we need a simple mechanism through which we can map all the possible information in a symbol interval $0 \leq t \leq T_M$ back to a number. That number should be an estimate of the symbol and consequently can be used to detect the bit pattern as discussed in the next section.

3.4 Demodulation: From Signals Back to Numbers

Remember that in Section 1.6, we claimed that *correlation is the heart of a digital communication system*. We now apply the concept of correlation to figure out the transmitted message which in this case is embedded in the amplitude of the received signal. Similarly later in the text, we will encounter numerous applications of the correlation concept to solve synchronization and equalization problems.

It was also discussed there that correlation of a signal with proper normalization is maximum with itself and lesser for all the other signals. Since the number of possible signals is limited in a digital communication system, we can use the correlation between incoming signal $r(nT_S)$ and possible choices $s_0(nT_S)$ and $s_1(nT_S)$ in a digital receiver. Consequently, a decision can be made in favor of the one with highest correlation. It turns out that the theory of maximum likelihood detection formalizes this conclusion that it is the optimum receiver for

equal energy signals.

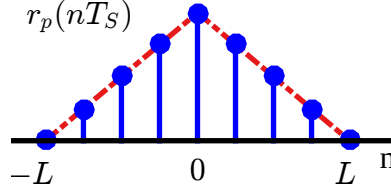


Figure 3.8: Auto-correlation $r_p(nT_S)$ of a rectangular pulse shape

Before correlating the received signal with possible choices, it is helpful to see the correlation of a rectangular pulse shape with itself – called the auto-correlation $r_p(nT_S)$ and drawn earlier in Figure 2.13. Here, Figure 3.8 illustrates the auto-correlation $r_p(nT_S)$ of the rectangular pulse shape whose time support is equal to a symbol duration, i.e., from 0 to $L - 1$ samples. As a consequence of one rectangle sliding over another sample-by-sample, its auto-correlation is a triangular pulse shape with duration clearly from $-(L - 1)$ to $L - 1$. From the definition of correlation in Eq (1.38),

$$r_p(nT_S) = p(nT_S) \heartsuit p(nT_S) = \sum_i p(iT_S)p(iT_S - nT_S) \quad (3.6)$$

Since correlation is similar to convolution with one signal flipped and flipping does not alter the length of a signal, the length of the correlation output is the same as convolution output, i.e., $L + L - 1 = 2L - 1$.

Eq (3.6) is extremely important in the design of communication systems and this term, $r_p(nT_S)$, will appear over and over again in any communications text.

Note 3.3 Convenience of linearity

Notice that our focus is on linear modulation, i.e., the signal is generated by linear scaling of a basic pulse shape to vary its amplitude as $\pm A, \pm 3A$, etc. as expressed in Eq (3.5). Therefore, the effect of scaling the pulse at the Tx will appear as the same scaling at the Rx without affecting the pulse shape or its auto-correlation. This leads to the conclusion that we have two choices available:

- correlate the incoming signal with all possible transmitted signals such as $s_0(nT_S)$ and $s_1(nT_S)$ here, or
- correlate the incoming signal with the pulse shape only, since $s_0(nT_S)$ and $s_1(nT_S)$ are just the scaled by $-A$ or $+A$ versions of the same pulse shape. With this strategy, the peak of the auto-correlation will bear the amplitude scaling $-A$ or $+A$ done at the Tx.

If you have some confusion regarding this point, don't worry. We will shortly see it graphically in Section 3.5 during detection of pulse amplitude modulated signals.

Now adopting the latter strategy above, let us compute the correlation between the received signal $r(nT_S)$ and the pulse shape $p(nT_S)$ as

$$z(nT_S) = \sum_{i=-\infty}^{\infty} r(iT_S)p(iT_S - nT_S) \quad (3.7)$$

for all possible lags n . The question then is to find the optimum sampling instant: the value of n that gives the best result through maximizing the correlation output. For both $r(nT_S) = s_0(nT_S)$ and $r(nT_S) = s_1(nT_S)$ with zero noise, Figure 3.9 draws the correlations for many different lags n .

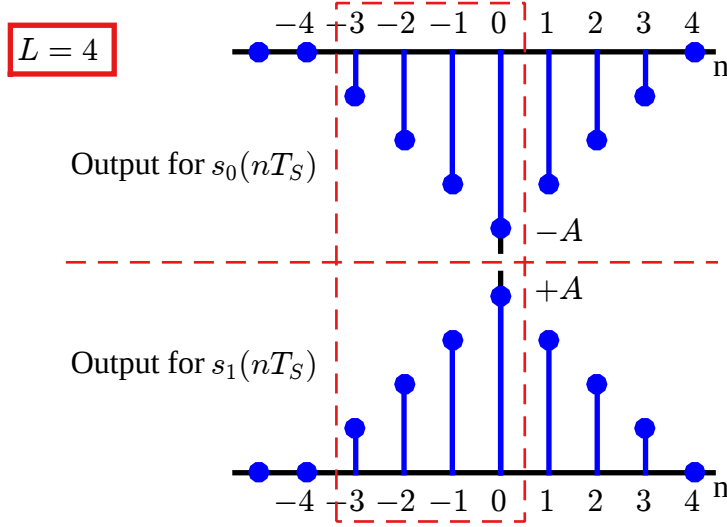


Figure 3.9: Correlation outputs of pulse shape $p(nT_S)$ for different lags n

For $L = 4$, imagine a rectangular pulse normalized by $1/\sqrt{4}$ sliding sample-by-sample over the Rx signal. This Rx signal is another rectangular pulse with a scaling factor of $-A/\sqrt{4}$ or $+A/\sqrt{4}$. Then, the correlation outputs $z(nT_S)$ vary as

$$\begin{aligned} & \vdots \\ n = -4 & \rightarrow z(-4T_S) = 0 \\ n = -3 & \rightarrow z(-3T_S) = \pm A/4 \\ n = -2 & \rightarrow z(-2T_S) = \pm 2A/4 \\ n = -1 & \rightarrow z(-1T_S) = \pm 3A/4 \end{aligned}$$

$$n = 0 \rightarrow z(0T_S) = \pm 4A/4 = \pm A$$

$$\begin{aligned} n = +1 & \rightarrow z(+1T_S) = \pm 3A/4 \\ n = +2 & \rightarrow z(+2T_S) = \pm 2A/4 \\ n = +3 & \rightarrow z(+3T_S) = \pm A/4 \\ n = +4 & \rightarrow z(+4T_S) = 0 \end{aligned}$$

$\vdots$

It is evident that at time instant $n = 0$, the output for $r(nT_S) = s_0(nT_S)$ is $-A$ and the output for $r(nT_S) = s_1(nT_S)$ is $+A$. This time instant is the point of maximum overlap where the received signal $r(nT_S)$ is completely aligned with the pulse shape. For symbol detection, only this particular sample of correlation output is needed.

$$\hat{a}[m] = \begin{cases} -A, & z(0T_S) < 0 \\ +A, & z(0T_S) > 0 \end{cases}$$

The rest of the samples can simply be discarded. This is the process of *demodulation* that maps a signal back to a number.

3.4.1 Matched Filter in Time Domain

Observe from Figure 3.9 that the time base of our demodulator starts at $n = -L$ samples or $-T_M$ seconds (the point where the upward slope of auto-correlation begins). However, a real receiver cannot begin processing a signal before it arrives! That would require having an ability to perceive events in the future: ask *the Oracle*.

Therefore, a Rx can only start processing the received signal at time $n = 0$. The peak or maximum of the auto-correlation shifts by the same amount of time which is $n = L$ samples or T_M seconds.

We just found in the last section that the maximum overlap is the only sample within a symbol time T_M that is required out of $L = T_M/T_S$ samples. Let us substitute $n = L = T_M/T_S$ in Eq (3.7) as

$$\begin{aligned} z(T_M) &= z(nT_S) \Big|_{n=L=T_M/T_S} = \sum_i r(iT_S) p(iT_S - nT_S) \Big|_{n=L=T_M/T_S} \\ &= \sum_i r(iT_S) p\left(-(nT_S - iT_S)\right) \Big|_{n=L=T_M/T_S} \\ &= r(nT_S) * p(-nT_S) \Big|_{n=L=T_M/T_S} \end{aligned} \quad (3.8)$$

We conclude that the process of correlation can be implemented as *convolution with a filter* whose impulse response $h(nT_S)$ is a flipped version of the actual pulse shape $p(nT_S)$. Most texts say that this is true only as far as the point of maximum overlap is concerned but they define the processing of a correlator in a specific manner. We discuss this point in detail during a comparison of a matched filter with a correlator.

Combining the above two facts, not only that this filter $h(nT_S)$ starts at $n = 0$ but also the filter output reaches its maximum at time T_M seconds. This flipped and delayed version of the template pulse shape is called *the matched filter* given by

$$h(nT_S) = p(-nT_S + T_M)$$

Above statement is true because the pulse has real coefficients. For complex signals, complex conjugation is also required which follows from how the correlation of complex signals is defined.

$$h(nT_S) = p^*(-nT_S + T_M)$$

(3.9)

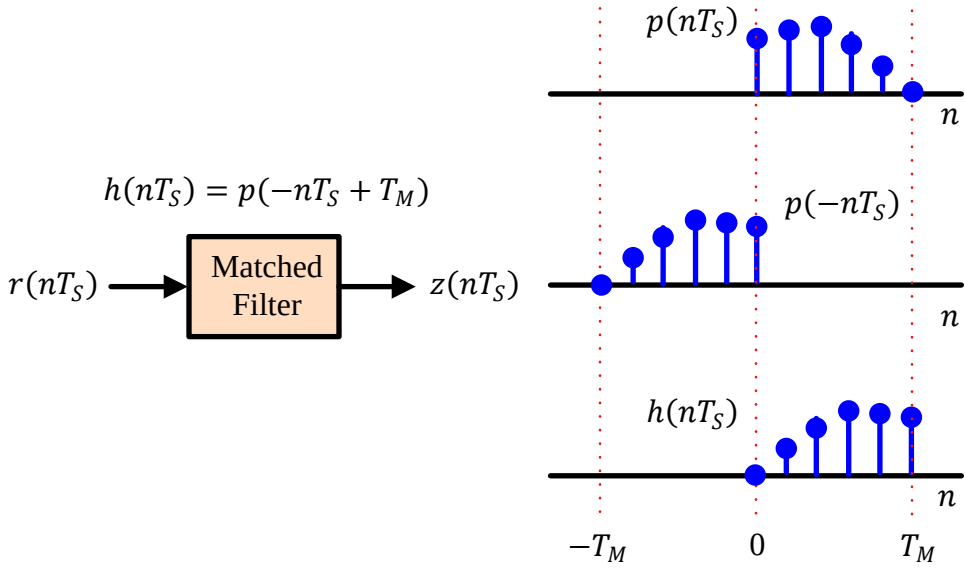


Figure 3.10: A matched filter

Figure 3.10 draws a matched filter for an example pulse shape. To find the matched filter output, consider that in the absence of noise, the received signal is

$$r(nT_S) = \pm A \cdot p(nT_S)$$

From Eq (3.8),

$$\begin{aligned} z(T_M) &= \pm A \cdot p(nT_S) * p(-nT_S) \Big|_{n=L=T_M/T_S} \\ &= \pm A \cdot 1 = \pm A \end{aligned}$$

where we utilized the fact that convolution of a signal with its flipped version is its auto-correlation which is unity at the middle point (L^{th} sample) as shown in Figure 3.8. Thus, the output $z(T_M)$ reflects the information embedded in the amplitude of the signal. Keep in mind that the above procedure describes the demodulation of a single symbol. When a whole symbol sequence is pulse shaped and transmitted (discussed in the later sections), L samples delay of the matched filter is implicitly assumed and not necessarily mentioned every time.

A question that immediately arises from the above derivation is the following: how do we know for sure that the remaining $L - 1$ samples are of little use, as far as the process of demodulation is concerned? We answer this question in the next subsection during the matched filter viewpoint in frequency domain.

Note 3.4 A natural approach

For an intuitive understanding of matched filtering approach, consider the following argument: we know that correlation can be implemented through convolution if one of the signals is flipped beforehand. During convolution with the received signal, this flipped version is folded

again, hence bringing the original expected signal back and ready to be correlated.

By definition, a matched filter is a linear filter that maximizes the output signal-to-noise ratio (SNR) if the signal is buried in AWGN. *Since the incoming signal structure is already known*, the matched filter can be easily constructed through time-reversing and delaying this known signal for maximizing the output correlation.

As an example in everyday life, carefully watch out for the opinions you hold. In general, we are always defending the viewpoints we already have and rejecting (filtering out) either the unknown or what we decided to oppose in the past. That is matched filtering^a!

^aPaul Graham has written a great article on this topic titled “Keep your identity small”

3.4.2 Matched Filter in Frequency Domain

In frequency domain, the matched filter has an interesting view as well. Using $L = T_M/T_S$, we can write the matched filter expression as

$$\begin{aligned} h(nT_S) &= p^*(-(nT_S - T_M)) = p^*\left(-\left(n - \frac{T_M}{T_S}\right)T_S\right) \\ &= p^*(-(n - L)T_S) \end{aligned}$$

In terms of discrete-time, this signal can then be given as

$$h[n] = p^*[-(n - L)]$$

To view this signal in frequency domain, we need two pieces of information.

- In Section 2.9, we proved that the DFT of a time-reversed and complex conjugated signal $s^*[(-n) \bmod N]$ is given by the complex conjugate of its DFT, i.e.,

$$s^*[(-n) \bmod N] \xrightarrow{F} S^*[k] \quad (3.10)$$

So the magnitude of the DFT stays the same while the phase becomes negative due to conjugate operation in k .

- The time shift of an input signal $s[(n - n_0) \bmod N]$ results in a corresponding phase shift $\Delta\theta(k) = -2\pi(k/N)n_0$ *at each frequency* of its DFT $S[k]$ with no change in magnitude.

Combining the above two facts and implicitly assuming a circular shift, the N -point DFT of the matched filter impulse response $h[n]$ can be deduced.

$$\begin{aligned} |H[k]| &= |P[k]| \\ \angle H[k] &= -\angle P[k] - 2\pi \frac{k}{N}L \end{aligned} \quad (3.11)$$

So the DFT of the matched filter has a magnitude that is the same as the magnitude of the underlying pulse while a phase shift that is negative of the phase of the pulse DFT

(which arises from taking the complex conjugate), plus an additional factor proportional to the symbol time T_M seconds, or L samples.

What happens when the received signal is passed through the matched filter? First, in the presence of zero noise, the Rx signal is expressed as

$$r(nT_S) = \pm A \cdot p(nT_S)$$

whose spectrum can be written as

$$\begin{aligned} |R[k]| &= A \cdot |P[k]| \\ \angle R[k] &= \angle A + \angle P[k] \end{aligned}$$

where $\angle A$ is 0 if A is positive and π if A is negative. At the output of the matched filter, the convolution in time domain is multiplication in frequency domain leading to the following results.

Magnitude and Phase

Utilizing Eq (3.11), the magnitude of the matched filter output $Z[k]$ can be written as

$$|Z[k]| = |R[k]| \cdot |H[k]| = A \cdot |P[k]| \cdot |P[k]| = A \cdot |P[k]|^2 \quad (3.12)$$

Consider that for a number b ,

$$\begin{aligned} b^2 &> b & \text{if } b > 1 \\ b^2 &< b & \text{if } 0 < b < 1 \end{aligned}$$

Therefore, the matched filter has high gain at frequencies where $P[k]$ is large and low gain at frequencies where $P[k]$ is small. It can be deduced that the matched filter enhances the strong spectral components and reduces the weak spectral components in $P[k]$.

On the other hand, the phase of $Z[k]$ takes the form

$$\begin{aligned} \angle Z[k] &= \angle R[k] + \angle H[k] = \angle A + \angle P[k] - \angle P[k] - 2\pi \frac{k}{N} L \\ &= \angle A - 2\pi \frac{k}{N} L \end{aligned} \quad (3.13)$$

where we can notice the following.

Phase cancellation: The phase of the incoming signal $\angle P[k]$ is canceled by the matched filter phase $-\angle P[k]$. This phase adjustment enables all the spectral components to align at the sampling time instant.

Tx time reference: The additional phase of $-2\pi(k/N)L$ comes due to a delay of L samples from the Tx time reference 0. Since the Rx is only interested in the value of the symbol and not the Rx location (in reality, there are several additional delays that are added to the incoming signal such as the propagation delay and the Tx/Rx frontend processing in both analog and digital domains), this delay is irrelevant and it can set its time reference 0 at the peak of the auto-correlation. Then, the phase term $-2\pi(k/N)L$ disappears.

Figure 3.11 illustrates an example signal template and a corresponding matched filter where the solid blue lines represent the signal template while the dotted red lines show the matched filter. Notice the same magnitude on each spectral line but exactly the opposite phase. This is the reason complex conjugation is required for designing a matched filter for complex signals.

Conjugate operation aligns the phases in frequency domain

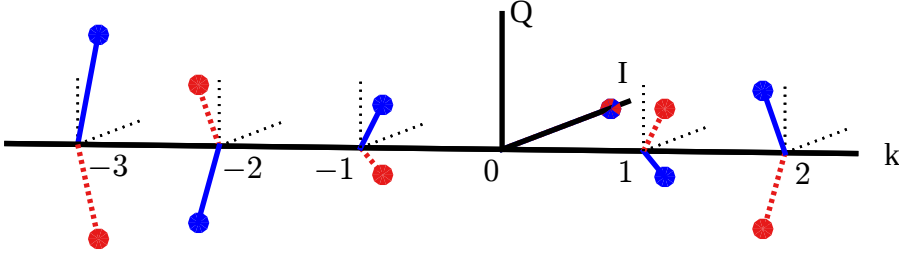


Figure 3.11: Conjugate operation of a matched filter aligns all spectral components by canceling the phase. The solid blue line is the signal template and dotted red line is the matched filter

I and Q Components

From the magnitude and phase expressions above in Eq (3.12) and Eq (3.13), respectively, and incorporating $\angle A$ in $\pm$ form, $Z[k]$ can be written in terms of IQ as

$$Z_I[k] = \pm A \cdot |P[k]|^2 \cos 2\pi \frac{k}{N} L$$

$$Z_Q[k] = \mp A \cdot |P[k]|^2 \sin 2\pi \frac{k}{N} L$$

where the difference of $\pm$ and $\mp$ is due to the identities $\cos(-B) = \cos B$ and $\sin(-B) = -\sin B$.

To find the output in time domain $z(nT_S)$, iDFT of $Z[k]$ needs to be taken.

$$I \rightarrow \quad z_I(nT_S) = \frac{1}{N} \sum_{k=-N/2}^{N/2-1} \left[Z_I[k] \cos 2\pi \frac{k}{N} n - Z_Q[k] \sin 2\pi \frac{k}{N} n \right]$$

$$Q \uparrow \quad z_Q(nT_S) = \frac{1}{N} \sum_{k=-N/2}^{N/2-1} \left[Z_Q[k] \cos 2\pi \frac{k}{N} n + Z_I[k] \sin 2\pi \frac{k}{N} n \right]$$

Plugging $Z_I[k]$ and $Z_Q[k]$ in iDFT above and using identities $\cos A \cos B + \sin A \sin B = \cos(A - B)$ and $\sin A \cos B - \cos A \sin B = \sin(A - B)$, we get

$$I \rightarrow \quad z_I(nT_S) = \frac{1}{N} \sum_{k=-N/2}^{N/2-1} \pm A \cdot |P[k]|^2 \cos \left[2\pi \frac{k}{N} (n - L) \right]$$

$$Q \uparrow \quad z_Q(nT_S) = \frac{1}{N} \sum_{k=-N/2}^{N/2-1} \pm A \cdot |P[k]|^2 \sin \left[2\pi \frac{k}{N} (n - L) \right] \quad (3.14)$$

Sampling this matched filter output at the instant $n = T_M/T_S = L$ yields

$$\begin{aligned}
 I \rightarrow \quad z_I(nT_S) \Big|_{n=T_M/T_S} &= z_I(T_M) = \frac{1}{N} \sum_{k=-N/2}^{N/2-1} \pm A \cdot |P[k]|^2 \\
 Q \uparrow \quad z_Q(nT_S) \Big|_{n=T_M/T_S} &= z_Q(T_M) = 0
 \end{aligned}$$

Figure 3.12 verifies the above finding by illustrating the frequency terms on the right hand side of the above equation. Notice that there is no Q component and all frequency domain samples are perfectly aligned to contribute towards I term.

Matched filter output at correct sampling instant

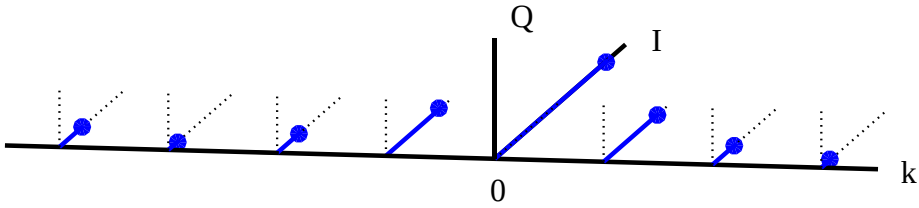


Figure 3.12: At correct sampling instant, the spectral components have all matching phase and contribute maximally towards the sampled output

Using Parseval's relation from Section 2.9 that relates the signal energy in time domain to that in frequency domain,

$$E_s = \sum_{n=0}^{N-1} |s[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |S[k]|^2$$

the matched filter output in time domain can be written as

$$\begin{aligned}
 I \rightarrow \quad z_I(T_M) &= \pm A \sum_{n=0}^{N-1} |p[n]|^2 = \pm A \cdot E_p \\
 Q \uparrow \quad z_Q(T_M) &= 0
 \end{aligned}$$

Owing to the unit energy pulse $E_p = 1$ from Eq (3.3),

$$\begin{aligned}
 I \rightarrow \quad z_I(T_M) &= \pm A \\
 Q \uparrow \quad z_Q(T_M) &= 0
 \end{aligned}$$

and hence the modulated information is retrieved through the amplitude of the matched filter output.

In fact, this is the purpose of auto-correlation at lag 0: finding the energy of the signal which is *a real number* with zero Q part.

$$\text{corr}[0] = \sum_{m=-\infty}^{\infty} s[m]s^*[m] = \sum_{m=-\infty}^{\infty} |s[m]|^2 = E_s$$

That is how the SNR is maximized by the matched filter output sampled at $n = L$.

For the case where an all 1 sequence is transmitted (no $-A$ in the received sequence) and signal in Eq (3.14) is sampled at a different time instant instead of $n = L$, say at $n = L - 1$, we have from Eq (3.14),

$$I \rightarrow z_I(nT_S) = \frac{1}{N} \sum_{k=-N/2}^{N/2-1} |P[k]|^2 \cos 2\pi \frac{k}{N}$$

$$Q \uparrow z_Q(nT_S) = -\frac{1}{N} \sum_{k=-N/2}^{N/2-1} |P[k]|^2 \sin 2\pi \frac{k}{N}$$

which is nothing but a linear phase shift arising in all frequency bins due to sampling the output one unit of time earlier. Recall by going back to Figure 1.55 that a phase shift of $2\pi(k/N)n_0$, as a function of k , is actually a frequency domain complex sinusoid that modulates the spectrum.

Now, the Q component of the output $z_Q(nT_S)$ is not zero. We can say that a part of the actual signal energy – that should have appeared in I branch – has leaked into the Q arm. The I arm with reduced energy is not optimal anymore for symbol detection. This is drawn in Figure 3.13. Notice that the spectral components have been rotated by symmetrical phase $-2\pi k/N$ due to a time difference of 1 sample. This non-zero Q sample also reveals the

Matched filter output at incorrect sampling instant

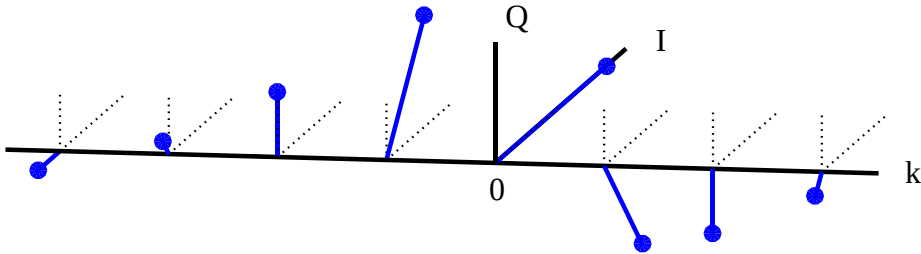


Figure 3.13: At incorrect sampling instant, the spectral components are rotated symmetrically which reduces their contribution towards I term as well as generating correlated noise samples through the filtering effect

following.

- In time domain, all the samples of the correlation result are less in magnitude than the peak due to a partial overlap instead of the maximum overlap.
- Going into frequency domain, this disappeared energy can be found in Q arm associated with phase rotations. The same frequency response is very similar for all time shifts – the only difference being the scattered phases for $n \neq L$ resulting in misaligned summation.

Also remember that in an actual receiver, *noise* is also added to the transmitted signal and has to pass through the same filter (not matched for noise). For noise $w(nT_S)$, the

filtered noise samples $w'(nT_S)$ are given by

$$w'(nT_S) = \sum_i w(iT_S)h(nT_S - iT_S)$$

Therefore, this filtering on the noise generates correlation in noise samples at its output that is directly proportional to the auto-correlation of the pulse shape (such as the triangular auto-correlation of a rectangular pulse).

$$r_{w'}(nT_S) \propto h(nT_S) \heartsuit h^*(nT_S) \quad (3.15)$$

For this reason, this correlation among noise samples is zero only when the underlying pulse auto-correlation is zero, which occurs at a spacing of L samples either side of the peak, as is evident from Figure 3.8.

Here, we concentrated on the matched filter in relation to the underlying pulse shape due to the modulation being linear. This pulse shape is real and symmetric as well. In the field of signal processing, the matched filter has a much broader meaning that works well for non-linear modulations, distinct communication and radar waveforms and other specialized areas, where the matched filter is designed in relation to the expected signal itself. The fundamental concept is still the same: *utilize the knowledge of what is expected* and suitably invert, delay and take its complex conjugate.

So far, we have learned how to map a symbol to a signal and then a signal back to a symbol. All of this was focused within each symbol duration T_M . Next, we discuss the practical case of a bit stream and corresponding processes of modulation, waveform generation and detection.

3.5 Pulse Amplitude Modulation (PAM)

In Section 3.3, we said that the Pulse Amplitude Modulation (PAM) is an amplitude scaling of the pulse $p(nT_S)$ according to the symbol value. What happens when this process of scaling the pulse amplitude by symbols is repeated for every symbol during each interval T_M ? Clearly, a series of bits $\underline{b}$ (1010 in our initial example) can be transmitted by choosing a rectangular pulse and scaling it with appropriate symbols.

$m = 0 \rightarrow b = 1$	$\rightarrow a[0] = +A$
$m = 1 \rightarrow b = 0$	$\rightarrow a[1] = -A$
$m = 2 \rightarrow b = 1$	$\rightarrow a[2] = +A$
$m = 3 \rightarrow b = 0$	$\rightarrow a[3] = -A$

Our next step is forming a cumulative waveform from these individual symbol-scaled pulses. Remember that a signal $p(nT_S)$ delayed by an amount T_M or L samples is given as $p(nT_S - LT_S) = p(nT_S - T_M)$, where T_M is the symbol duration and L is samples/symbol defined as T_M/T_S . Since the same pulse is scaled by the symbol value during each T_M , we observe the following.

- At time instant $0T_M$, the output is $a[0]p(nT_S - 0T_M)$.
- At time instant $1T_M$, the output is $a[1]p(nT_S - 1T_M)$.

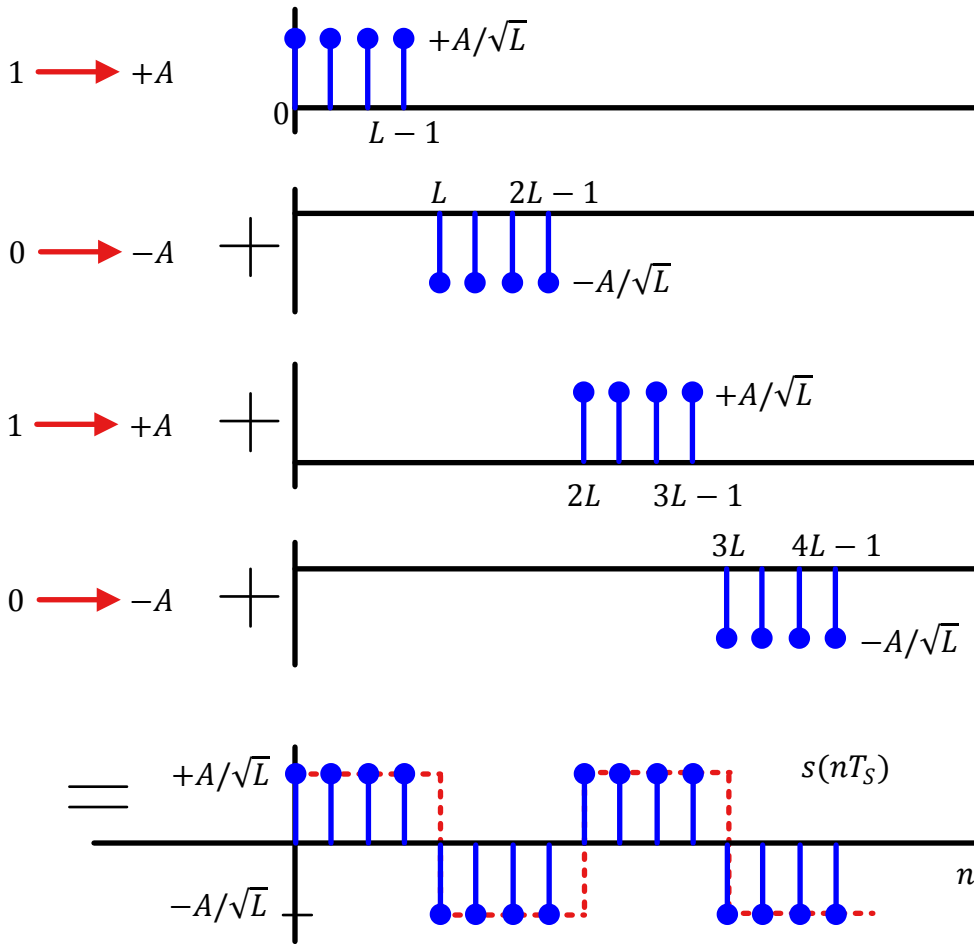


Figure 3.14: A 2-PAM waveform

■ At time instant $2T_M$, the output is $a[2]p(nT_S - 2T_M)$.

■ At time instant $3T_M$, the output is $a[3]p(nT_S - 3T_M)$.

And so on. Finally, their addition gives the expression for a general PAM waveform:

$$\begin{aligned}
 s(nT_S) &= a[0]p(nT_S - 0T_M) + a[1]p(nT_S - 1T_M) + \\
 &\quad a[2]p(nT_S - 2T_M) + a[3]p(nT_S - 3T_M) + \dots \\
 &= \sum_m a[m]p(nT_S - mT_M)
 \end{aligned} \tag{3.16}$$

After Digital to Analog Conversion (DAC), the continuous-time signal $s(t)$ can be expressed as

$$s(t) = \sum_m a[m]p(t - mT_M) \tag{3.17}$$

As an example, a 2-PAM waveform is illustrated in Figure 3.14 with red dashed curve being the underlying continuous-time signal.

In a similar manner to 2-PAM, a 4-PAM waveform based on 4 symbols can be constructed by scaling the pulse amplitude by different symbol values during each T_M , as illustrated in Figure 3.15.

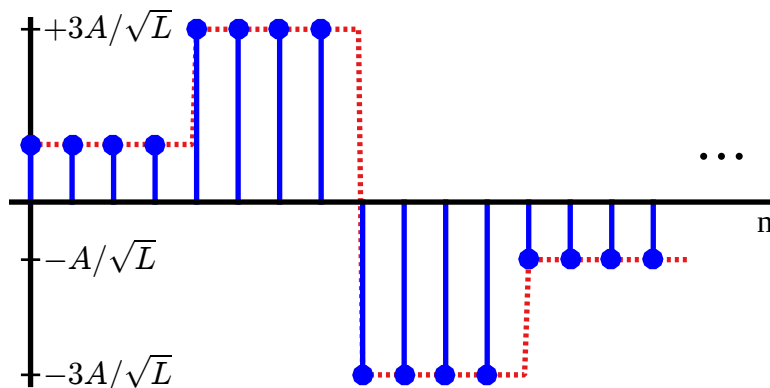


Figure 3.15: A 4-PAM waveform

Note 3.5 Orthogonality in frequency domain

There is something interesting about the PAM data generation above. The symbols are orthogonal in time due to the time shift mT_M of a basic pulse shape. This time difference of a symbol time T_M between two symbols implies that no symbol interferes with the other. How is the corresponding orthogonality maintained in frequency domain?

Recall from Eq (1.61) that for a time shift mT_M of the basic pulse shape, the DFT of this pulse shape is multiplied with a frequency domain complex sinusoid with inverse period mT_M . All data symbols in frequency domain are riding on the pulse shape DFT multiplied with the orthogonal complex sinusoids (overlapping within the pulse bandwidth) with increasingly large inverse periods as m increases. This is interesting since just a passing of time is giving rise to spiraling sinusoids in frequency domain. Only if it could solve the world's energy problems!

Constellation Diagram

Just like a constellation of stars, a *constellation diagram* shows the actual symbol values representing a set of $\log_2 M$ bits. We have already encountered constellation diagrams before (e.g., in Figure 3.1 for $M = 2$ and in Figure 3.3 for $M = 4$). Some example symbol sets are as follows.

$$2\text{-PAM Constellation} = \{-A, +A\}$$

$$4\text{-PAM Constellation} = \{-3A, -A, +A, +3A\}$$

$$M\text{-PAM Constellation} =$$

$$\{- (M-1)A, \dots, -3A, -A, +A, +3A, \dots, +(M-1)A\}$$

A general constellation diagram for M -PAM is shown in Figure 3.16.

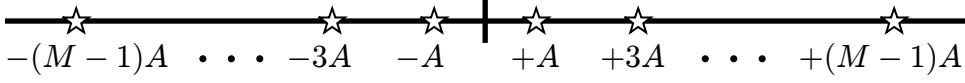


Figure 3.16: General constellation diagram for M -PAM

Average Symbol Energy

The *average symbol energy* in a constellation is given by the average of all individual symbol energies. For $M = 2$,

$$E_{M\text{-PAM}} = \frac{A^2}{2} \{ (+1)^2 + (-1)^2 \} = A^2$$

And for $M = 4$,

$$E_{M\text{-PAM}} = \frac{A^2}{4} \{ (-3)^2 + (-1)^2 + (+1)^2 + (+3)^2 \} = 5A^2$$

For a general M ,

$$\begin{aligned} E_{M\text{-PAM}} &= \frac{A^2}{M} \{ (-M+1)^2 + \dots + (-3)^2 + (-1)^2 + \\ &\quad (+1)^2 + (+3)^2 + \dots + (M-1)^2 \} \\ &= 2 \frac{A^2}{M} \{ 1^2 + 3^2 + \dots + (M-1)^2 \} \end{aligned}$$

The term in the brackets is the sum of squares of first $(M-1+1)/2 = M/2$ odd integers. Using the formula $k(2k-1)(2k+1)/3$ for the first k odd integers squared, we get

$$\begin{aligned} E_{M\text{-PAM}} &= 2 \frac{A^2}{M} \cdot \frac{M/2 \cdot (M-1) \cdot (M+1)}{3} \\ &= \frac{M^2 - 1}{3} A^2 \end{aligned} \tag{3.18}$$

The main purpose of a communications system is to solve the following problem: which operations are necessary to perform at the Tx and Rx sides such that we can detect that symbol with the highest probability? To explore the answer, we start with a simple PAM modulator and detector.

PAM Modulator

At this stage, we are ready to build a conceptual PAM modulator. The block diagram is drawn in Figure 3.17 in which Tx signal is generated in the following way.

- Every T_b seconds, a new bit arrives at the input forming a serial bit stream.
- A serial-to-parallel (S/P) converter collects $\log_2 M$ such bits every $T_M = \log_2 M \times T_b$ seconds that are used as an address to access a Look-Up Table that stores M symbol values specified by the constellation.

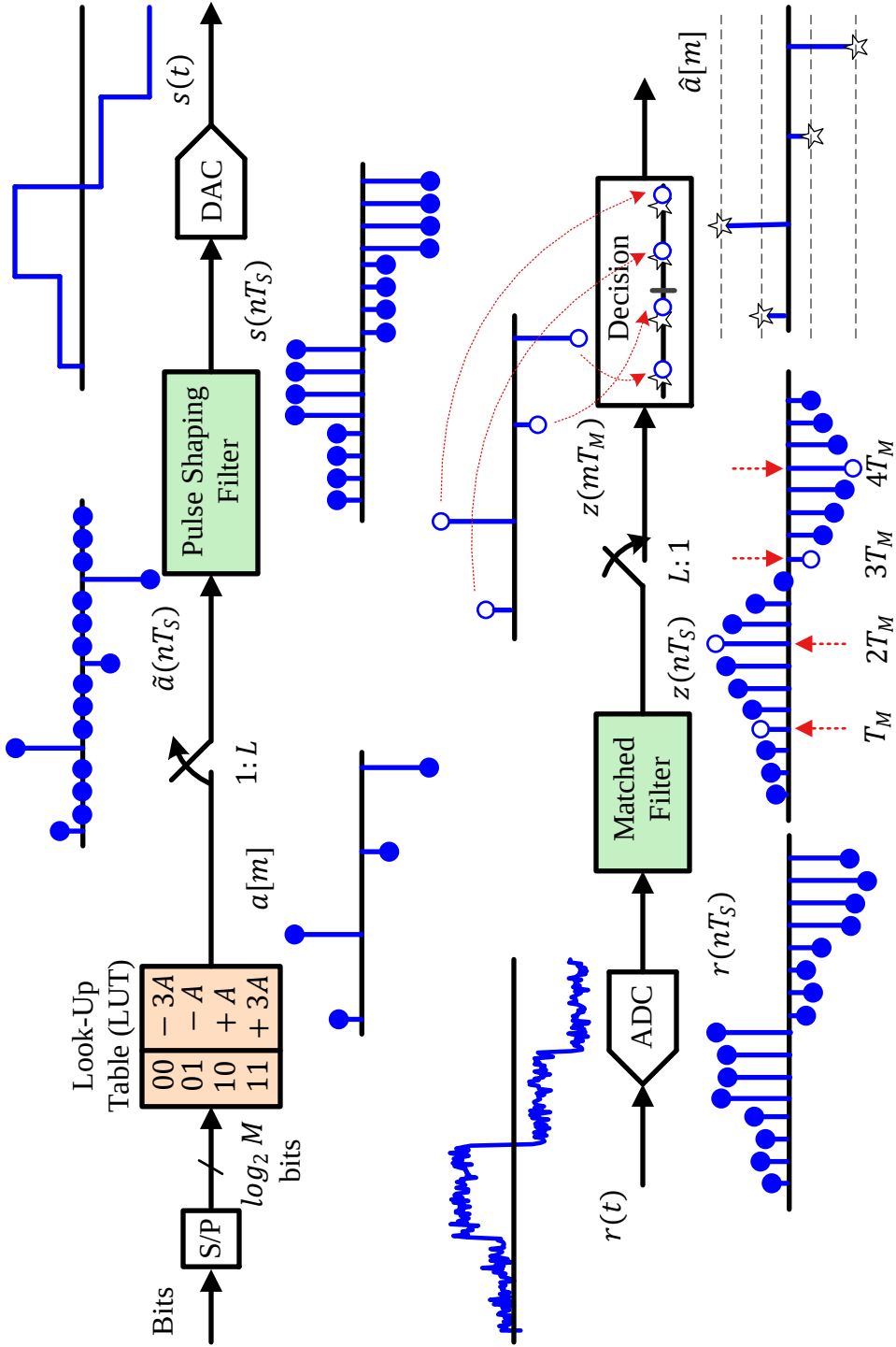


Figure 3.17: A general PAM modulator and detector

- To produce a PAM waveform, the symbol sequence $a[m]$ is converted to a discrete-time impulse train through upsampling by L where L is samples/symbol, i.e., the ratio of symbol time to sample time T_M/T_S , or equivalently, sample rate to symbol rate F_S/R_M . Let us denote such a sequence by $\tilde{a}(nT_S)$.
- The upsampling operation inserts $L - 1$ zeros between each set of two symbols after which the intermediate samples can be raised from dead with the help of a lowpass filter that suppresses all the spectral replicas except the primary one. We will see in the next section that a proper pulse shaping filter $p(nT_S)$ is also a lowpass filter and hence an extra lowpass filter is not actually required.

$$s(nT_S) = \tilde{a}(nT_S) * p(nT_S)$$

- The generated discrete-time signal $s(nT_S)$ is converted to a continuous-time signal $s(t)$ by a DAC.

PAM Detector

The received signal $r(t)$ is the same as the transmitted signal $s(t)$ but with the addition of additive white Gaussian noise (AWGN) $w(t)$. We will consider other distortions in the later chapters. The symbols are detected through the following steps illustrated in Figure 3.17.

- Through an Analog to Digital Converter (ADC), $r(t)$ is sampled at a rate of F_S samples/s to produce a sequence of T_S -spaced samples $r(nT_S)$.
- Next, $r(nT_S)$ is processed through a matched filter at the Rx side to generate $z(nT_S)$. As discussed earlier, the output of the matched filter $z(nT_S)$ is the correlation of the symbol-scaled pulse shape with an unscaled and time-reversed pulse shape $p(-nT_S)$.
- This output is downsampled by L at optimal sampling instants

$$n = mL = m \frac{T_M}{T_S} \quad (3.19)$$

to produce T_M -spaced numbers $z(mT_M)$ back from the signal. Take a special note of Eq (3.19). It will be employed over and over again.

- The minimum distance decision rule is employed to find the symbol estimates $\hat{a}[m]$.

Notice that a symbol is the basic building block of a digital communication system. Consequently, symbol time T_M is the basic unit of measurement along the time axis of such a system. While Figure 3.9 depicted sampling the output just once at the optimum instant 0 out of $-(L - 1) \leq n \leq 0$, it is true for all integer multiples of symbol time in the case of a symbol sequence, i.e., the output is sampled just once for every symbol duration at optimum instants $T_M, 2T_M, \dots$.

Note 3.6 Key samples

Carefully examine the key samples at $T_M, 2T_M, \dots$. These are the samples we are looking for in the waveform for the detection purpose. Even when the waveform has suffered from all the distortions the real world has to offer, locating these samples and mapping them back to the constellation is a beautiful process, actual details of which we will encounter throughout this text.

Let us discuss the mathematical details of this process. The matched filter output is written as

$$z(nT_S) = r(nT_S) * h(nT_S) = \sum_l r(lT_S)h(nT_S - lT_S) \quad (3.20)$$

Now in a noiseless case, Rx signal $r(nT_S)$ is the same as $s(nT_S)$.

$$s(nT_S) = \sum_i a[i]p(nT_S - iT_M)$$

Moreover, the matched filter $h(nT_S)$ is the same as $p(-nT_S)$. The output of Eq (3.20) becomes

$$\begin{aligned} z(nT_S) &= \sum_l \left[\sum_i a[i]p(lT_S - iT_M) \right] p\left\{ -(nT_S - lT_S) \right\} \\ &= \sum_i a[i] \cdot \sum_l p(lT_S - iT_M)p\left\{ lT_S - iT_M - (nT_S - iT_M) \right\} \\ &= \sum_i a[i]r_p(nT_S - iT_M) \end{aligned} \quad (3.21)$$

where $r_p(nT_S)$ comes into play from the definition of auto-correlation function in Eq (3.6). To generate symbol decisions, T_M -spaced samples of the matched filter output are required at $n = mL = mT_M/T_S$. Downsampling the matched filter output generates

$$\begin{aligned} z(mT_M) &= z(nT_S) \Big|_{n=mL=mT_M/T_S} \\ &= \sum_i a[i]r_p(mT_M - iT_M) = \sum_i a[i]r_p\left\{ (m - i)T_M \right\} \\ &= a[m] \end{aligned}$$

This is because for a rectangular pulse shape, the matched filter output is triangular with maximum occurring at $r_p[0]$, i.e., when

$$m - i = 0, \quad m = i$$

and zero at the next symbol location $r_p[L]$, see Figure 3.8.

Observe that the system shown in Figure 3.17 is a multirate system. In the PAM detector, for example, the ADC and the matched filter operate at the sample rate F_S . After the output of the matched filter is downsampled by L , the symbol decisions are made at the symbol rate $1/T_M$. Furthermore, there are some hidden assumptions in the PAM detector:

Resampling The ADC in general does not produce an integer number of samples per symbol, i.e., T_M/T_S is not an integer. A resampling system is required in the Rx chain that changes the sample rate from the ADC rate to a rate that is an integer multiple of the symbol rate.

Symbol Timing Synchronization The peak sample at the end of each symbol duration is not known in advance at the Rx and in fact does not necessarily coincide with a generated sample as well. This is because ADC just samples the incoming continuous waveform without any information about the symbol boundaries. This is a symbol timing synchronization problem which we will solve in Chapter 7.

Equalization: The above methodology works only in an AWGN channel. Multipath reflections from a wireless channel introduce ISI and distortion in the signal that need to be recovered through an equalizer. This is what we study in Chapter 8.

Matched Filter vs Correlator

We derived the concept of the matched filter in Section 3.4 starting with the definition of correlation. However, in communications jargon, a correlator is not the same as the matched filter and instead it is a block used in detectors with a slightly different functionality. Here, we elaborate the difference between a matched filter and a correlator in a communications Rx.

A correlator is a device that performs the correlation of a received signal with its template *within a given window of time*. In our context, that window of time is the symbol duration, T_M . So a correlator performs the following operations.

- It takes this sample-by-sample product and sums them together.
- Next, it samples the output of this accumulation at the optimal instant T_M .
- Finally, it *resets* itself at each sampling instant T_M , i.e., it starts computing the correlation again from zero for the next symbol. This is shown in Figure 3.18.

The matched filter, on the other hand, is just a filter that keeps convolving the input signal with the time reversed template while supplying the output at each multiple of T_M . Due to this reason, operating at L samples/symbol, both the matched filter and the correlator generate the same output at L^{th} sample but different outputs for the remaining $L - 1$ samples (that were going to be discarded anyway). This is plotted in Figure 3.18.

During the days when most of the signal processing was implemented in analog domain, the design of a correlator only required an analog mixer (multiplication) and an integrator (summation), the output of which could be sampled at multiples of T_M . On the other hand, the matched filter – while looking simpler on paper due to just a filtering operation – was more complicated due to the difficulty in designing an analog filter whose impulse response is some complicated pulse shape (we will see the examples of better pulse shapes than a rectangular one in Section 3.6) or a sequence. Hence, the correlator receiver was the preferred choice.

After the great advancement in computational power, most of the analog circuits capable of performing analog signal processing (e.g., by using resistors, capacitors, inductors, op amps and so on – *basically physics and devices*) have been replaced by powerful digital processors that can perform the required number crunching (*basically algorithms run by computer programs*) at a much better price vs performance point. Now, design and implementation of matched filters for any kind of applications is more convenient, requires less

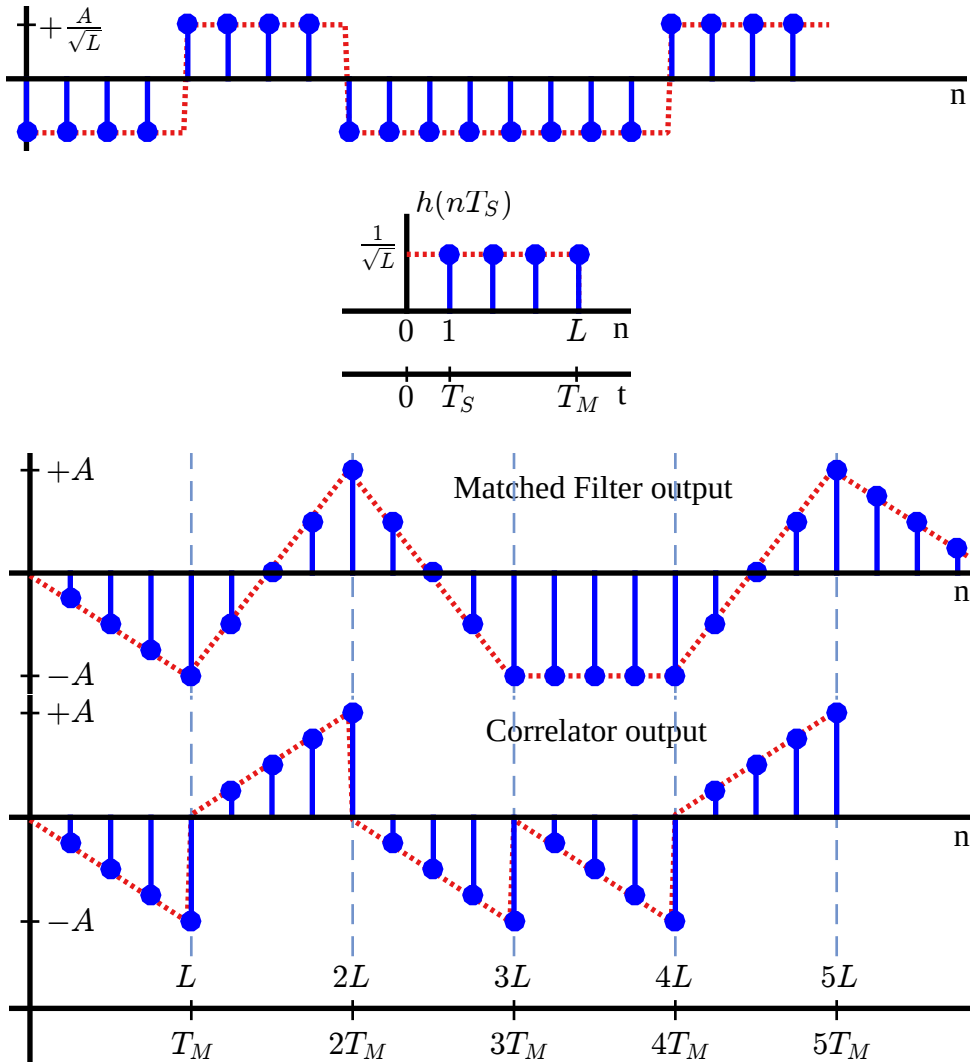


Figure 3.18: Matched filter vs correlator. The outputs are same at optimal sampling instants mT_M but different for the remaining symbol interval

bookkeeping and combines efficiently with other receiver blocks such as synchronizers and equalizers[†].

To conclude, the matched filter computes *true correlation* of the received signal with the template signal for the duration of the whole symbol sequence, while the correlator resets itself to zero every symbol time.

Having understood the basic functionality of a PAM modulator and detector, we can discuss the important topic of a pulse shaping filter.

[†]Sometimes I wonder how amazing our world looked like if a revolution like scaling down of a transistor size would have occurred in the mechanical domain as well.

3.6 Pulse Shaping Filter

We have seen above how a Tx generates a signal from the numbers and the Rx recovers those numbers from that signal. In time domain, everything looks nice and perfect.

Let us investigate the system characteristics in frequency domain. As is clear from the block diagram of a PAM system, the main component that defines the spectral contents of the signal is the pulse shape $p(nT_S)$ at the Tx. We start with our attention towards the rectangular pulse shape used so far.

Note 3.7 Frequency domain view

It is straightforward to find the DFT of many signals from definition and examples in Chapter 1, or a software routine. The discrete frequency plot as a function of k , however, can only show a limited amount of frequency content due to distinct frequency domain samples. In the spirit of simplicity we have adopted in this text, we will interpolate many of the future graphs for a finer resolution. All we have to do is choose a large value of N when plotting.

As discussed in Section 1.8, for a reasonably band-limited signal, we can approximate the continuous frequency F by discrete frequency k/N when scaled by F_S . Then, does a larger N imply that we can cover a larger frequency range in reality? No, recall from Section 1.7 that $F = F_S \cdot k/N$ and

$$-0.5 \leq \frac{k}{N} = \frac{F}{F_S} < +0.5 \quad (3.22)$$

A large N just makes the resolution k/N finer and finer. The range k/N stays within -0.5 to $+0.5$ and so does F between $-0.5F_S$ and $+0.5F_S$, until the sample rate F_S itself changes. When we illustrate the frequency domain beyond this range, then we want to look into aliasing replicas of the spectrum as well.

Spectrum of a Rectangular Pulse

In time domain, a rectangular pulse has a clear benefit that Tx symbols do not naturally invade into each other's territory. In case there are no reflections or dispersions in the channel, there are no adjacent symbols interfering into each other at the Rx as well – a phenomenon called *Inter-Symbol Interference (ISI)*. ISI simplifies the detector in Rx design because everything related to a particular m^{th} symbol remains within the symbol interval $(m-1)T_M \leq t \leq mT_M$. Or in other words, *what happens in a symbol duration stays within that symbol duration*.

As for the frequency domain, we derived an analytical expression for the DFT of a rectangular sequence in Eq (1.68) of Section 1.10.1 as

$$|P_{\text{rect}}[k]| = \frac{\sin \pi Lk/N}{\sin \pi k/N}$$

In the graph of this sinc function, nulls should occur wherever the numerator is zero, except at $k/N = 0$ where the peak exists. Now $\sin(\cdot) = 0$ when its argument is an integer multiple of π . Thus, equating the argument of the numerator to π returns the first null position.

$$\pi L \left(\frac{k}{N} \right)_{\text{null}} = \pi \quad \Rightarrow \quad \left(\frac{k}{N} \right)_{\text{null}} = \frac{1}{L}$$

Thus, the positions of nulls are $\pm 1/L, \pm 2/L, \pm 3/L, \dots$ with respect to discrete frequency k/N . In Hz, with $F_S = L/T_M$, the null positions are $\pm 1/T_M, \pm 2/T_M, \pm 3/T_M, \dots$ because

$$F_{\text{null}} = F_S \cdot \left(\frac{k}{N} \right)_{\text{null}} = F_S \cdot \frac{1}{L} = \frac{1}{T_M}$$

Finally, the range is

$$-0.5F_S = -0.5 \frac{L}{T_M} \leq F < +0.5 \frac{L}{T_M} = +0.5F_S$$

which for $L = 8$ samples/symbol produces

$$-\frac{4}{T_M} \leq F < +\frac{4}{T_M}$$

The spectrum of a rectangular pulse shape shows these deductions in Figure 3.19.

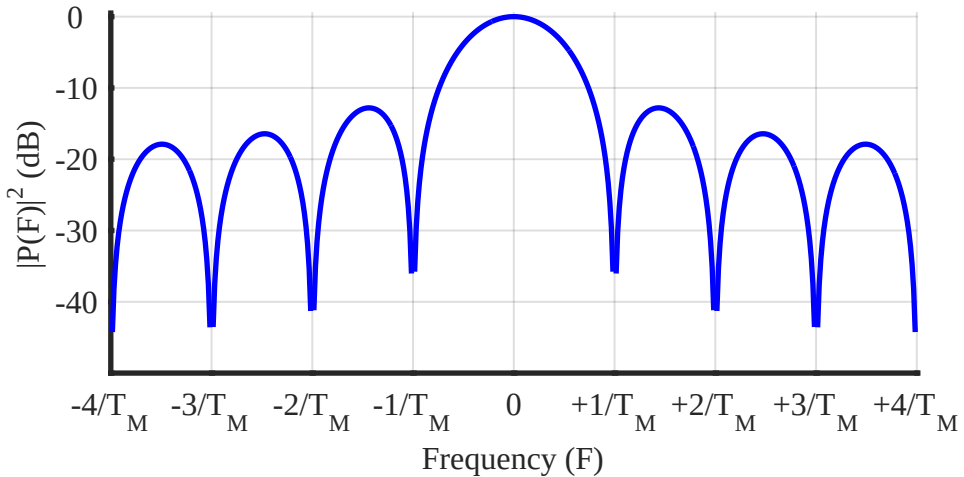


Figure 3.19: Spectrum of a rectangular pulse shape with $L = 8$ samples/symbol

A great disadvantage of a rectangular pulse shape is now visible: the sidelobe level is just 13 dB below the mainlobe, a very high value. Furthermore, the subsequent sidelobes are not decaying fast enough. If there is a communications link which occupies the drawn spectrum and the neighboring spectrum is allocated to someone else, then there will be a lot of interference between the two parties due to their sidelobe energies interfering with each other. In frequency domain, this is known as adjacent channel interference.

Just like real estate, radio spectrum is a very precious resource and must be utilized judiciously. This is why there are spectrum regulatory authorities in every country who impose strict restrictions to comply with a *spectral mask*. Even for wired channels, there is always a natural bandwidth of the medium (copper wire, coaxial cable, optical fiber) that imposes upper limits on its utilization.

Not only that a rectangular pulse shape is a poor choice due to its large spectral occupancy, another side effect of having a wider bandwidth pulse shape is that the matched filter at the Rx also has a similar bandwidth. Viewed in frequency domain, the larger the bandwidth of the Rx filter, the more noise and interference it allows into the system.

Note 3.8 Randomness of bit stream

It is important to remember that such a spectrum approximately appears at the output of the transmitter only if there is a sufficient randomness in the bit stream, i.e., bits 0 and 1 occur with equal probability and so there are enough transitions between them. Any pattern in the bit stream can alter the output spectrum significantly.

As an example, imagine a constant bit stream of 1's that is then modulated to a symbol level $+A$. After pulse scaling, the output is nothing but $+A$ times an all-ones sequence that has a Fourier Transform of an impulse at bin 0 (DC) and nothing else. In real applications, there is enough randomness in the modulated sequence and hence the spectrum will closely resemble the spectrum shown in Figure 3.19.

Reducing the bandwidth

Having known the disadvantages of a rectangular pulse shape, a superior pulse shaping filter needs to be looked for that should help in placing multiple channels adjacent to each other while minimizing inter-channel interference between them as well as noise bandwidth at the Rx. However, such advantages should not come at a price of introducing ISI in the system.

Nyquist No-ISI Criterion in Time Domain

To get that superior pulse shape, we trace the following sequence of steps through the help of Figure 3.20[†] that plots the auto-correlation $r_p(nT_S)$ of both a rectangular and a conceptual pulse shape. Auto-correlation of a pulse shape was defined in Eq (3.6) as

$$r_p(nT_S) = \sum_i p(iT_S)p(iT_S - nT_S)$$

and plots are shown in continuous-time to make the related steps below more understandable.

- Recall that signals that are narrow in time domain are wide in frequency domain, and vice versa. Accordingly, a consequence of narrow time span of the pulse shape is that it gets expanded in frequency. That is why a rectangular pulse has a large bandwidth. To decrease the bandwidth, the length of the pulse shape needs to be extended in time domain – much more than a single symbol duration T_M .
- The effect of extending the length of the pulse shape is that every symbol now interferes with a number of symbols occurring both before and after that symbol. The dreaded ISI appears! Assuming a usual even shaped pulse, every symbol contributes towards ISI in symbols towards its left and symbols towards its right on the time axis.
- To strike peace with this ISI, we exploit a key property of digital communication waveforms explained before: the critical samples at the output of matched filter occur at the end of integer multiples of symbol duration T_M . For symbol detection, only these

[†]Thanks to Eric Jacobsen for his wonderful article Ref. [8], Figure 4 of which provided the inspiration behind this figure.

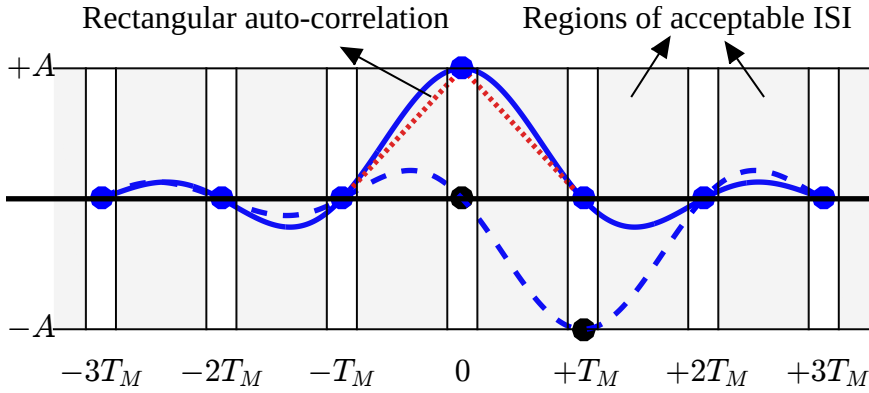


Figure 3.20: Auto-correlation functions $r_p(nT_s)$ of a rectangular pulse and a conceptual pulse extended over time

T_M -spaced samples are needed and the rest can be discarded. This is what we saw at the output of downsampler in Figure 3.17.

If all but one sample during each symbol interval T_M are discarded, they can be assigned any value without any effect on the detector output. These are the “don’t care” samples we give up for ISI to play with in the expectation that we get to shape our desired spectrum in return, and with the result that our desired T_M -spaced samples remain intact. Keep in mind that *we are talking about the output of the matched filter – and hence auto-correlation of the pulse shape* – here. The reason is to elaborate the nature of signal that is eventually downsampled to yield symbol estimates. The actual pulse shape will be extracted from this auto-correlation, as we will shortly see.

■ In the light of above discussion, three pulse shape auto-correlations are illustrated in Figure 3.20.

1. A rectangular pulse shape has a duration of T_M seconds, or L samples. Hence, its auto-correlation extending over $2T_M$ seconds is shown as a dotted red line. Notice that it has a maximum value at the desired current sampling instant and zero for adjacent two symbols. The time span, however, is too short giving rise to an unreasonably large spectrum.
2. A pulse shape auto-correlation that is expanded over many symbols is shown in blue curve, where it is scaled by a symbol $+A$. Note that at all the sampling instants (integer multiples of T_M), the effect of this particular symbol is zero. We intend to assign values to its “don’t care” samples in shaded regions to achieve a compact spectrum.
3. An adjacent symbol $-A$ scales the same pulse shape at time T_M . Observe the same zero-ISI effect on the current symbol at time 0 and on all other symbols.

Not shown in the above figure are all other symbols shaped by pulses at $2T_M, 3T_M$, and so on. But it is clear that sum total of ISI from all adjacent symbols is zero at sampling instant of every single symbol. For a concrete formulation of this zero-ISI criterion, we turn towards its mathematical foundation.

Just as in the case of rectangular pulse before, the input signal is

$$s(nT_S) = \sum_i a[i]p(nT_S - iT_M)$$

When this signal[†] is input to a matched filter $h(nT_S) = p(-nT_S)$, the output from Eq (3.21) is written as

$$\begin{aligned} z(nT_S) &= \left(\sum_i a[i]p(nT_S - iT_M) \right) * p(-nT_S) \\ &= \sum_i a[i]r_p(nT_S - iT_M) \end{aligned}$$

To generate symbol decisions, T_M -spaced samples of the matched filter output at $n = mL = mT_M/T_S$ are

$$\begin{aligned} z(mT_M) &= z(nT_S) \Big|_{n=mL=mT_M/T_S} \\ &= \sum_i a[i]r_p(mT_M - iT_M) = \sum_i a[i]r_p\{(m-i)T_M\} \\ &= \underbrace{a[m]r_p(0T_M)}_{\text{current symbol}} + \underbrace{\sum_{i \neq m} a[i]r_p\{(m-i)T_M\}}_{\text{ISI}} \end{aligned} \quad (3.23)$$

In the above equation, the first term is the currently desired symbol and the second term is Inter-Symbol Interference (ISI). It is obvious that ISI can be zero if the pulse auto-correlation satisfies the condition

$$r_p(mT_M) = \begin{cases} 1, & m = 0 \\ 0, & m \neq 0 \end{cases} \quad (3.24)$$

This is Nyquist no-ISI criterion in time domain – a mathematical expression of the same concept we explained above: to obtain zero ISI, the pulse auto-correlation should pass through zero for all integer multiples of T_M before and after the current symbol.

Note 3.9 Nyquist filter

A filter with impulse response coefficients satisfying Nyquist no-ISI criterion is called a *Nyquist filter*. This is also usually called a Nyquist pulse but the Nyquist filter is a better term, as this is actually not the pulse shape but the auto-correlation of the underlying pulse

[†]Although this relation was derived in Eq (3.17) for a pulse with duration T_M , it is also true for longer pulses because

$$\begin{aligned} \sum_i a[i]p(nT_S - iT_M) &= a[0]p(nT_S) + a[1]p(nT_S - T_M) + \\ &\quad a[2]p(nT_S - 2T_M) + \dots \\ &= \left(a[0]\delta[nT_S] + a[1]\delta[nT_S - T_M] + \right. \\ &\quad \left. a[2]\delta[nT_S - 2T_M] + \dots \right) * p(nT_S) \end{aligned}$$

This is because convolution between any signal and a unit impulse results in the same signal. Therefore, even for longer pulses, scaling each individual symbol with its own pulse shape is equivalent to convolution of upsampled symbol stream with a pulse shaping filter.

shape. The coefficients of the pulse shape itself will be derived later.

When Nyquist no-ISI criterion in time domain is satisfied, plugging Eq (3.24) into Eq (3.23) gives

$$z(mT_M) = a[m]$$

i.e., the downsampled matched filter output maps back to the Tx symbol in the absence of noise. This is the crux of digital communications theory.

Nyquist No-ISI Criterion in Frequency Domain

Throughout the text, we have been avoiding complicated mathematical derivations. We will do the same here and instead of proving Nyquist no-ISI theorem which connects time and frequency domain properties of pulse auto-correlation, we again take the intuitive route.

Remember that the point of this whole exercise was to limit the enormous bandwidth occupied by a time domain rectangular pulse. In other words, the quest is to design a pulse shape such that it complies with the spectral mask within a channel bandwidth B . Nyquist no-ISI criterion in frequency domain helps here.

As long as we operate at a sample rate F_S , the spectral replicas are spaced at $F_S = L/T_M$ apart. But since we downsample $r_p(nT_S)$ by $L = T_M/T_S$ to get $r_p(mT_M)$, our sample rate changes from F_S at the output of matched filter to $1/T_M$ after the downsampler. To see what happens in frequency domain, observe the following.

- In discrete domain, $r_p(mT_M)$ in Eq (3.24) is nothing but a single impulse at time 0. This can be seen as the blue dots in Figure 3.21a. Hence, its DFT is an all-ones rectangular sequence in frequency domain as

$$R_p[k] = 1$$

In terms of continuous frequency, we say that it is equal to a constant 1.

- Moreover, we know from Section 2.7.1 that a consequence of downsampling by L is appearance of new spectral aliases $F_S/L = 1/T_M$ apart from each other.

Combining the above two facts produces Figure 3.21b illustrating this relationship between symbol-spaced pulse auto-correlation and its Fourier Transform.

This is exactly what Nyquist in Ref. [9] showed: a necessary and sufficient condition for zero-ISI condition of Eq (3.24) is that the Fourier transform $R_p(F)$ of the pulse auto-correlation satisfies the condition

$$\begin{aligned} \sum_{i=-\infty}^{\infty} R_p\left(F + \frac{i}{T_M}\right) &= \cdots + R_p\left(F + \frac{2}{T_M}\right) + R_p\left(F + \frac{1}{T_M}\right) + \\ &R_p(F) + R_p\left(F - \frac{1}{T_M}\right) + R_p\left(F - \frac{2}{T_M}\right) + \cdots \\ &= T_M \end{aligned} \quad (3.25)$$

where the factor T_M is reminiscent of the factor N in iDFT definition. Since Figure 3.21b depicts that *the spectrum of the pulse auto-correlation should be a constant T_M* , we perform the following steps.

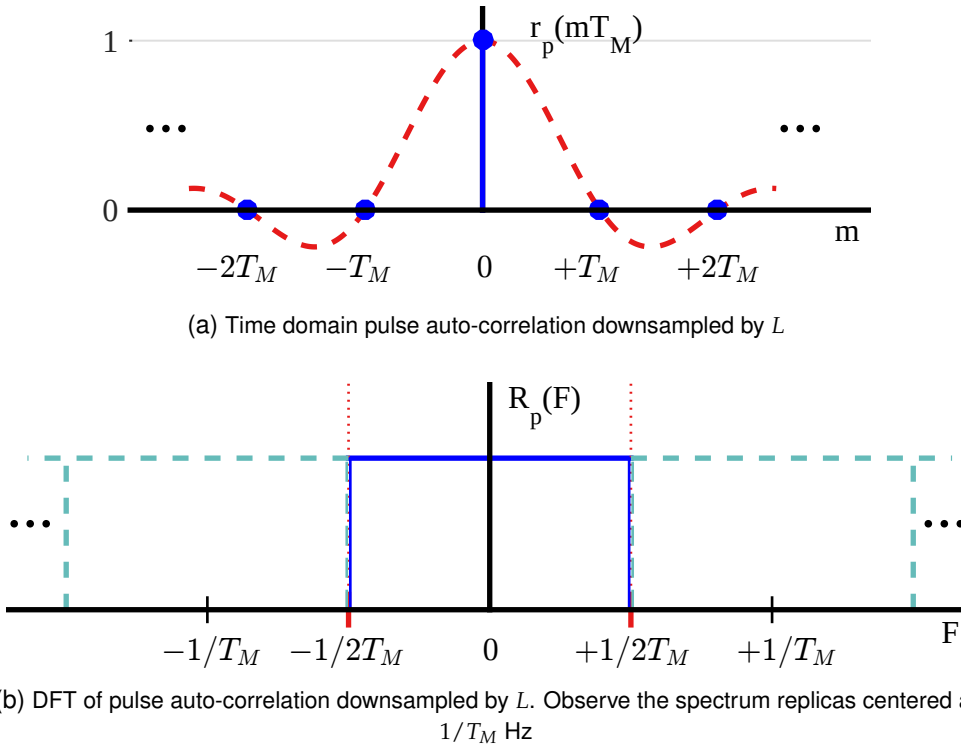


Figure 3.21: Time and frequency domain interpretations of Nyquist no-ISI criterion

- Draw the primary spectrum $R_p(F)$ whose range is $-0.5F_s \leq F < +0.5F_s$.
- Draw its shifted replicas at frequencies $\pm 1/T_M$. Repeat the same for all integer multiples of $\pm 1/T_M$ from $-\infty$ to $+\infty$.
- Add all these replicas together.
- If this sum results in a constant T_M , only then the time domain pulse auto-correlation will satisfy the no-ISI condition (3.24).

Figure 3.22 shows two spectra, one of which satisfies the Nyquist no-ISI criterion while the other does not. The spectrum on the left is constant because the bandwidth is as wide as allowed by the sampling theorem (which is $B_{\max} = 1/2T_M$ for sample rate $1/T_M$). This not only avoids aliasing but a flat spectrum within $-0.5/T_M \leq F < +0.5/T_M$ produces a flat spectrum in the range $-1.5/T_M \leq F < -0.5/T_M$, $+0.5/T_M \leq F < +1.5/T_M$, and so on. Consequently, the cumulative spectrum satisfies Eq (3.25) as a constant function of frequency from $-\infty$ to $+\infty$.

We can also see that the signal bandwidth B in this particular case is

$$B = \frac{1}{2T_M}$$

A word of caution: Never confuse a rectangular pulse shape in time domain with a pulse auto-correlation that is a rectangle in frequency domain. The above expression is the maximum bandwidth after which aliasing will occur from spectral replicas. This also sets the

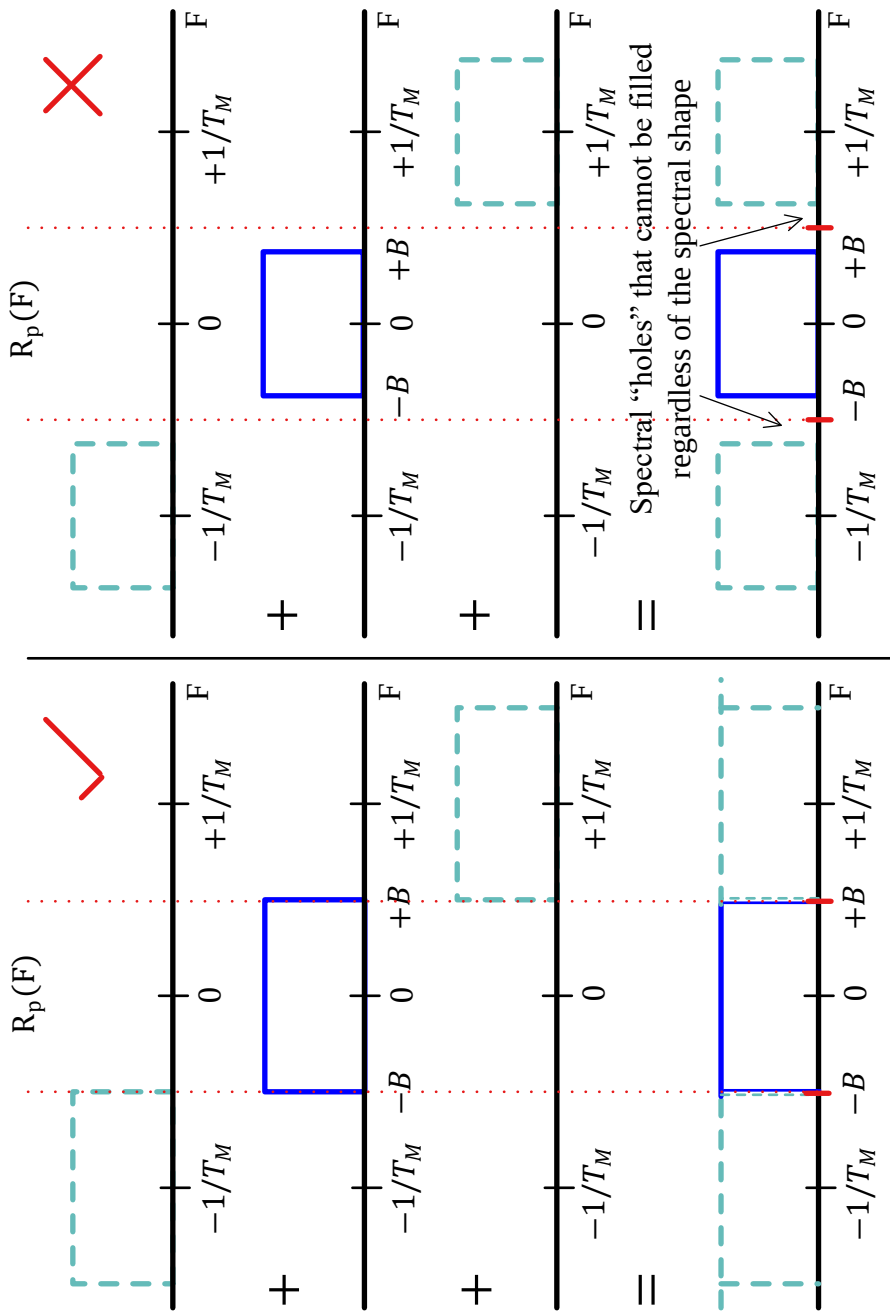


Figure 3.22: The spectrum on the left satisfies the Nyquist no-ISI criterion in frequency domain, while the one on the right does not

fundamental limit between the bandwidth B and symbol rate $1/T_M$ as

$$\left\{ \frac{1}{T_M} \right\}_{\max} \leq 2B \quad (3.26)$$

Hence, with zero ISI, a maximum symbol rate $1/T_M$ equal to $2B$ symbols/second can be supported within a bandwidth $B^\dagger$.

Interestingly, *this maximum bandwidth $1/2T_M$* allowed by sampling theorem is also the *minimum bandwidth* allowed by Nyquist no-ISI criterion in frequency domain through Eq (3.25). Any smaller than this, and it becomes impossible to obtain an overall flat spectrum. That is shown on the right of Figure 3.22 where there are spectral “holes” that cannot be filled regardless of the spectral shape.

Coefficients of an Ideal Pulse Auto-correlation

Above, we said that a filter whose impulse response coefficients satisfy Nyquist no-ISI criterion is a Nyquist filter, which is actually the auto-correlation $r_p(nT_S)$ of the underlying pulse shape. From Figure 3.22 and the discussion in the last section, *we find our ideal Nyquist filter*: a pulse auto-correlation $r_p(nT_S)$ whose spectrum $R_p(F)$ is a rectangle from $-1/2T_M$ to $+1/2T_M$. This filter is shown in Figure 3.21 and left of Figure 3.22.

To derive the time domain representation of such a filter, remember that a rectangle in frequency domain is a sinc in time domain due to time frequency duality. This rectangle covers the entire continuous frequency range from $F = -1/2T_M$ to $+1/2T_M$ (or discrete frequency range $k/N = -0.5$ to $+0.5$) yielding $L = N$ in Eq (1.68). Note that this L is not samples/symbol, just the sequence length in that equation. There is a scaling factor of $1/N$, furthermore, from the definition of iDFT. Combining these facts,

$$s[n] = \frac{1}{N} \frac{\sin \pi n}{\sin \pi n/N}$$

For our purpose with a sample rate of $1/T_M$ so far,

$$r_p[m] = \frac{1}{N} \frac{\sin \pi m}{\sin \pi m/N}$$

It satisfies Nyquist no-ISI criterion in Eq (3.24) because it is equal to 1 at $m = 0$ and zero for $m \neq 0$ and shown as blue dots in Figure 3.21.

Using the identity $\sin \theta \approx \theta$ for small θ , the term in the denominator is quite small as compared to numerator due to division by N . Then, the above equation can be written as

$$r_p[m] \approx \frac{1}{N} \frac{\sin \pi m}{\pi m/N} = \frac{\sin \pi m}{\pi m}$$

This is the time domain waveform sampled at rate $1/T_M$. As we are interested in time domain coefficients $r_p[n]$ of this ideal pulse auto-correlation, we sample the waveform at

[†]An extension of such a framework is Faster Than Nyquist (FTN) signaling in which this fundamental limit is broken by design to send more symbols within the same bandwidth. The cost of such a setup is a loss of orthogonality, i.e., occurrence of ISI due to closer packing of symbols than T_M which is removed at the Rx through more computationally complex algorithms, usually by employing an iterative error correcting code.

rate $F_S = L/T_M$ instead of just $1/T_M$, an increase by a factor of L . Correspondingly, a decrease in sample time implies we plug $m = n/L$ in the above equation.

$$r_p[n] = \frac{\sin \pi n / L}{\pi n / L} \quad (3.27)$$

These are the coefficients of our ideal pulse auto-correlation, which extends from $-\infty$ to $+\infty$ in time domain as shown in Figure 3.23.

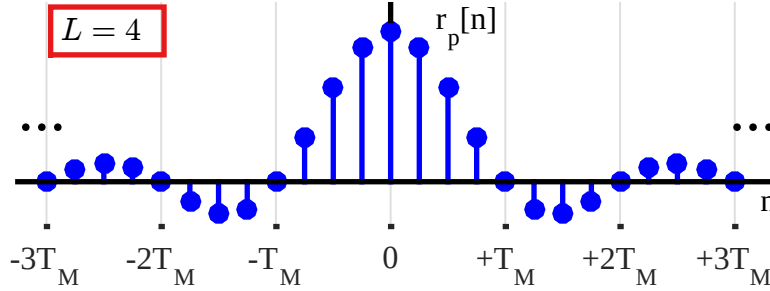


Figure 3.23: Coefficients of an ideal pulse auto-correlation with $L = 4$ samples/symbol

There are two main problems with this ideal pulse auto-correlation, *both arising due to a very slow rate of decay for the tail*.

Truncation Error Any practical system cannot use a filter with an infinite number of coefficients like the one shown in Figure 3.23. Hence, the filter impulse response must be truncated at a few symbols to the left and a few symbols to the right of the current symbol. To ensure that this truncated filter closely approximates the ideal impulse response, only very small values can be ignored. That generates a large filter length N .

Timing Errors We know that an optimal timing instant coincides with symbol boundaries. Timing synchronization block in a Rx is responsible for extracting this symbol-aligned clock. However, even a small error in timing synchronization output causes a significant increase in ISI due to large value of samples at the tail interacting with neighbouring modulated pulses. This is demonstrated later in Section 7.1. We want a quicker tail decay as it leads to less errors arising from the timing jitter when sampling adjacent pulses.

To address these issues, we take the reverse route now. We started with the aim of reducing the bandwidth of a rectangular pulse shape. We found an ideal pulse auto-correlation with a rectangular spectrum that satisfies Nyquist no-ISI criterion. To overcome the limitations in time domain associated with it, we seek to trade-off some bandwidth with a desirable time domain behaviour.

Raised Cosine (RC) Filter

So the target is to (slightly) increase the bandwidth now. Remember that we observed regions of acceptable ISI in Figure 3.20 and deduced that these “don’t care” samples can be given up for ISI to play with in the expectation that we get to shape our desired spectrum in

return, while our T_M -spaced samples remain intact. *If there is a relaxation and a restriction in time domain, then intuitively there should be a corresponding relaxation and restriction in frequency domain as well.* We have found the restriction to be a flat spectrum across the whole frequency range. The relaxation is that how this spectrum becomes flat is a matter of no concern. The actual spectral shape is a “don’t care” case, as long as the sum of all shifted replicas is flat.

It is clear that we cannot reduce the bandwidth below the fundamental limit $1/2T_M$, see Figure 3.22. The only way to alter the spectral shape is by expanding the bandwidth beyond $1/2T_M$ in such a way that the sum total of the replicas again becomes flat. For this purpose, an odd symmetry around the point $1/2T_M$ is required. The advantage of such odd symmetry is that the spectral magnitudes before and after $1/2T_M$ are 180° rotated versions of each other. In other words, the spectral copies at $\pm 1/T_M$ fold around $\pm 1/2T_M$ into the original bandwidth. That supplies the additional amplitude required to bring the spectrum in a flat shape, as illustrated in Figure 3.24.

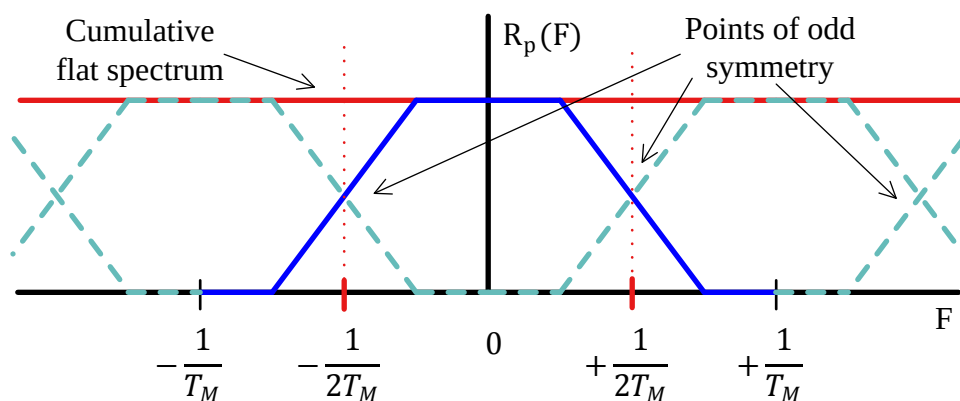


Figure 3.24: Odd symmetry around frequency points $\pm 1/2T_M$ adding up to a flat spectrum

Higher frequency components of a signal arise from abrupt changes in time domain, which was the reason of large spectral occupancy of a rectangular pulse shape. Similarly, long tails in a time domain signal arise from abrupt transition in the flat spectrum as in Figure 3.21b. Also recall that the length of an FIR filter depends on its transition bandwidth.

Having located the abrupt transition bandwidth around $\pm 1/2T_M$ as the root cause, we can extend the bandwidth of the pulse auto-correlation in any shape as long as it has odd symmetry around the points $\pm 1/2T_M$. The purpose is to smooth out its spectrum so that it has a short time domain support.

The smoothest spectral shape one can imagine is a sinusoid. If such a spectral taper is convolved with the ideal rectangular spectrum, the discontinuity in the spectrum can be removed. Since a half-cosine is an even symmetric shape, it is shown to be convolved with the rectangular spectrum in Figure 3.25. The width of the half-cosine is α/T_M where $0 \leq \alpha \leq 1$ which forms the transition bandwidth of the resultant spectrum and its even symmetric shape preserves the odd symmetry around $\pm 1/2T_M$. As a consequence of this odd symmetry, this is also a Nyquist filter. The resultant spectrum out of this convolution is discussed soon.

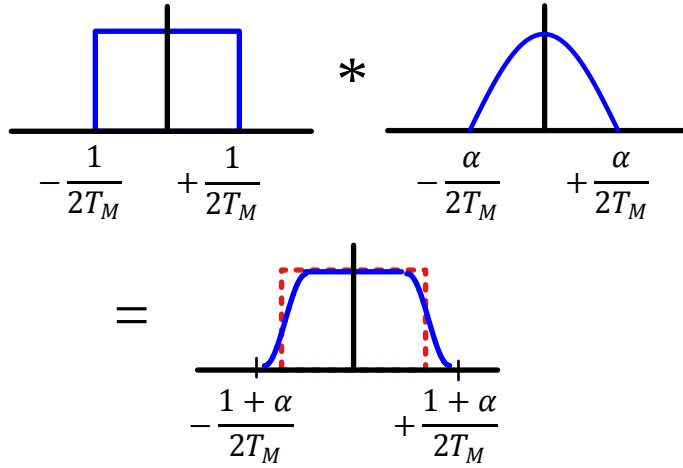


Figure 3.25: Spectral convolution of ideal Nyquist spectrum with even symmetric half-cosine

It can be observed that the effect of spectral convolution is an increase in bandwidth from $1/2T_M$ to $(1 + \alpha)/2T_M$ (convolution length in continuous domain is the sum of lengths of the two signals). Since α specifies the bandwidth beyond the minimum bandwidth $1/2T_M$ (Nyquist frequency or folding frequency), it is called *excess bandwidth* or *roll-off factor*. For example, when $\alpha = 0.25$, the total bandwidth is 25% more than the minimum and when $\alpha = 1$, the total bandwidth is exactly 100% more than, or twice, the minimum. Typical values of α range from 0.1 to 0.3.

In time domain, this is equivalent to product of a sinc signal with an even signal of time[†]. The sinc signal guarantees the zero crossings as they cannot be moved by multiplication, and the even signal dampens the long tails in time. The higher the bandwidth, the narrower this signal in time and hence faster the decay. The exact diagram of this process is drawn soon in Figure 3.27a.

First, consider $\alpha = 1$, the widest bandwidth case. Here, a rectangle of spectral width $1/T_M$ gets convolved with a half-cosine of width $1/T_M$. The convolution of the resultant spectrum again generates a cosine shape from $F = -1/T_M$ to $+1/T_M$ for a total width of $2/T_M$. This is plotted in Figure 3.26, where smooth frequency transition can be seen with an odd symmetry around $\pm 1/2T_M$. It is commonly known as a *Raised Cosine (RC)* filter and has nothing to do with an RC circuit consisting of a resistance and a capacitor. The name raised cosine comes from the fact that *the transition bandwidth of the spectrum consists of a cosine raised by a constant to make it non-negative* as shown in Figure 3.26. Keep in mind that it is a raised cosine in frequency domain, not time domain.

Let us explore its mathematical expression. Due to the way it is usually written in literature, the formula looks more complicated than it actually is. Recall that a sinusoid in time domain is written as

$$s(t) = \cos(2\pi \underbrace{F}_{=1/T, \text{ inverse period}} \underbrace{t}_{\text{independent variable}})$$

Inverse period by definition is frequency; however, I devised this expression since we are

[†]Since the spectrum is real and even, the time domain signal is also real and even.

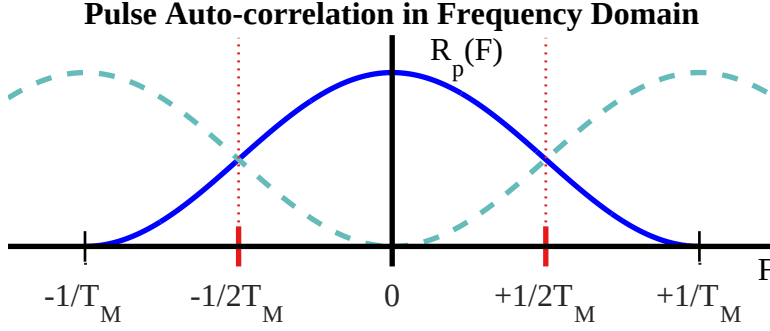


Figure 3.26: Raised Cosine spectrum with excess bandwidth $\alpha = 1$

going to use it into frequency domain. A frequency domain sinusoid will just interchange these roles, with inverse period and independent variable defined in terms of frequency. We change the above equation as follows.

- A cosine from $F = -1/T_M$ to $+1/T_M$ implies that the period in frequency is $2/T_M$ (the inverse of which is $T_M/2$).
- It is raised by a constant 1 to make it non-negative, as the smallest value of a cosine is -1 .
- Then, $1 + \cos(\cdot)$ will become 2 at $F = 0$, so it is scaled by $1/2$ to get a unity gain.
- Finally, a scaling factor of T_M comes from Nyquist no-ISI criterion in frequency domain, see Eq (3.25).

Thus, the expression for this cosine spectrum on the positive half is given as

$$\begin{aligned}
 R_p(F) &= \frac{T_M}{2} \left[1 + \cos\left(2\pi \underbrace{\frac{1}{(2/T_M)}}_{\text{inverse period}} \underbrace{F}_{\text{independent variable}}\right) \right] \\
 &= \frac{T_M}{2} \left[1 + \cos\left(2\pi \cdot \frac{T_M}{2} \cdot F\right) \right] \quad 0 \leq F \leq +\frac{1}{T_M}
 \end{aligned}$$

The overall spectral shape from $-1/T_M$ to $+1/T_M$ can be given as

$$R_p(F) = \frac{T_M}{2} \left[1 + \cos\left(2\pi \cdot \frac{T_M}{2} \cdot |F|\right) \right] \quad -\frac{1}{T_M} \leq F \leq +\frac{1}{T_M} \quad (3.28)$$

where the negative half of the spectrum is the same as the positive half, and hence the term $|F|$.

We obtain its time domain expression through the following steps:

- In frequency domain, the constant term and cosine in the above expression do not have a frequency support from $-\infty$ to ∞ , but only from $-1/T_M$ to $+1/T_M$.
- In frequency domain, that is equivalent to multiplication with a rectangular window of the same width.

- Multiplication in frequency domain is convolution in time domain.
- In time domain, the constant term translates to an *impulse* at time location 0, and $\cos(\cdot)$ results in *two impulses* with half that amplitude at time locations (inverse period) $\pm T_M/2$.
- The rectangular window of width $2/T_M$ that defines its support in frequency is a sinc signal in time domain with zero crossings at integer multiples of $\pm T_M/2$.
- Convolution of a signal with an impulse is the signal itself. When convolved with sinc arising from the window, this generates three sinc signals: one at time 0 (convolution of the above sinc with impulse from constant term) while two at time $\pm T_M/2$ (convolution of the above sinc with impulses from the cosine).

The above sequence of steps result in time domain signal of an extended bandwidth pulse auto-correlation shown in Figure 3.27. The reason of choosing a frequency support

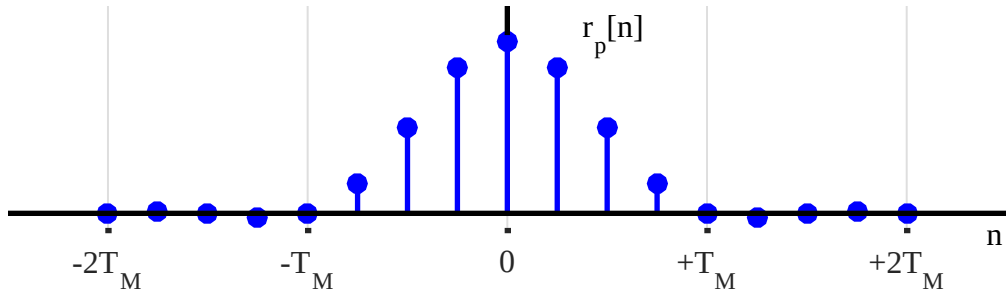
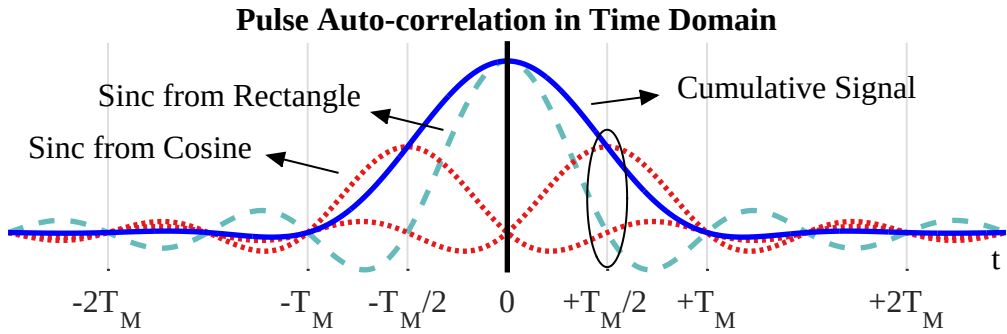


Figure 3.27: Time domain view of pulse auto-correlation of Figure 3.26 and its sampled coefficients

$-1/T_M \leq F \leq +1/T_M$ is now clear. With support $B = 2/T_M$ in frequency, the sinc resulting from cosine adds out of phase at $t = T_M/2$ with the sinc from the constant term. The signs of their tails are opposite to each other and that is what brings down the levels of long tails of

an ideal pulse auto-correlation. Furthermore, since from both constant term and cosine have zero crossings at integer multiples of $\pm T_M/2$, as marked through the ellipse in the figure. Nevertheless, a time shift of $T_M/2$ for the sinc from cosine causes the cumulative signal to pass through zero at integer multiples of $\pm T_M$. This is how this pulse auto-correlation fulfills the Nyquist no-ISI criterion in time domain. Finally, sampling the pulse auto-correlation at $L = 4$ samples/symbol produces the discrete-time coefficients that can be used for pulse shaping in digital domain.

Compare this discrete-time signal with the ideal Nyquist filter in Figure 3.23. Now we can truncate the filter to a small length without much loss in accuracy. In addition, a sampling time misalignment at the Rx will have a relatively less influence on the current symbol. Remember that the cost of this relief is the wider bandwidth $1/T_M$. Observing this trend, we can think of striking a trade-off between the filter length and bandwidth expansion. The excess bandwidth α comes into play here.

For this purpose, the excess bandwidth α can be reduced from $\alpha = 1$ to a lower value. By reducing the bandwidth, the following sequence occurs.

- When $\alpha < 1$, the period in frequency of the half-cosine (on each side of zero) in Figure 3.26 decreases, while still maintaining odd symmetry around $\pm 1/2T_M$.
- As shown in Figure 3.28, a decrease in frequency period results in three distinct regions of the spectrum: a half-cosine in the positive part of spectrum

$$+\frac{1-\alpha}{2T_M} \leq F \leq +\frac{1+\alpha}{2T_M},$$

a half-cosine in the negative portion of the spectrum

$$-\frac{1+\alpha}{2T_M} \leq F \leq -\frac{1-\alpha}{2T_M},$$

and a constant term

$$-\frac{1-\alpha}{2T_M} \leq F \leq +\frac{1-\alpha}{2T_M}.$$

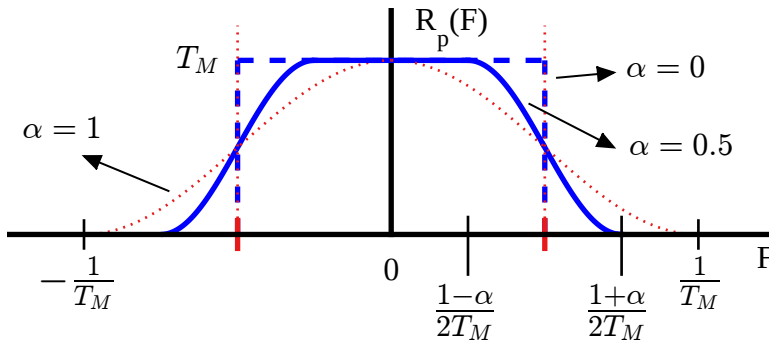


Figure 3.28: Raised Cosine spectrum with different excess bandwidths α . Note that $\alpha = 1$ is the maximum bandwidth case, while $\alpha = 0$ coincides with a rectangular spectrum of ideal Nyquist filter as in Figure 3.21 and Figure 3.22

Note that when $\alpha = 1$, the bandwidth was $1/T_M$ which can be written as $(1 + 1)/2T_M$. A decrease in α results in new bandwidth $(1 + \alpha)/2T_M$ as drawn in Figure 3.28. Also observe that $\alpha = 0$ case coincides with a rectangular spectrum of ideal Nyquist filter as in Figure 3.21 and Figure 3.22. This was the minimum bandwidth allowed by Nyquist no-ISI criterion.

- A decrease in bandwidth results in multiple windowed cosines and a constant. To avoid complication, its detailed analysis is not necessary. We just note that as α gets smaller and smaller, the sincs in time domain resulting from windowed cosine in frequency move away from each other, thus causing less tail suppression as compared to $\alpha = 1$ case. In conclusion, the excess bandwidth α gives the system designer a trade-off between reduced bandwidth and time domain tail suppression.

Taking clue from Eq (3.28), the general expression for a Raised Cosine can be written as

$$R_p(F) = \begin{cases} T_M & 0 \leq |F| \leq \frac{1-\alpha}{2T_M} \\ \frac{T_M}{2} \left[1 + \cos \left\{ 2\pi \cdot \frac{T_M}{2\alpha} \left(|F| - \frac{1-\alpha}{2T_M} \right) \right\} \right] & \frac{1-\alpha}{2T_M} \leq |F| \leq \frac{1+\alpha}{2T_M} \\ 0 & |F| \geq \frac{1+\alpha}{2T_M} \end{cases} \quad (3.29)$$

The above equation looks intimidating but it is not. There are only three minor differences from Eq (3.28).

1. The first term is a constant occupying the bandwidth left behind by the half-cosine. It is equal to T_M due to plugging $F = 0$ in the expression.
2. $T_M/2\alpha$ in the middle term shows a reduction in bandwidth and a spectral shift of cosine center from $F = 0$ to $F = (1 - \alpha)/2T_M$. This is the edge of the passband now and $(1 - \alpha)/2T_M$ is called the *passband frequency*.
3. The final term shows no bandwidth occupied from $(1 + \alpha)/2T_M$ onwards. This is the start of the stopband and $(1 + \alpha)/2T_M$ is called the *stopband frequency*.

Now it is clear why an oversampling factor of L samples/symbol – a sampling rate of $L \cdot 1/T_M$ – is used in digital communication systems. Applying sampling theorem to the stopband frequency $(1 + \alpha)/2T_M$, the sampling theorem sets the minimum sampling rate at $(1 + \alpha)/T_M$. So at least $L = 2$ (the closest integer to any $1 + \alpha$) or more samples are required per symbol.

To obtain its time domain expression, recall Figure 3.25 where we found that the resultant waveform in time domain is equivalent to product of a sinc signal with an even signal of time. Without exploring the expression for that even signal, we just state the final form that can be derived from the inverse transform of Eq (3.29).

$$r_p[n] = \frac{\sin(\pi n/L)}{\pi n/L} \cdot \frac{\cos(\pi \alpha n/L)}{1 - (2\alpha n/L)^2} \quad (3.30)$$

This is drawn in Figure 3.29 for different values of $\alpha = 0, 0.5$ and 1 . Note the simultaneous zero crossings of all waveforms at integer multiples of T_M . Also, plugging $\alpha = 0$ above produces the coefficients of a sinc signal, the same ideal Nyquist filter in Eq (3.27).

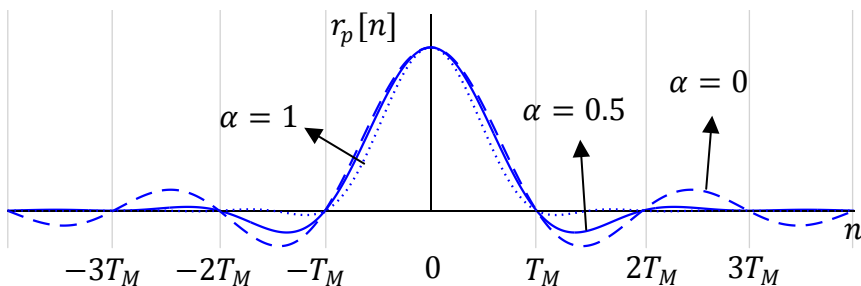


Figure 3.29: Raised Cosine waveform in time domain with different excess bandwidths α with $L = 64$ samples/symbol. Notice the simultaneous zero crossings of all waveforms at integer multiples of T_M

The above equation becomes indeterminate for $n = 0$ and $n = \pm L/(2\alpha)$. It can be shown that

$$r_p[0] = 1$$

$$r_p\left[\pm \frac{L}{2\alpha}\right] = \frac{\alpha}{2} \sin \frac{\pi}{2\alpha}$$

This is done by using a mathematical technique called L'Hôpital's rule in which the derivative of both the numerator and the denominator is taken before plugging in those values for n .

Note 3.10 A Summary

Figure 3.30 summarizes our sequence of thoughts so far. A rectangular pulse has large spectral sidelobes due to which we constrain the spectrum into a brickwall shape but that raises long tails in time domain. Consequently, introducing an excess bandwidth α is a compromise between the two extremes.

Resulting Pulse Shape

Until now in this section, everything we have discussed so far was about pulse auto-correlation that satisfies Nyquist no-ISI criteria and is also called a Nyquist filter. However, *how to derive the actual pulse shape is still not known*. This is what we intend to find out next.

We start with a simple question: where should a Raised Cosine filter be placed in the system? There are only three possible choices.

Receiver The most straightforward way to incorporate it is to have it at the Rx. However, then a significant disadvantage is that there would be no mechanism to control the spectral sidelobes at the Tx. The spectral shaping required to minimize the out-of-band energy cannot be avoided.

Transmitter If the spectrum is fully shaped at the Tx side, then any additional filtering to eliminate noise and interference at the Rx will not be “matched” to the incoming signal causing ISI. In the imaginary case of no filtering at all at the Rx, the noise and energy from adjacent channels will enter in the Rx in addition to unavoidable in-band noise. This significantly reduces the SNR as well as causing other issues such as increasing

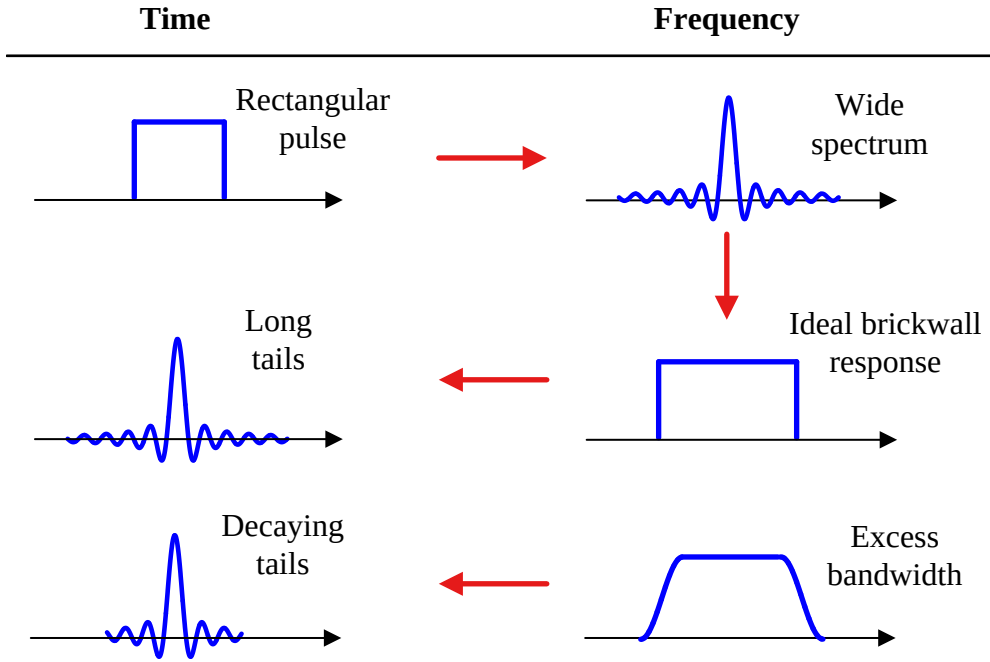


Figure 3.30: A sequence of thoughts for pulse shape design procedure

the interference sensitivity and dynamic range (adjacent channel energy can be quite higher than the desired band). So the Rx filter must be as compact as possible around the Tx spectrum so that maximal amounts of noise and adjacent channel interference can be eliminated.

Both The solution then is to split the Raised Cosine spectrum into two parts, one at the Tx to control the spectrum and the other at the Rx to reject noise and adjacent channels but still have zero-ISI cumulative response. Remember that the pulse auto-correlation can be given by convolution formula as

$$r_p(nT_S) = p(nT_S) * p^*(-nT_S)$$

The pulse shape $p(nT_S)$ can be placed at the Tx and its matched filter $p^*(-nT_S)$ at the Rx (the same as $p(nT_S)$ as the pulse shape is real). Above, the response of the two filters in cascade is the convolution of their impulse responses, which implies multiplication of their frequency responses.

$$R_p(F) = P(F) \cdot P^*(F) = |P(F)|^2$$

From above equation, the frequency response of the actual pulse shape can be derived as

$$P(F) = \sqrt{R_p(F)} \quad (3.31)$$

A significant advantage of this arrangement is that the matched filter simultaneously maximizes the Rx SNR.

In the case of Raised Cosine pulse auto-correlation, the underlying pulse shape is called *Square-Root Raised Cosine (SRRC)* pulse or filter, where the square-root is in frequency domain. Referring to Eq (3.29), taking the square-root does not affect 1 and 0 values. Using the identity $0.5(1 + \cos 2\theta) = \cos^2 \theta$ and taking the square-root on both sides adjusts the middle term as

$$P(F) = \begin{cases} \sqrt{T_M} & 0 \leq |F| \leq \frac{1-\alpha}{2T_M} \\ \sqrt{T_M} \cos \left\{ 2\pi \cdot \frac{T_M}{4\alpha} \left(|F| - \frac{1-\alpha}{2T_M} \right) \right\} & \frac{1-\alpha}{2T_M} \leq |F| \leq \frac{1+\alpha}{2T_M} \\ 0 & |F| \geq \frac{1+\alpha}{2T_M} \end{cases} \quad (3.32)$$

For various values of excess bandwidth α , this is shown in Figure 3.31. For $\alpha = 0$, there is no difference between a Raised Cosine and a Square-Root Raised Cosine filter due to a rectangular spectrum. Also notice from Eq (3.32) that *the transition band of a Square-Root Raised Cosine pulse is a quarter cycle of a cosine as compared to a half-cosine of a Raised Cosine filter*, which can be seen as follows. From the equation, the inverse period of the cosine is $T_M/4\alpha$ and hence the period in frequency domain is $4\alpha/T_M$. However, it exists in the region

$$\frac{1+\alpha}{2T_M} - \frac{1-\alpha}{2T_M} = \frac{\alpha}{T_M}$$

Compared to $4\alpha/T_M$, the transition band is a quarter cycle of a cosine. It has implications which we shortly see.

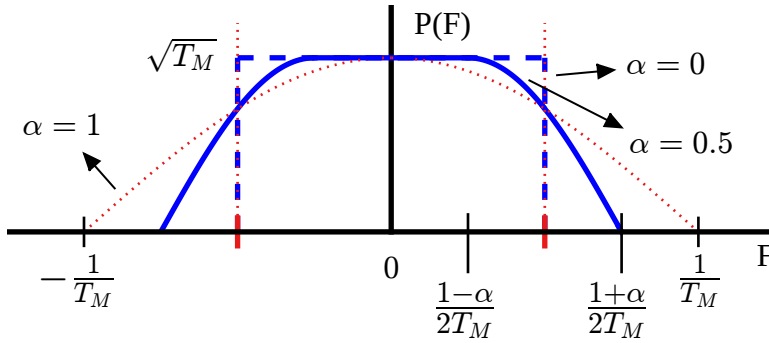


Figure 3.31: Square-Root Raised Cosine (SRRC) spectrum with different excess bandwidths α .

Without mathematical derivation, the time domain waveform is given from inverse transform of Eq (3.32) as

$$p[n] = \frac{1}{\sqrt{L}} \cdot \frac{\sin [\pi(1-\alpha)n/L] + 4\alpha n/L \cdot \cos [\pi(1+\alpha)n/L]}{\pi n/L [1 - (4\alpha n/L)^2]} \quad (3.33)$$

Figure 3.32 plots these time domain waveforms of Square-Root Raised Cosine for different values of α . Observe again that plugging $\alpha = 0$ produces a sinc signal, the same ideal Nyquist filter in Eq (3.27). Through the red ellipse in the figure, the zero crossings are seen at integer multiples of T_M only for $\alpha = 0$. For all other α , the Square-Root Raised Cosine

pulse does not satisfy Nyquist no-ISI criterion. This is not surprising because it has to fulfill that criterion only after matched filtering with another Square-Root Raised Cosine at the Rx to form the cumulative Raised Cosine shape.

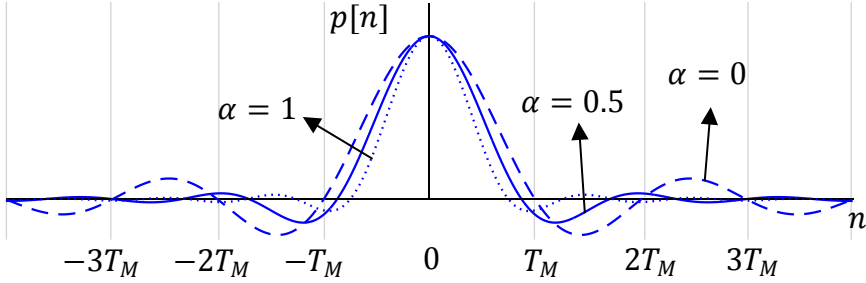


Figure 3.32: Square-root Raised Cosine (SRRC) waveform in time domain with different excess bandwidths α for $L = 64$ samples/symbol. Observe that zero crossings do not necessarily coincide with integer multiples of T_M

In a software routine, the above equation can be used to find the coefficients of the Square-Root Raised Cosine pulse shape used for Tx filtering. Just like in Raised Cosine case, the denominator in this equation becomes zero for $n = 0$ and $n = \pm L/(4\alpha)$. Pulse coefficients at these values can be given as

$$p[0] = \frac{1}{\sqrt{L}} \cdot \left(1 - \alpha + 4\frac{\alpha}{\pi}\right) \quad (3.34)$$

$$p[\pm \frac{L}{4\alpha}] = \frac{\alpha}{\sqrt{2L}} \left\{ \left(1 + \frac{2}{\pi}\right) \sin \frac{\pi}{4\alpha} + \left(1 - \frac{2}{\pi}\right) \cos \frac{\pi}{4\alpha} \right\} \quad (3.35)$$

Again, this is found by using L'Hôpital's rule in which the derivative of both the numerator and the denominator is taken before plugging in those values for n .

Being a band-limited signal, the Square-Root Raised Cosine is infinitely long in time that cannot be implemented in real systems. Therefore, it must be truncated to G symbols to the left and G symbols to the right. This is equivalent to $-LG \leq n \leq LG$ in samples, thus resulting in total filter length $N = 2LG + 1$. This time span of G symbols or LG samples is called **group delay**. The concept of group delay was discussed in detail in Section 2.5.

Exercise 3.1

In GNU Radio, an SRRC filter can be generated using the 'Root Raised Cosine Filter' block which is a wrapper for an [firdes](#) taps generating function. This function [firdes.root_raised_cosine\(\)](#) can also be employed to assign the generated taps to a variable. The required parameters are as follows.

Gain: Different filter gains are set according to different requirements such as maintaining a similar amplitude or energy. See Table 2.3 in Section 2.7.2 for a detailed discussion.

Sample Rate: Sample rate F_S of the system.

Symbol Rate: Symbol rate $1/T_M$ is defined through the number of samples/symbol L , i.e., F_S/L .

Alpha: Excess bandwidth α .

Num Taps: Total number of taps is given by choosing the group delay G first as in the above

discussion. Then, the filter length becomes $2LG + 1$.

In Matlab, the corresponding function is `rcosdesign()` that takes the input parameters α , $2G$, L and an argument 'sqrt'. Replacing 'sqrt' with 'normal' generates the taps for a Raised Cosine pulse instead.

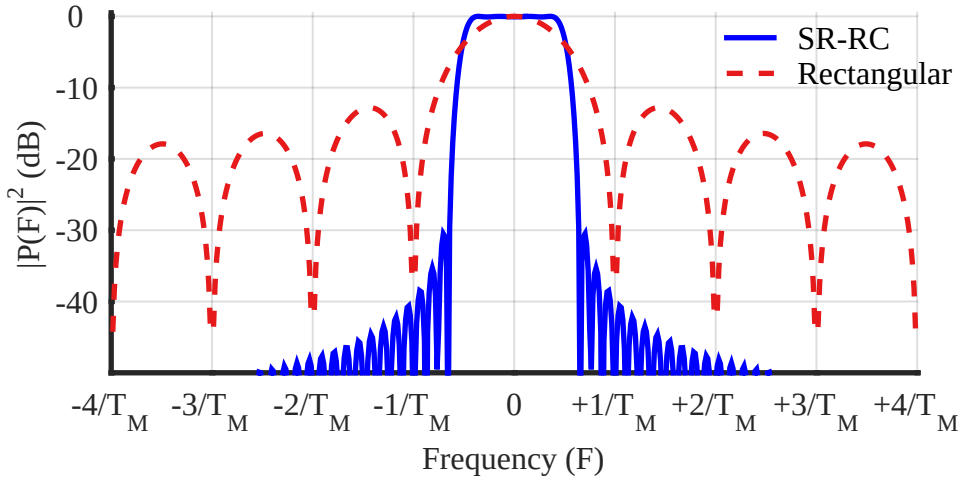


Figure 3.33: Spectrum of a Square-Root Raised Cosine pulse shape compared with that of a rectangular pulse. The pulse was generated for excess bandwidth $\alpha = 0.25$, $L = 8$ samples/symbol and length $N = 2LG + 1 = 65$

Figure 3.33 compares the spectra of a Square-Root Raised Cosine pulse with that of a rectangular pulse (rectangular in time, not frequency). The Square-Root Raised Cosine pulse is generated using $\alpha = 0.25$, $L = 8$ samples/symbol and $G = 4$ for a total filter length of $N = 2LG + 1 = 65$ samples. The duration of rectangular pulse is obviously 8 samples. A huge improvement in sidelobe suppression is fairly visible.

Example 3.1

As an example, Wideband Code-Division Multiple-Access (WCDMA) - the main technology behind 3rd-generation (3G) cellular systems - implements a Square-Root Raised Cosine pulse shape with excess bandwidth $\alpha = 0.22$, which translates the signaling rate of 3.84 MHz to a bandwidth of $0.5 \times (1 + \alpha) \times 2 \times 3.84 = 4.68$ MHz (the factor 2 arises for the RF bandwidth - we will discuss that in Section 3.7). Accounting for the guard-bands to minimize interference between neighboring channels, the signal bandwidth in WCDMA systems is 5 MHz.

PAM Revisited

Let us now replace the Square-Root Raised Cosine filter above into our basic PAM system of Figure 3.17 in place of a rectangular pulse. For a 2-PAM modulation system with symbols $\pm A$, the process unfolds as follows. Keep comparing the signals thus generated with those from rectangular pulse shape in Figure 3.17.

- A source generates Tx bits to be sent to a destination. These bits are input to a Look-Up Table that maps bits to symbols according to a chosen modulation scheme. Next, the Tx symbols $a[m]$ are upsampled by $L = 2$ samples/symbol. An example created of 10 bits is illustrated in Figure 3.34.

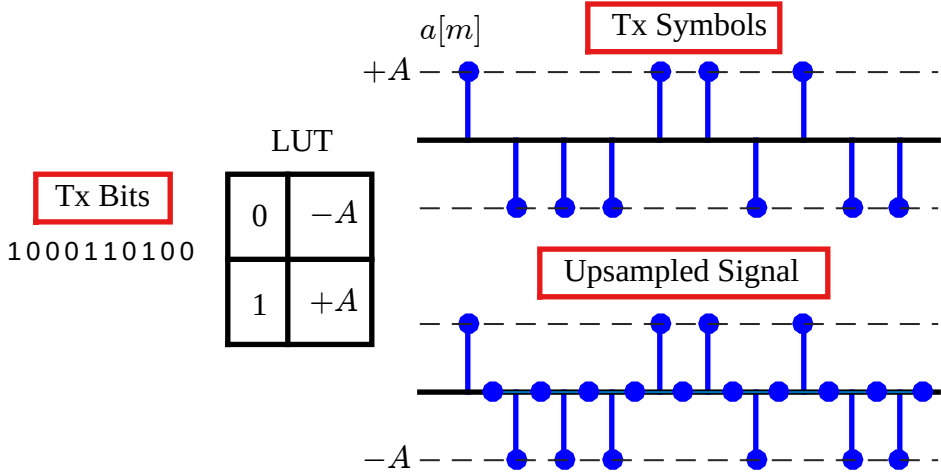


Figure 3.34: Tx bits, Look-Up Table and Tx symbols for 2-PAM modulation. The bottom plot shows the Tx symbols upsampled by $L = 2$ samples/symbol

- The pulse shaping block with $L = 2$ samples/symbol, excess bandwidth $\alpha = 0.5$ and group delay $G = 5$ symbols filters the waveform to output a smooth waveform $s(nT_S)$ which then is passed to the analog portion of the Tx. A DAC produces the continuous output $s(t)$ illustrated as dashed red line in Figure 3.35. The bit information behind the waveform is also shown.

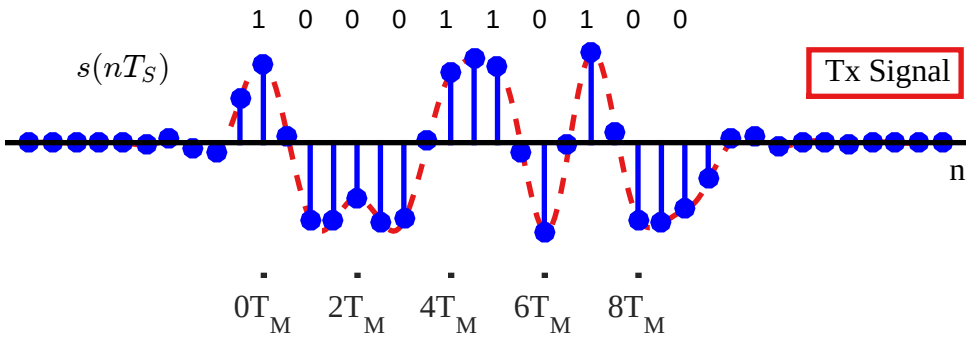


Figure 3.35: Upsampled signal shaped by a Square-Root Raised Cosine filter with $L = 2$ samples/symbol, excess bandwidth $\alpha = 0.5$ and group delay $G = 5$ symbols

- In the best case scenario, i.e., a channel that only adds AWGN to the Tx signal, the Rx signal $r(t)$ is drawn as in Figure 3.36.

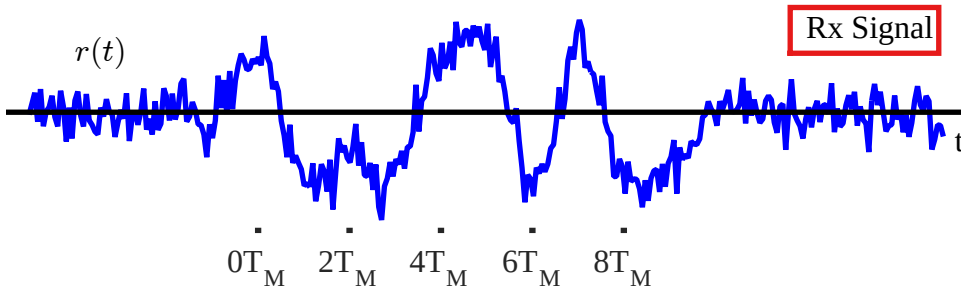


Figure 3.36: Rx signal in an AWGN channel

- The Rx signal $r(t)$ is sampled by an ADC to produce $r(nT_S)$ and then input to a matched filter which is the same pulse shaping filter as at the Tx. The matched filter output $z(nT_S)$ at $L = 2$ samples/symbol is shown in Figure 3.37. As long as the combination of Tx and Rx filters (i.e., overall Raised Cosine filter) obeys Nyquist criterion, there is no ISI and one out of every L samples at $n = mL = mT_M/T_S$ can be preserved to form a symbol estimate while the remaining $L - 1$ samples are thrown away. If there was zero noise, these downsampled values directly map to the transmitted symbols without any error, from which the sequence of Rx bits can be found with the help of an LUT.

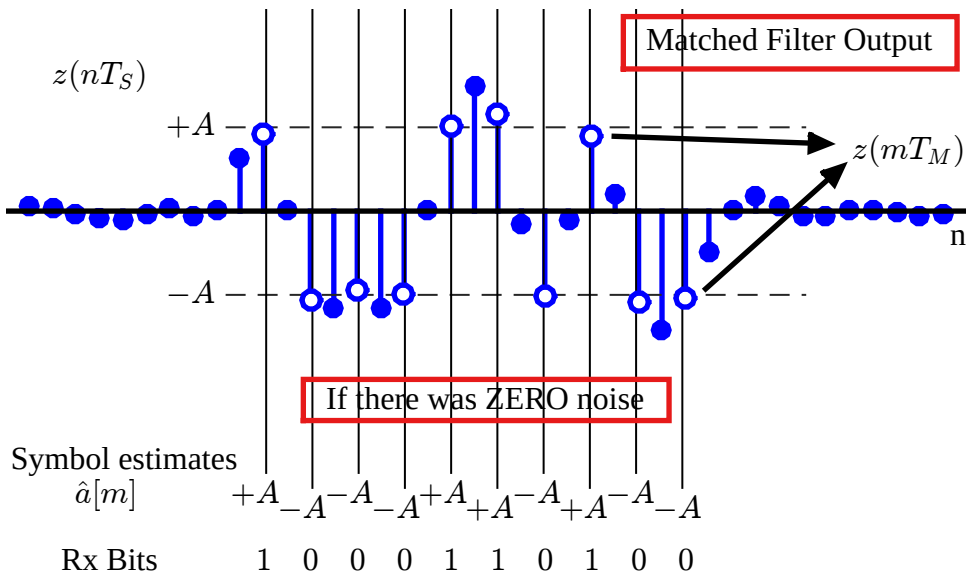


Figure 3.37: Matched filter output. White samples yield the symbol estimates

To decide which one sample to keep out of every L samples is the job of timing synchronization subsystem. We discuss symbol timing synchronization in Chapter 7.

- When the noise is present which is always the case, the minimum distance rule is employed to produce symbol estimates $\hat{a}[m]$ according to its shortest distance from a

constellation point, as illustrated in Figure 3.38. Depending on numerous factors in system design, a proportion of bits can eventually end up in error, i.e., they can be different than Tx bits. The ratio of number of Rx bits in error to the total number of bits is called *Bit Error Rate (BER)*.

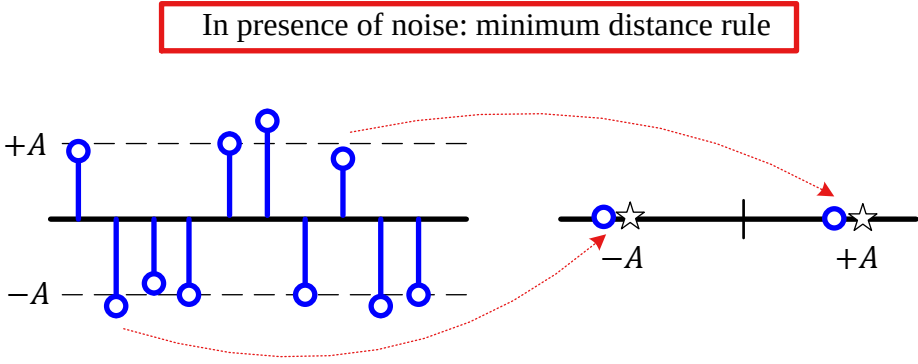


Figure 3.38: Downsamped matched filter output is mapped back to the constellation for decisions

In summary, when the source information bits are filtered by a square-root Nyquist filter, the sharp edges visible for a rectangular pulse are smoothed out by a considerable margin, as shown in Figure 3.39 for 20 2-PAM symbols and excess bandwidth $\alpha = 0.5$. This smoothness in time domain actually limits the bandwidth. Also observe that in the absence of noise, the values at optimum sampling locations are not all the same and exhibit ISI in a square-root Nyquist case. This is because the zero crossings of a square-root Nyquist shape are not necessarily at integer multiples of symbol times, see Figure 3.32. However, after complete Nyquist filtering and no noise, all values coincide with the same optimum symbol amplitudes shown as red asterisks in RC filtered PAM symbols of Figure 3.39.

Coming back to our linear modulation framework where the pulse shape is scaled by a symbol and matched filter by the Rx, we have the following expression similar to Eq (3.17) for the matched filter output.

$$z(t) = \sum_m a[m]r_p(t - mT_M)$$

We can break this relation down to get two equivalent representations.

$$\begin{aligned} \sum_i a[i]r_p(t - iT_M) &= a[0]r_p(t) + a[1]r_p(t - T_M) + a[2]r_p(t - 2T_M) + \dots \\ &= \left(a[0]\delta(t) + a[1]\delta(t - T_M) + \right. \\ &\quad \left. a[2]\delta(t - 2T_M) + \dots \right) * r_p(nT_S) \end{aligned}$$

This is because convolution between any signal and a unit impulse results in the same signal. So far, we utilized the second expression involving the convolution in the above description. Now we have a look at the equivalent first relation for a deeper understanding.

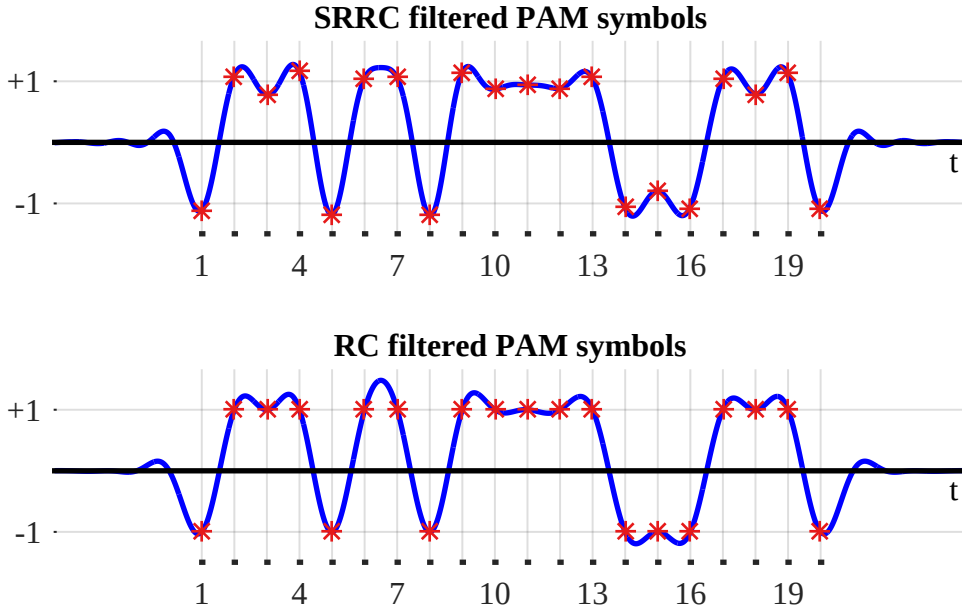


Figure 3.39: 20 binary PAM symbols filtered by a Square-Root Raised Cosine and a Raised Cosine filter with excess bandwidth $\alpha = 0.5$. Observe that values at optimum sampling locations exhibit ISI in the former but coincide with symbol values in the latter

The unmodulated Nyquist pulse shapes (not square-root Nyquist) are drawn in Figure 3.40a. Notice the zero crossings of the 0^{th} pulse passing through

$$T_M, 2T_M, 3T_M, \dots$$

and so on, right at the peaks of the other pulse locations $t = mT_M$.

When a linear modulation scheme, such as BPSK, is utilized, the Nyquist pulse amplitudes are $a[m] = +1$ or $a[m] = -1$. The clock at the Rx then needs to sample at exactly the same instants mT_M , exhibiting zero ISI – maximum contribution comes from the desired pulse and zero contribution from all the rest. This is plotted in Figure 3.40b in the form of grid lines. Later in Section 7.1, we will see how the presence of a timing offset causes the sampling in time domain at the wrong instants thus introducing ISI in the system.

Drawbacks of Square-Root Raised Cosine Pulse

As discussed above, Square-Root Raised Cosine pulse is much better than a rectangular pulse in shaping the spectrum but it has two major drawbacks.

Insufficient Sidelobe Attenuation There is a limit to the sidelobe suppression that a Square-Root Raised Cosine pulse can achieve. The sidelobe levels are reasonably higher than realistic spectral mask requirements imposed by regulatory authorities of attenuating out-of-band energy up to 60 to 80 dB. Looking back, recall that the transition band of a Raised Cosine pulse is half cycle of a cosine. Therefore, *the transition band of a*

Cosine spectrum and a sinc signal in frequency domain[†].

$$p_{\text{Truncated}}[n] = p[n] \cdot w_{\text{Rect}}[n]$$

$$P_{\text{Truncated}}(F) = P(F) * \text{sinc}(F)$$

As a result of this truncation in time domain and subsequent convolution in frequency domain, the half amplitude values are moved away from the odd symmetry points of half symbol rate, or $F = \pm 1/2T_M$ as illustrated in Figure 3.41b where the filter's 1/2 (or -3 dB) point has moved away from the nominal band edge. Compare this to the case in Figure 3.24 where the pulse shape is passing through 1/2 value at exactly half the symbol rate. For this reason, the spectral replicas now do not add up to exactly a constant value. In other words, Nyquist no-ISI criterion is only approximately satisfied by the truncated pulse shape, thus giving rise to increased ISI. *The maximum magnitude of $r_p(mT_M)$ for $m \neq 0$ is called peak ISI*, shown in Figure 3.41a to be equal to -0.0115 for excess bandwidth $\alpha = 0.5$ and group delay $G = 3$.

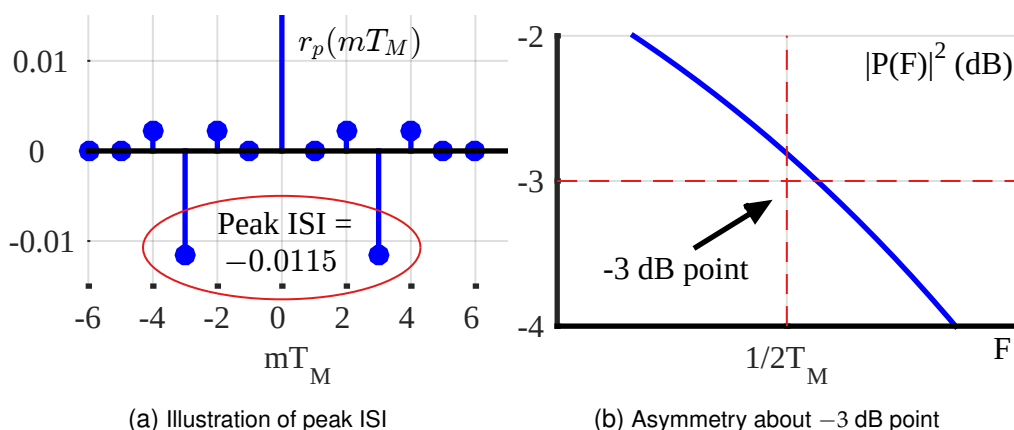


Figure 3.41: Raised Cosine generated by convolving two truncated Square-Root Raised Cosines with excess bandwidth $\alpha = 0.5$ and group delay $G = 3$ and downsampling by L samples/symbol. Due to asymmetry, spectral replicas do not add up to a constant value thus violating Nyquist no-ISI criterion

If we attempt to control the sidelobes by applying a second window to the already rectangular windowed prototype, these ISI levels rise even more significantly. The reader is encouraged to plot the spectra for different values of α and G to appreciate their role in spectral shaping.

Square-Root Raised Cosine and Raised Cosine are good starting points for pulse shape design and capture the involved details fairly well. Moreover, they have closed-form mathematical expressions that are good for analytical purposes. Nevertheless, the quarter cycle cosine of the Square-Root Raised Cosine results in high in-band ripples and insufficient out-of-band attenuation levels. There are other pulse shape design procedures that produce a Nyquist filter with high sidelobe attenuation and preserve Nyquist no-ISI criteria as well.

[†]The sidelobes and in-band ripple are inherited from that oscillating sinc signal and are a function of excess bandwidth α and the pulse extension G in symbols.

A Frequency Domain Window Based Pulse

A spectral shape is a discrete-time signal and hence just a sequence of numbers. This sequence of numbers in time domain can be generated through iDFT of a carefully designed discrete-time frequency response. Recall that a Raised Cosine filter was generated in frequency domain through convolution of an ideal rectangular spectrum with a half-cosine taper, see Figure 3.25. Here, we replace the half-cosine with an alternative taper that is an improved spectral window. The span of this spectral window is determined by the excess bandwidth α and the iDFT length.

The criteria for this spectral window design are the following.

1. Narrow width of the mainlobe to affect a smoother transition band in resulting pulse spectrum (the width of the transition band should remain unchanged).
2. Small sidelobe levels that get inherited in the pulse spectrum.

Based on a technique devised by fred harris in Ref. [10], one such candidate is a Kaiser window in frequency domain which has a minimum mainlobe spectral width for specified sidelobe levels. For a Kaiser window of a particular length, the sidelobe height is controlled by a parameter β . In Matlab, a Kaiser window of length N with parameter β can be generated through the command `kaiser(N,beta)`. Figure 3.42 compares this taper with a cosine taper with the same transition bandwidth $\alpha = 0.25$ and highlights the difference of smoothness between the two.

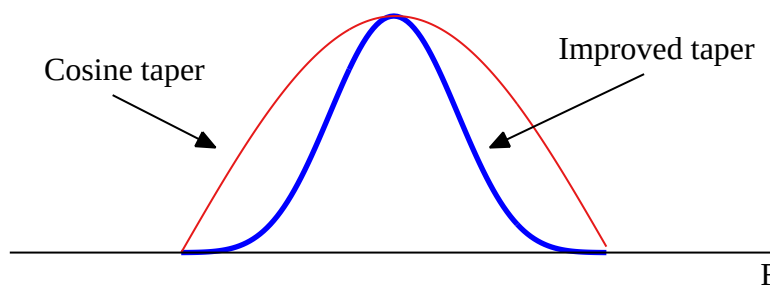
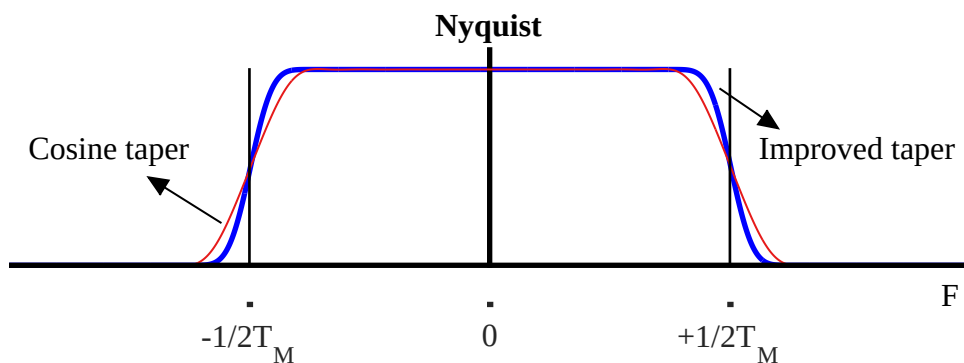


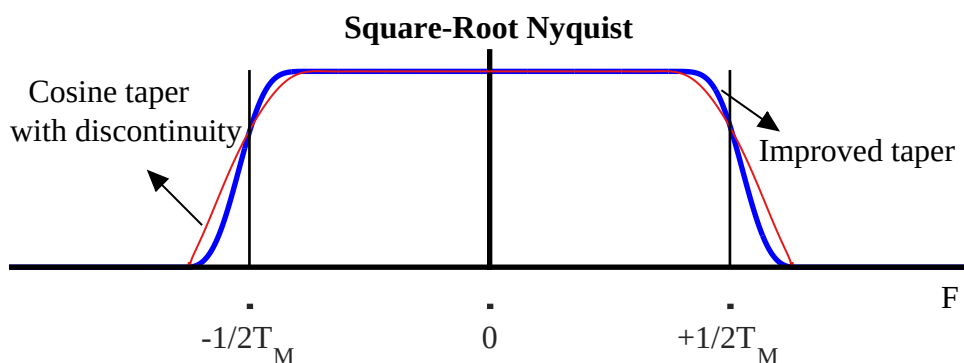
Figure 3.42: Comparison of cosine and improved spectral tapers for transition bandwidth $\alpha = 0.25$

In frequency domain, the convolution of this Kaiser window taper with an ideal rectangular spectrum generates the desired pulse shape. This frequency domain result is shown in Figure 3.43 for excess bandwidth $\alpha = 0.25$ where it is compared with a Square-Root Raised Cosine pulse with the same excess bandwidth. Figure 3.43a depicts both the improved Nyquist filter and a Raised Cosine prior to square-root operation.

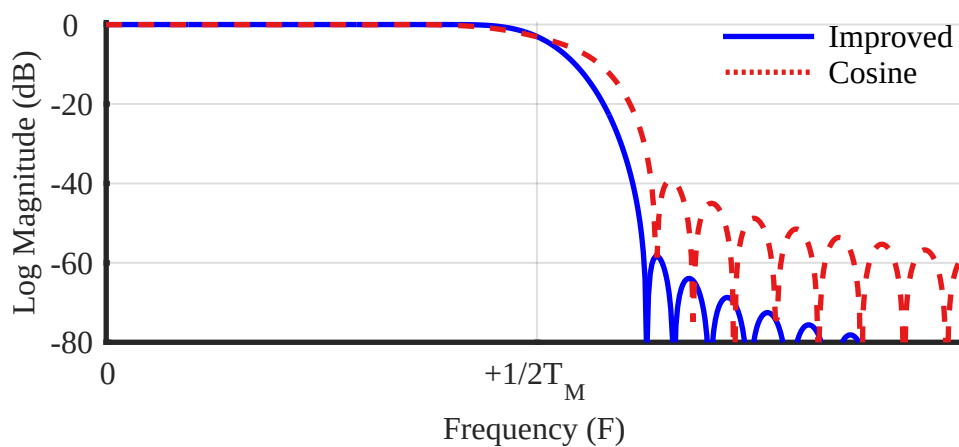
Notice that a narrower mainlobe width of the improved taper has produced a smoother transition band in the improved Nyquist filter as compared to a Raised Cosine. This smoothness gets inherited by square-root Nyquist filter and it does not exhibit a similar kind of discontinuity as a Square-Root Raised Cosine, both of which are shown in Figure 3.43b. This discontinuity gives rise to high levels of time response even after many symbol intervals. When such an impulse response is truncated, it generates high sidelobes as well as



(a) Nyquist filter with smoother transition band



(b) Square-root Nyquist filter without discontinuity at band edge



(c) Resulting better sidelobe suppression

Figure 3.43: A comparison of improved pulse with Square-Root Raised Cosine

in-band ripple in the spectrum of a Square-Root Raised Cosine. This is evident from Figure 3.43c where an order of magnitude improvement in sidelobe suppression is visible. For

example, an attenuation of 60 dB is obtained through this procedure as compared to 40 dB through Square-Root Raised Cosine.

Not shown here is the fact that low levels of in-band ripple accompany the low levels of sidelobes because they are always equal in a window based design (arising from convolution of the same spectral taper sliding through both stopband and passband). For reasons that are beyond the scope of this text, the low in-band ripple generates less ISI after matched filtering of the square-root Nyquist filter. Therefore, *the additional advantage of sidelobe suppression comes with a benefit, not at a cost.*

Finally, there are other pulse shape design procedures as well that employ iterative techniques to convert an initial lowpass filter to a Nyquist filter with high sidelobe attenuation while preserving Nyquist no-ISI criteria as well. For the purpose of this text, we continue using the Square-Root Raised Cosine pulse and Raised Cosine pulse auto-correlation to keep things simple and assume that a reader who goes on to implement the system will use the better alternatives discussed above[†].

Until now, we saw only pulse amplitude modulation, a 1-dimensional modulation scheme. A wireless system achieves higher spectral efficiency when it implements a modulation scheme with 2 dimensions, namely I and Q . One example of such a scheme is Quadrature Amplitude Modulation Scheme (QAM) which we cover next.

3.7 Quadrature Amplitude Modulation (QAM)

We discussed earlier that Pulse Amplitude Modulation (PAM) transmits information through amplitude scaling of the pulse $p(nT_s)$ according to the symbol value. To understand QAM, two routes need to be traversed.

We start the first route with differentiating between baseband and passband signals. A *baseband signal* has a spectral magnitude that is non-zero only for frequencies around origin ($F = 0$) and negligible elsewhere. An example spectral plot for a PAM waveform (e.g., in Figure 3.35) is shown in Figure 3.44 for 200 2-PAM symbols shaped by a Square-Root Raised Cosine pulse with excess bandwidth $\alpha = 0.5$ and samples/symbol $L = 8$. The PAM signal has its spectral contents around zero and hence it is a baseband signal.

For wireless transmission, this information must be transmitted through space as an electromagnetic wave. Also observe that once this spectrum is occupied, no other entity in the surrounding region can share the wireless channel for the purpose of communications. For these reasons (and a few others as well), a wireless system is allocated a certain bandwidth around a higher carrier frequency and the generated baseband signal is shifted to that specific portion of the spectrum. Such signals are called *passband signals*. That is why wireless signals in everyday use such FM radio, WiFi, cellular like 4G, 5G, and Bluetooth are all passband signals which execute their transmissions within their respective frequency bands sharing the same physical medium.

The easiest method to shift the spectrum to a designated carrier frequency is by multiplying, or mixing, it with a sinusoidal waveform due to the following reason.

[†]For non-linear modulations, a Gaussian pulse shape is a popular choice, e.g., in 2G (GSM) systems. A Gaussian pulse shape is not Nyquist and hence introduces ISI in neighboring symbols. This is a cost paid in exchange for excellent spectral properties of the pulse.

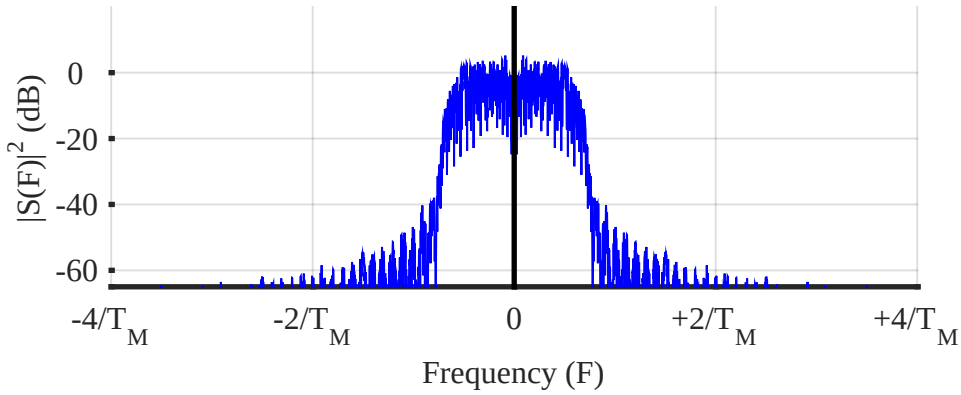


Figure 3.44: Spectrum of a 2-PAM modulated signal consisting of 200 symbols shaped by a Square-Root Raised Cosine pulse of excess bandwidth $\alpha = 0.5$ and samples/symbol $L = 8$

- In time domain, a PAM waveform can be multiplied with a carrier sinusoid at frequency F_C .
- Time domain multiplication between two signals is frequency domain convolution between their spectra.
- The spectrum of a pure sinusoid at any frequency is composed of two impulses at that frequency (one positive and one negative, both with half the amplitude).
- Hence in frequency domain, the convolution of a signal with a unit impulse results in the same signal “parked” at its allocated slot.

This is drawn in Figure 3.45.

After this *spectral upconversion*, both positive and negative portions of the baseband spectrum appear at $+F_C$ and $-F_C$. The bandwidth – positive portion of the spectrum – hence becomes double. However, a relief comes from the fact that both $\cos(\cdot)$ and $\sin(\cdot)$ can be used to carry independent waveforms due to their orthogonality (having phases 90° apart) to each other over a complete period, i.e.,

$$\sum_{n=0}^{N-1} \cos 2\pi \frac{k}{N} n \cdot \sin 2\pi \frac{k}{N} n = 0$$

Obviously, this is true over a complete period. However, this is approximately true for very high carrier frequencies (typical case in wireless communications) as well because $2 \cos A \sin A = \sin 2A$ that doubles the frequency. Due to positive and negative halves of a sine, the area under the curve (sum of samples) is then zero for all integer number of cycles. This area under any fraction of a cycle outside the last integer cycle then remains negligible at a high carrier frequency.

The second route starts with amplitude modulation, where all the amplitudes still extend on the same axis of the PAM constellation diagram. On the other hand, for a phase modulated system, one axis is not enough because an actual angle is formed by *two lines*, not one. This is shown in Figure 3.46. In other words, an amplitude modulation scheme has a 1-dimensional constellation diagram while a phase modulation scheme requires a 2-dimensional constellation diagram.

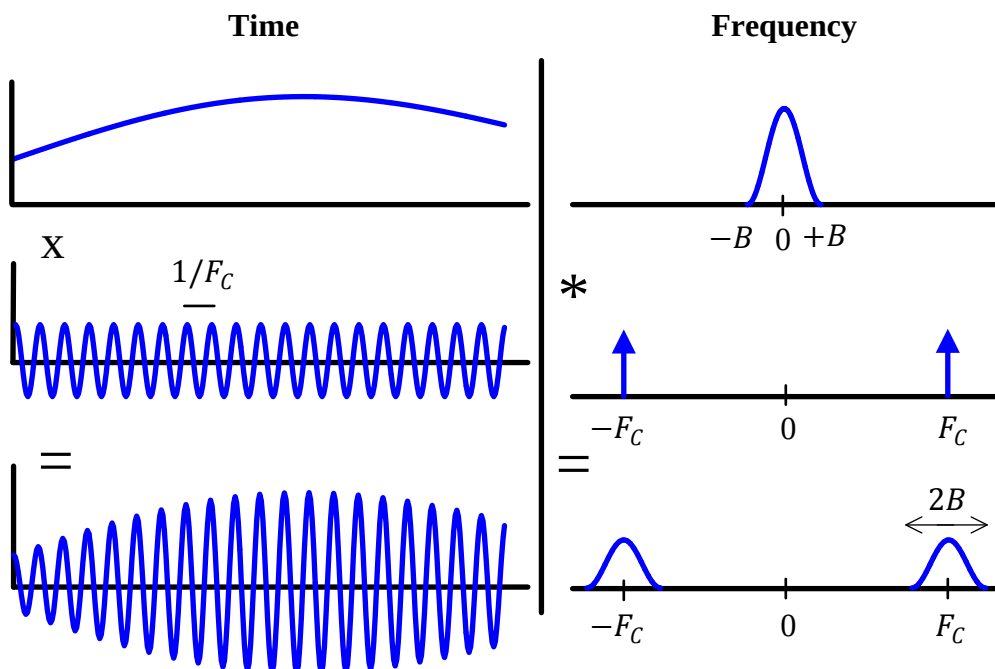


Figure 3.45: Multiplication in time domain of a signal with a sinusoid of frequency F_C shifts the spectrum at F_C due to its convolution with two impulses in frequency domain

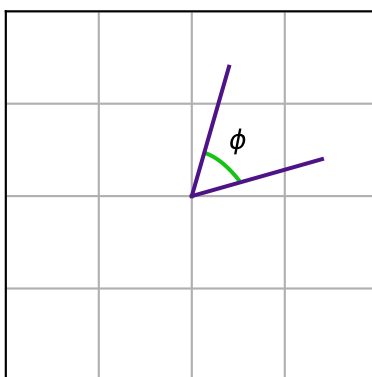


Figure 3.46: An angle is formed by two lines, not one

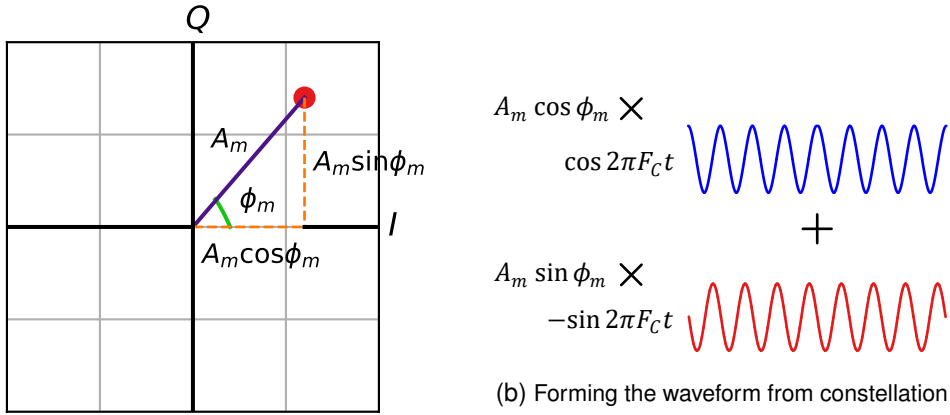
If amplitude modulation sends the information on discrete amplitudes in one dimension and phase modulation chooses a set of discrete phases in two dimensions but on the same circle, both of these ideas can be combined together in the form of *Quadrature Amplitude Modulation (QAM)*. For a practical implementation of this idea, we start with introducing *both* amplitude and phase modulation in a carrier waveform.

$$s(t) = A_m \cdot \cos(2\pi Ft + \phi_m) \quad (3.36)$$

Using the identity $\cos(A + B) = \cos A \cos B - \sin A \sin B$, we open this up as

$$\begin{aligned} s(t) &= A_m \cdot \cos(2\pi Ft) \cdot \cos \phi_m - A_m \cdot \sin 2\pi Ft \cdot \sin \phi_m \\ &= \underbrace{(A_m \cos \phi_m)}_{\text{I part}} \cdot \cos(2\pi Ft) - \underbrace{(A_m \sin \phi_m)}_{\text{Q part}} \cdot \sin 2\pi Ft \end{aligned} \quad (3.37)$$

Here, the amplitude $A_m \cos \phi_m$ is the projection of the modulation point on x axis, while the amplitude $A_m \sin \phi_m$ is the projection of the modulation point on y axis. This is plotted in Figure 3.47a. These amplitudes are then modulated on two **quadrature** carriers, namely $\cos(2\pi Ft)$ and $-\sin(2\pi Ft)$ according to Eq (3.37) and summed together to be sent over the air as plotted in Figure 3.47b. Why are there two waveforms as opposed to one (before the summation)? Because constructing or measuring an angle requires two lines, and hence constructing or measuring a phase requires two waveforms, one for each line.



(a) I/Q plane and modulation parameters A_m and ϕ_m

Figure 3.47: QAM waveform generation

To explain I and Q notations further, recall that each phase, like each angle, needs a reference. For an angle, this reference is provided by the horizontal line connecting point $(0,0)$ to point $(1,0)$. For a phase, a cosine wave with phase 0° , i.e., $\cos(2\pi Ft)$, is taken as the reference (an arbitrary choice) that is the first waveform in Eq (3.37). Acting as an amplitude of this waveform $\cos(2\pi Ft)$, the **symbol level** $A_m \cos \phi_m$ on the horizontal axis is known as the **In-Phase** component (in phase with a cosine), from which the notation I is extracted.

To understand the Q part, we need another waveform to represent the amplitude on the vertical axis. If the horizontal axis links angle 0° on a 2-D plane to a cosine wave $\cos(2\pi Ft)$ with the starting phase 0° , then the vertical axis at an angle 90° on a 2-D plane should be linked to which waveform? Clearly, this should be another cosine wave with the starting phase of 90° . A cosine wave with a starting phase of 90° is found by

$$\cos(2\pi Ft + 90^\circ) = \cos 2\pi Ft \cdot \underbrace{\cos 90^\circ}_0 - \sin 2\pi Ft \cdot \underbrace{\sin 90^\circ}_1 = -\sin 2\pi Ft$$

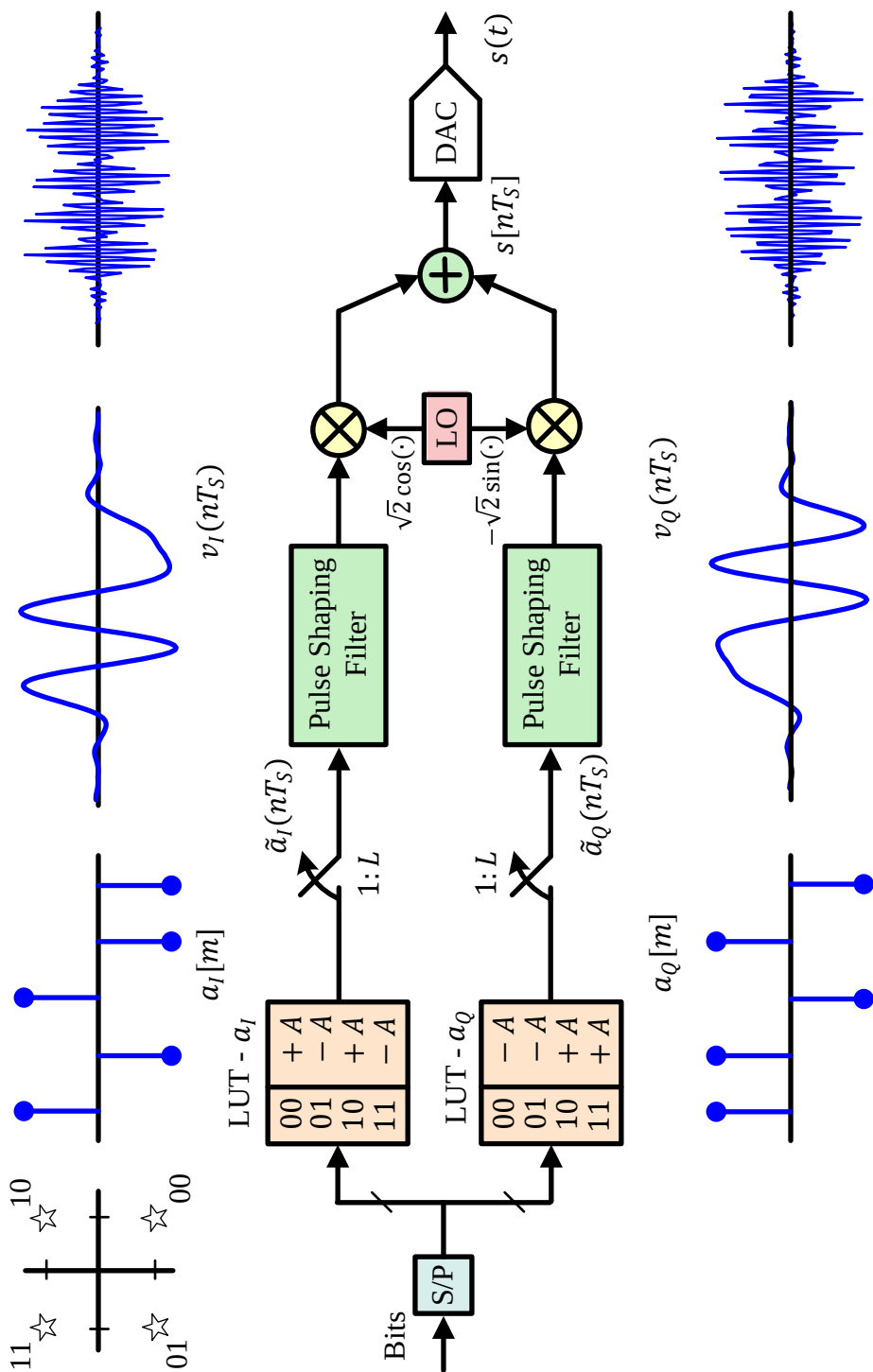


Figure 3.48: A general QAM modulator

This second cosine wave comes out to be a negative sine wave!

That is the birth of *Quadrature Amplitude Modulation (QAM)*, in which two independent PAM waveforms are communicated through mixing one with a cosine and the other with a sine. Just like constellation, the term quadrature also comes from astronomy to describe the position of two objects 90° apart.

QAM Modulator

To build a conceptual QAM modulator, we follow similar steps as in a PAM modulator in Section 3.5. The block diagram for a 4-QAM modulator is drawn in Figure 3.48.

- Every T_b seconds, a new bit arrives at the input forming a serial bit stream.
- A serial-to-parallel (S/P) converter collects $\log_2 M$ such bits every $T_M = \log_2 M \times T_b$ seconds that are used as an address to access *two Look-Up Tables (LUT)*. One LUT stores $\sqrt{M}$ symbol values a_I while the other stores $\sqrt{M}$ symbol values a_Q specified by the constellation. The choice of assigning a particular symbol set (e.g., $\{+A, -A\}$) to a particular bit set (e.g. 00) is completely arbitrary, as long as the Rx agrees on the same assignment.

For example, the two bits 00 form the address 0 in the Look-Up Tables which selects $a_I = +A$ and $a_Q = -A$. This comes from the assignment on the left hand side $\{+A, -A\}$ shown in Figure 3.49.

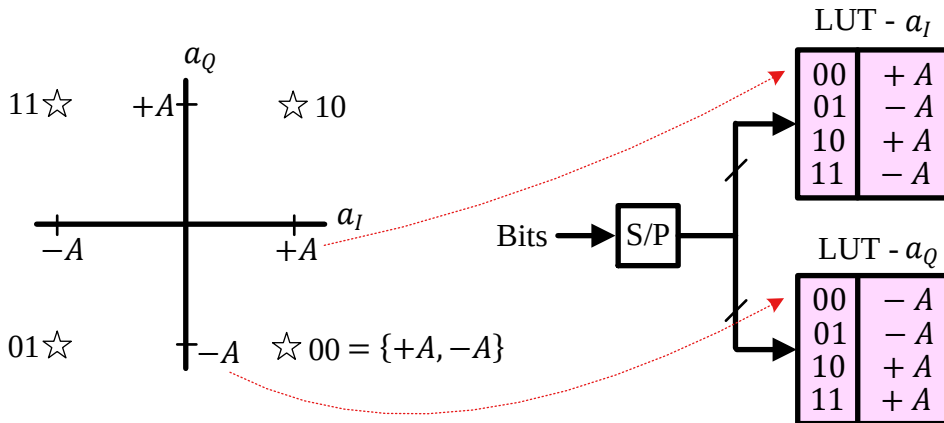


Figure 3.49: An example of selecting a_I and a_Q from a pair of Look-Up Tables. The choice of assigning a bit sequence to a particular symbol set is arbitrary

- To produce a QAM waveform, the symbol sequences $a_I[m]$ and $a_Q[m]$ are converted to discrete-time impulse trains in separate arms (one I and the other Q) through up-sampling by L , where L is the number of samples per symbol (the ratio of symbol time to sample time T_M/T_S , or equivalently sample rate to symbol rate F_S/R_M). Let us denote these sequences by $\tilde{a}_I(nT_S)$ and $\tilde{a}_Q(nT_S)$, respectively.
- As explained in Section 2.7, upsampling inserts $L - 1$ zeros between each symbol after which the interpolated intermediate samples can be raised from dead with the

help of a pulse shaping filter in each arm that – in addition to shaping the spectrum – suppresses all the spectral replicas arising from the upsampling process except the desired one. These are the two I and Q PAM waveforms.

$$\begin{array}{ll} I & \rightarrow v_I(nT_S) = \tilde{a}_I(nT_S) * p(nT_S) \\ Q & \uparrow v_Q(nT_S) = \tilde{a}_Q(nT_S) * p(nT_S) \end{array}$$

■ Next, I PAM waveform $v_I(nT_S)$ is upconverted by mixing with the inphase carrier $\sqrt{2}\cos 2\pi F_C nT_S$ while the Q PAM waveform $v_Q(nT_S)$ is mixed with quadrature carrier $-\sqrt{2}\sin 2\pi F_C nT_S$, which are then summed to form the QAM signal $s(nT_S)$. The reason for choosing the scaling factor $\sqrt{2}$ as well as a negative sign with $\sin(\cdot)$ will shortly become clear. Note that this carrier is generated through a Local Oscillator (LO) at the Tx.

■ This discrete-time signal $s(nT_S)$ is converted to a continuous-time signal $s(t)$ by a DAC.

For mathematical expression for QAM, first consider the above two independent PAM waveforms with symbol streams $a_I[m]$ and $a_Q[m]$, respectively.

$$\begin{array}{ll} I & \rightarrow v_I(nT_S) = \sum_m a_I[m] p(nT_S - mT_M) \\ Q & \uparrow v_Q(nT_S) = \sum_m a_Q[m] p(nT_S - mT_M) \end{array} \quad (3.38)$$

Next, we multiply the resulting complex signal with a complex sinusoid at a frequency k_C/N , where k_C/N corresponds to the carrier frequency F_C . Here, we assume a conceptual up-conversion in discrete domain but in practice there are efficient techniques to accomplish this task that depend on the Tx architecture. Now from the multiplication rule of complex signals $I \cdot I - Q \cdot Q$ and $Q \cdot I + I \cdot Q$,

$$\begin{array}{ll} I & \rightarrow v_I(nT_S)\sqrt{2}\cos 2\pi \frac{k_C}{N}n - v_Q(nT_S)\sqrt{2}\sin 2\pi \frac{k_C}{N}n \\ Q & \uparrow v_Q(nT_S)\sqrt{2}\cos 2\pi \frac{k_C}{N}n + v_I(nT_S)\sqrt{2}\sin 2\pi \frac{k_C}{N}n \end{array}$$

Now the signals sent over the air or any medium are always real, so we can choose any one of the above arms, I or Q . Note that both arms contain all the information regarding $v_I(nT_S)$ and $v_Q(nT_S)$. As a convention, we choose the I arm for this purpose. Consequently, a general QAM waveform can be written as

$$\begin{aligned} s(nT_S) &= v_I(nT_S)\sqrt{2}\cos 2\pi \frac{k_C}{N}n - v_Q(nT_S)\sqrt{2}\sin 2\pi \frac{k_C}{N}n \\ &= v_I(nT_S)\sqrt{2}\cos 2\pi \frac{F_C}{F_S}n - v_Q(nT_S)\sqrt{2}\sin 2\pi \frac{F_C}{F_S}n \end{aligned} \quad (3.39)$$

where we have used the fundamental relation between continuous and discrete frequencies $k/N = F/F_S$. Observe that $s(nT_S)$ is a real signal with no Q component. And in a block diagram, it is obtained by multiplying v_I arm with $\sqrt{2}\cos 2\pi(k_C/N)n$ and v_Q arm with $-\sqrt{2}\sin 2\pi(k_C/N)n$. Plugging in the definitions of $v_I(nT_S)$ and $v_Q(nT_S)$,

$$\begin{aligned} s(nT_S) &= \sum_m a_I[m] p(nT_S - mT_M) \sqrt{2}\cos 2\pi F_C nT_S - \\ &\quad \sum_m a_Q[m] p(nT_S - mT_M) \sqrt{2}\sin 2\pi F_C nT_S \end{aligned}$$

After Digital to Analog Conversion (DAC), the continuous-time signal $s(t)$ can be expressed as

$$\begin{aligned} s(t) &= v_I(t)\sqrt{2}\cos 2\pi F_C t - v_Q(t)\sqrt{2}\sin 2\pi F_C t \\ &= \sum_m a_I[m]p(t - mT_M)\sqrt{2}\cos 2\pi F_C t - \\ &\quad \sum_m a_Q[m]p(t - mT_M)\sqrt{2}\sin 2\pi F_C t \end{aligned} \quad (3.40)$$

In Eq (3.40), symbols $a_I[m]$ determine the I signal of $v(t)$ while $a_Q[m]$ control the Q part of $v(t)$. Such representation of the QAM waveform as a sum of amplitude scaled and pulse shaped sinusoids is known as the **rectangular form**. Sometimes it is difficult to imagine how two different data streams can be combined into a single signal and sent over the air. For the ease of illustration, we have drawn zoomed in versions of I and Q arms in a QAM modulator in Figure 3.50.

We can convert it into a polar form by using the relation

$$A \cos 2\pi F_C t - B \sin 2\pi F_C t = \sqrt{A^2 + B^2} \cos \left(2\pi F_C t + \tan^{-1} \frac{B}{A} \right)$$

While the actual derivation is longer, you can use the trigonometric identity $\cos(A + B) = \cos A \cos B - \sin A \sin B$ on the right hand side along with $\tan A = \sin A / \cos A$ and the Pythagorean theorem. Applying this to the QAM waveform above, its polar form can be written as

$$\begin{aligned} s(t) &= \sum_m \sqrt{a_I^2[m] + a_Q^2[m]} \cdot p(t - mT_M) \cdot \sqrt{2} \cos \left(2\pi F_C t + \tan^{-1} \frac{a_Q[m]}{a_I[m]} \right) \\ &= \sum_m X[m] \cdot p(t - mT_M) \cdot \sqrt{2} \cos (2\pi F_C t + \phi[m]) \end{aligned} \quad (3.41)$$

where

$$\begin{aligned} X[m] &= \sqrt{a_I^2[m] + a_Q^2[m]} \\ \phi[m] &= \tan^{-1} \frac{a_Q[m]}{a_I[m]} \end{aligned}$$

In this **polar form**, we have a single sinusoid whose amplitude and phase are determined by the above combination of symbols $a_I[m]$ and $a_Q[m]$ during every symbol time. This is illustrated in Figure 3.51 for a slowly varying carrier wave.

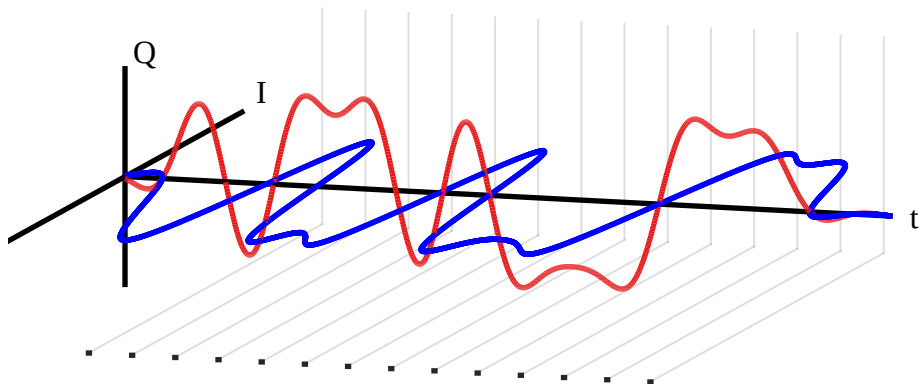
For example, the constellation points in BPSK are 0 and π . In QPSK, the constellation points are selected by $X[m]$ and $\phi[m]$ as

$$X[m] = 1, \quad \phi[m] = +\frac{\pi}{4}, +3\frac{\pi}{4}, -3\frac{\pi}{4}, -\frac{\pi}{4}$$

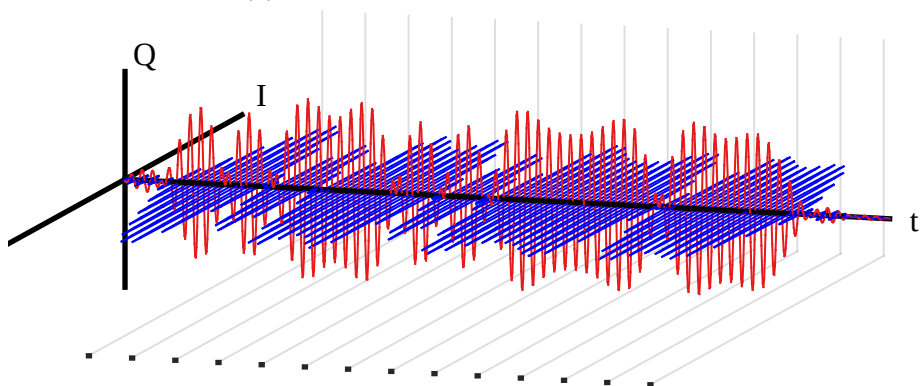
For a higher-order modulation scheme, the amplitude $X[m]$ also changes values according to the constellation points.

Constellation Diagram

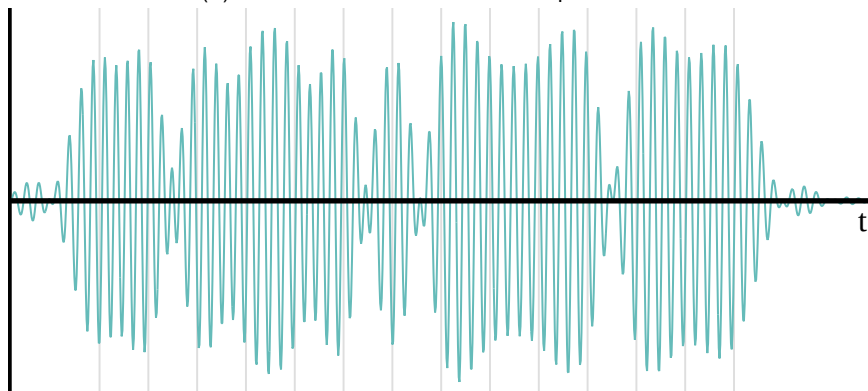
Some examples of QAM constellations are discussed now. For an even power of 2, **square QAM** is a constellation whose points are spaced on a grid in the form of a square. It is



(a) Two baseband PAM waveforms in I and Q arms



(b) Same waveforms after carrier upconversion



(c) Summation of the upconverted I and Q waveforms

Figure 3.50: QAM signal generation process

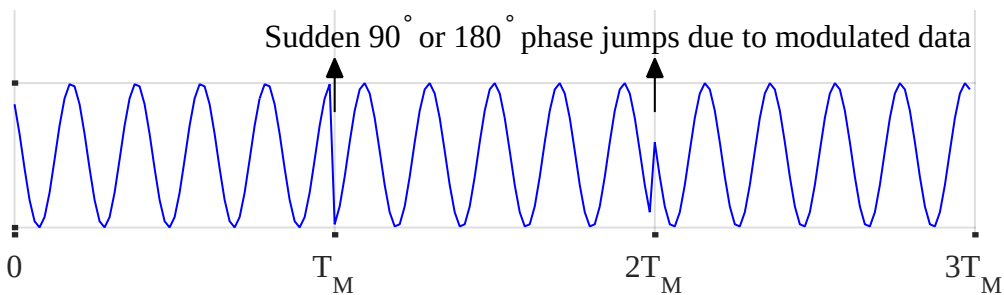


Figure 3.51: In polar form, a QPSK modulation can be seen to manifest 0° , $\pm 90^\circ$ or 180° phase jumps at each symbol time from the modulating data

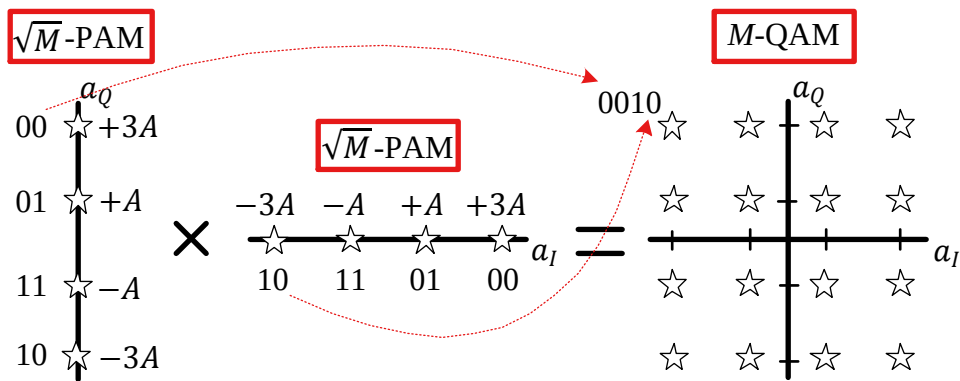


Figure 3.52: A 16-QAM constellation is formed by two $\sqrt{16} = 4$ PAM constellations

formed by a product of two $\sqrt{M}$ -PAM constellations, one on I -axis and the other on Q axis. For example, a 16-QAM constellation is formed by two $\sqrt{16} = 4$ PAM constellations as drawn in Figure 3.52, while some other square QAM constellations are also shown in Figure 3.53.

From Figure 3.53, notice how constellation points in higher-order QAM are closer to each other compared to lower-order QAM. A relatively lower noise power is then enough to cause a decision error by moving the received symbol over the decision boundary. This is the cost of increasing data rate by packing more bits in the same symbol. We will have more to say about it when we discuss Bit Error Rates (BER) for each constellation in Section 3.10.

For $M = 4$, the average symbol energy in square QAM is derived using Pythagoras theorem as

$$E_{4\text{-QAM}} = \frac{1}{4} \left\{ 4 \left(A^2 + A^2 \right) \right\} = 2A^2$$

For general M , a similar derivation as in the case of PAM studied earlier yields

$$E_{M\text{-QAM}} = \frac{2}{3} (M - 1) A^2 \quad (3.42)$$

A special case of M -QAM is M -PSK, which stands for Phase Shift Keying. As the name PSK implies, the amplitude remains constant in this configuration while the information is conveyed by different phases. Examples of 2-PSK, 4-PSK and 8-PSK are drawn in Figure 3.54. Notice that

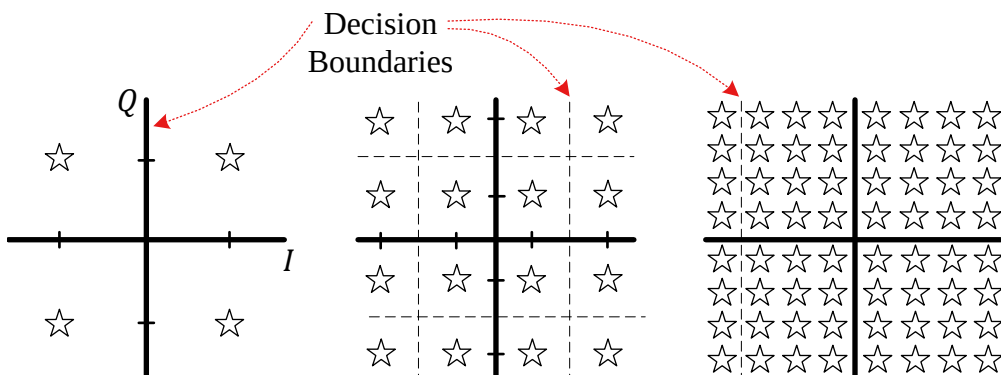


Figure 3.53: QAM constellation diagrams for $M = 4$, $M = 16$ and $M = 64$. Observe how points get closer for higher-order QAM and become prone to noise errors

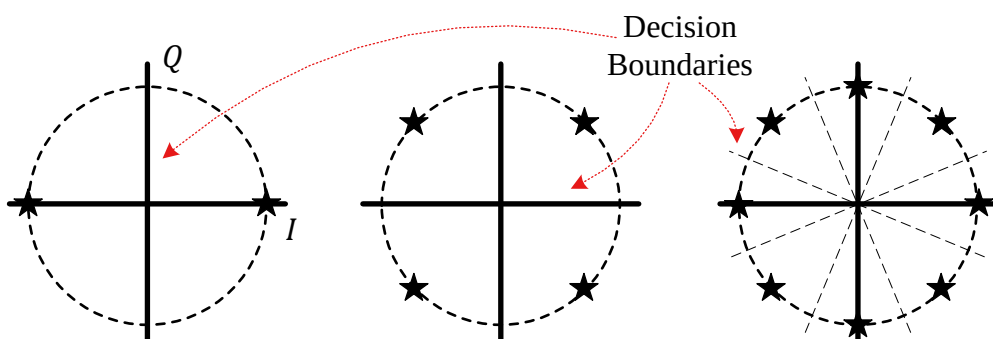


Figure 3.54: PSK constellation diagrams for $M = 2$, $M = 4$ and $M = 8$. Observe that all points lie on a circle. Also, they get closer for higher-order PSK and become prone to noise errors

- 2-PSK constellation (also known as *BPSK*) looks similar to 2-PAM. However, the difference is that there is no carrier upconversion in PAM systems.
- 4-PSK constellation (also known as *QPSK*) is exactly the same as square 4-QAM.
- For $M > 4$, square QAM packs the constellation points more efficiently than PSK and hence the modulation of choice in many wireless standards. These constellation points come relatively closer in PSK and the closer the points, the larger the probability of receiving a symbol in error due to a jump across the decision boundary. On the other hand, QAM heavily depends on overall system linearity due to information conveyed in the signal amplitude. A constant envelope of the modulated signal is much more suitable for transmission over nonlinear channels where PSK is the preferred choice. In such situations, the performance of PSK is quite insensitive to nonlinear distortion and heavy filtering, see Ref. [11].

Note 3.11 London Eye

Here is a little quiz for you: which PSK constellation is the London Eye shown in Figure 3.55? The answer is 32-PSK as the Eye has 32 pods but they are numbered from 1 to 33. The number 13 is skipped to avoid bad luck.



Figure 3.55: London Eye is a 32-PSK constellation

Since any constellation that uses both amplitude and phase modulation falls under the general category of QAM, there can be many other rearrangements of QAM constellation points. However, we focus on square QAM and PSK in this text which are widely used in wireless systems. Non-regular constellations can yield a marginal performance advantage at a cost of more complicated demodulation, some of which are utilized in wired media such as DSL and cable modems. Nevertheless, the more the points are in the constellation, the lesser is the difference.

QAM Detector

Let us consider the scenario where the received signal $r(t)$ is the same as the transmitted signal $s(t)$ but with the addition of additive white Gaussian noise (AWGN) $w(t)$. The symbols are detected through the following steps illustrated in Figure 3.56.

- Out of the infinite spectrum, the desired signal $r(t)$ is selected with the help of a Band-pass Filter (BPF).
- Through an Analog to Digital Converter (ADC), this signal is sampled at a rate of F_S samples/s to produce a sequence of T_S -spaced samples $r(nT_S)$.
- Next, *a complex signal $x(nT_S)$* is produced when $r(nT_S)$ is downconverted by mixing with the inphase and quadrature carriers which are generated through a Local Oscillator (LO) at the Rx. Just like upconversion, we assume a conceptual downconversion

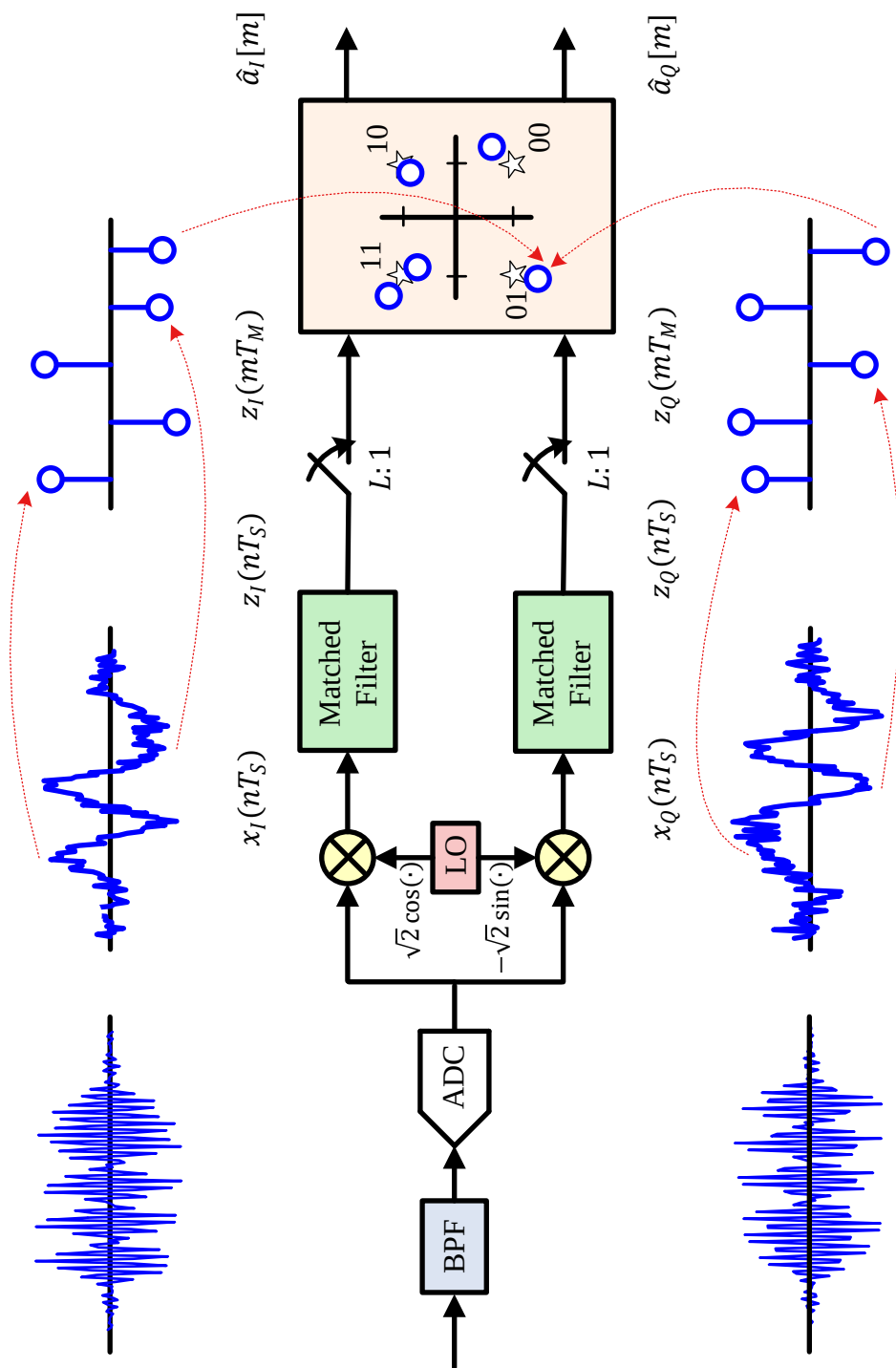


Figure 3.56: A general QAM detector

in discrete domain but in practice there are different techniques to accomplish this task that depend on the Rx architecture.

- The resulting complex waveform $x(nT_S)$ in I and Q arms is processed through two matched filters thus generating $z_I(nT_S)$ and $z_Q(nT_S)$. As discussed earlier, the outputs of the matched filter $z_I(nT_S)$ and $z_Q(nT_S)$ are the correlations of the symbol-scaled pulse shape with an unscaled and time-reversed pulse shape in their respective arms.
- These I and Q outputs are downsampled by L at optimal sampling instants

$$n = mL = m \frac{T_M}{T_S}$$

to produce T_M -spaced numbers $z_I(mT_M)$ and $z_Q(mT_M)$ back from the signal.

- The minimum distance decision rule is employed in IQ -plane to find the symbol estimates $\hat{a}_I[m]$ and $\hat{a}_Q[m]$ to decide on the final constellation point.

Let us discuss the mathematical details of this process for a noiseless Rx signal as in Eq (3.40).

$$r(t) = s(t) = v_I(t)\sqrt{2}\cos 2\pi F_C t - v_Q(t)\sqrt{2}\sin 2\pi F_C t \quad (3.43)$$

where F_C is the carrier frequency and $v_I(t)$ and $v_Q(t)$ are defined in Eq (3.38). After bandlimiting the incoming signal through a bandpass filter, it is sampled by the ADC operating at F_S samples/second to produce

$$\begin{aligned} r(nT_S) &= v_I(nT_S)\sqrt{2}\cos 2\pi F_C nT_S - v_Q(nT_S)\sqrt{2}\sin 2\pi F_C nT_S \\ &= v_I(nT_S)\sqrt{2}\cos 2\pi \frac{k_C}{N}n - v_Q(nT_S)\sqrt{2}\sin 2\pi \frac{k_C}{N}n \end{aligned}$$

where the relation $F/F_S = k/N$ is used. To produce a complex baseband signal from the Rx signal, the samples of this waveform are input to a mixer which multiplies them with discrete-time quadrature sinusoids $\sqrt{2}\cos 2\pi(k_C/N)n$ and $-\sqrt{2}\sin 2\pi(k_C/N)n$ yielding

$$\begin{aligned} I \rightarrow x_I(nT_S) &= r(nT_S) \cdot \sqrt{2}\cos 2\pi \frac{k_C}{N}n \\ &= 2v_I(nT_S)\cos^2 2\pi \frac{k_C}{N}n - 2v_Q(nT_S)\sin 2\pi \frac{k_C}{N}n \cos 2\pi \frac{k_C}{N}n \end{aligned}$$

$$\begin{aligned} Q \uparrow x_Q(nT_S) &= r(nT_S) \cdot -\sqrt{2}\sin 2\pi \frac{k_C}{N}n \\ &= 2v_Q(nT_S)\sin^2 2\pi \frac{k_C}{N}n - 2v_I(nT_S)\cos 2\pi \frac{k_C}{N}n \sin 2\pi \frac{k_C}{N}n \end{aligned}$$

Using the identities $2\cos^2 A = 1 + \cos 2A$, $2\sin^2 A = 1 - \cos 2A$ and $2\sin A \cos A = \sin 2A$,

$$I \rightarrow x_I(nT_S) = v_I(nT_S) + \underbrace{v_I(nT_S)\cos 2\pi \frac{2k_C}{N}n - v_Q(nT_S)\sin 2\pi \frac{2k_C}{N}n}_{\text{Double frequency terms}}$$

$$Q \uparrow x_Q(nT_S) = v_Q(nT_S) - \underbrace{v_Q(nT_S)\cos 2\pi \frac{2k_C}{N}n - v_I(nT_S)\sin 2\pi \frac{2k_C}{N}n}_{\text{Double frequency terms}}$$

Now we can observe why the two factors, a $\sqrt{2}$ with both sinusoids and a negative sign with $\sin 2\pi(k/N)n$, were inserted.

- A $\sqrt{2}$ at the modulator and later another $\sqrt{2}$ in the detector result in a gain of 2, which cancels the halving of sinusoid amplitudes in above trigonometric multiplications.
- When a complex signal is multiplied with a complex sinusoid, a negative sign appears with $v_Q(nT_S)$ in the I section of cumulative waveform. This is the real part of the complex product which is transmitted through the channel. Another negative sign with $\sin 2\pi(k/N)n$ at the Rx delivers a positive $v_Q(nT_S)$ at the output.

The matched filter output is written as

$$\begin{aligned}
 I \rightarrow z_I(nT_S) &= x_I(nT_S) * p(-nT_S) \\
 &= (v_I(nT_S) + \text{Double freq terms}) * p(-nT_S) \\
 &= \left(\sum_i a_I[i] p(nT_S - iT_M) + \frac{2k_C}{N} \text{ terms} \right) * p(-nT_S) \\
 &= \sum_i a_I[i] r_p(nT_S - iT_M) \\
 \\
 Q \uparrow z_Q(nT_S) &= x_Q(nT_S) * p(-nT_S) \\
 &= (v_Q(nT_S) + \text{Double freq terms}) * p(-nT_S) \\
 &= \left(\sum_i a_Q[i] p(nT_S - iT_M) + \frac{2k_C}{N} \text{ terms} \right) * p(-nT_S) \\
 &= \sum_i a_Q[i] r_p(nT_S - iT_M)
 \end{aligned}$$

where $r_p(nT_S)$ comes from applying the definition of the auto-correlation function in Eq (3.21). The double frequency terms in the above equation are filtered out by the matched filter $h(nT_S) = p(-nT_S)$, which also acts as a lowpass filter due to its spectrum limitation in the range $-1/2T_M \leq F \leq +1/2T_M$. To generate symbol decisions, T_M -spaced samples of the matched filter output are required at $n = mL = mT_M/T_S$. Downsampling the matched filter output generates

$$\begin{aligned}
 I \rightarrow z_I(mT_M) &= z_I(nT_S) \Big|_{n=mL=mT_M/T_S} \\
 &= \sum_i a_I[i] r_p(mT_M - iT_M) = \sum_i a_I[i] r_p\{(m-i)T_M\} \\
 &= \underbrace{a_I[m] r_p(0T_M)}_{\text{current symbol}} + \underbrace{\sum_{i \neq m} a_I[i] r_p\{(m-i)T_M\}}_{\text{ISI}}
 \end{aligned}$$

and

$$\begin{aligned}
 Q \uparrow z_Q(mT_M) &= z_Q(nT_S) \Big|_{n=mL=mT_M/T_S} \\
 &= \sum_i a_Q[i] r_p(mT_M - iT_M) = \sum_i a_Q[i] r_p\{(m-i)T_M\} \\
 &= \underbrace{a_Q[m] r_p(0T_M)}_{\text{current symbol}} + \underbrace{\sum_{i \neq m} a_Q[i] r_p\{(m-i)T_M\}}_{\text{ISI}}
 \end{aligned}$$

A square-root Nyquist pulse $p(nT_S)$ has an auto-correlation $r_p(nT_S)$ that satisfies no-ISI criterion of Eq (3.24).

$$r_p(mT_M) = \begin{cases} 1, & m = 0 \\ 0, & m \neq 0 \end{cases}$$

Thus,

$I \rightarrow$
 $Q \uparrow$

$$\begin{aligned} z_I(mT_M) &= a_I[m] \\ z_Q(mT_M) &= a_Q[m] \end{aligned}$$

In conclusion, the downsampled matched filter outputs map back to the Tx symbols in the absence of noise. If the world was simple, that would have been an end to it! But the world is complicated, and there are layers of issues that happen between the Tx information generation and Rx decision making. Anything we do after this chapter will be to combat a subset of signal distortions and towards recovering the original information.

Observe that the system shown in Figure 3.56 is a multirate system. In the QAM detector, for example, the ADC and the matched filters operate at the sample rate F_S . After the outputs of the matched filters are downsampled by L , the symbol decisions are made at the symbol rate $1/T_M$. Furthermore, there are some hidden assumptions in the QAM detector.

Symbol timing synchronization: The peak samples at the end of symbol durations in both I and Q arms are not known in advance at the Rx and in fact do not necessarily coincide with the generated samples as well. This is because ADC just samples the incoming continuous waveform without any information about the symbol boundaries. This is a symbol timing synchronization problem which we will study in Chapter 7.

Resampling: The ADC in general does not produce an integer number of samples per symbol, i.e., T_M/T_S is not an integer. As we will see later, a resampling system is required in the Rx chain that changes the sample rate from the ADC rate to a rate that is an integer multiple of the symbol rate.

Carrier frequency synchronization: The carrier frequency of the oscillator at the Tx and that at the Rx are not exactly the same. Instead, if the oscillator frequency at the Tx is denoted as F_C , then at the Rx, we have $F_C + F_\Delta$, where F_Δ is the difference between the two and can be either positive or negative. Any movement by the Tx, the Rx or within the environment between them also causes a shift in frequency (known as Doppler shift) that needs to be compensated. We will discuss carrier frequency synchronization in Chapter 6.

Carrier phase synchronization: Furthermore, the carrier phase at the Rx oscillators is not known beforehand and needs to be estimated which will be explained in Chapter 5.

Equalization: Only an AWGN channel has been assumed so far. Multipath reflections from a wireless channel introduce ISI and distortion in the signal that need to be recovered through an equalizer. This is what we cover in Chapter 8.

3.8 Stethoscopes for a Communication System

We are now in a position to devise some tools that help us diagnose problems with the communication system under study. I like to call them the *stethoscopes for a communication system* due to the crucial functionality they provide regarding the health of the communication system being analyzed. We discuss three such tools, namely an eye diagram, a transition diagram and a scatter plot below.

Eye Diagram

An eye diagram is an excellent summary of the signal behaviour in time domain, very similar to a spectrum in frequency domain. Imagine the samples of the matched filter output taken at a much higher rate, say $L = 64$ samples/symbol, instead of $L = 2$ samples/symbol so that the underlying plot looks continuous as in Figure 3.57.

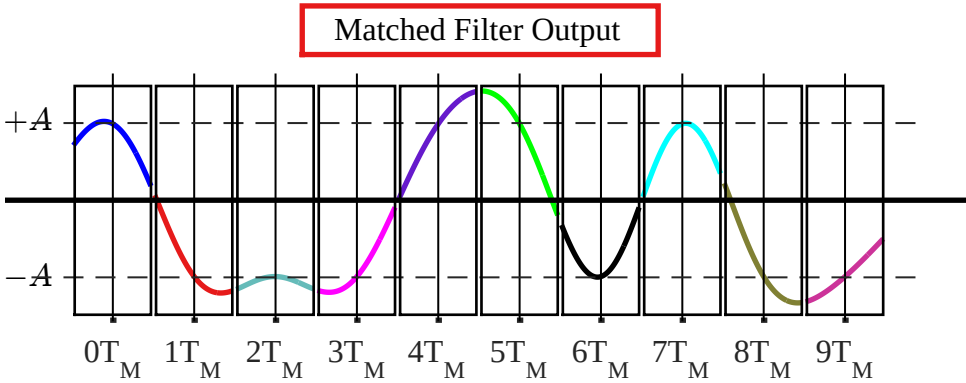


Figure 3.57: Matched filter output of Figure 3.37 as a continuous waveform

Next, the waveform is divided into black boxes of width T_M seconds such that each signal portion within a box starts half a symbol duration $T_M/2$ before the ideal sampling time iT_M , where i is an integer. Similarly, each signal portion within a box ends at half a symbol duration $T_M/2$ after the ideal sampling time iT_M .

Assume that such a stream of PAM symbols is printed on a paper and all black boxes are cut into separate pieces precisely at symbol boundaries. My daughter Varda just did that when I gave her a printed PAM sequence, as shown in Figure 3.58.

Now if they are placed on top of one another, we get a diagram drawn in Figure 3.59a. This is *eye diagram*, which is a modulo- T_M plot of matched filter output against time. Compare the color patterns between both figures and observe that time base in Figure 3.59a is shifted such that optimum sampling instant occurs in the middle of the plot. *This shift in time base can sometimes cause confusion* and hence it should be remembered that although the ideal sampling instants actually occur at the end of a symbol interval, an eye diagram shows that instant in the middle.

Therefore, it can be concluded in general that an eye diagram draws an overlapping symbol waveform within an interval

$$-0.5T_M \leq t \leq +0.5T_M$$

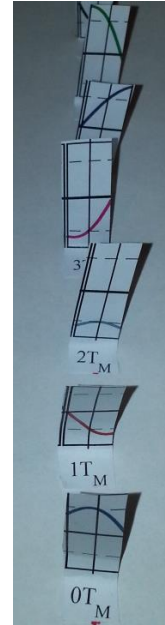
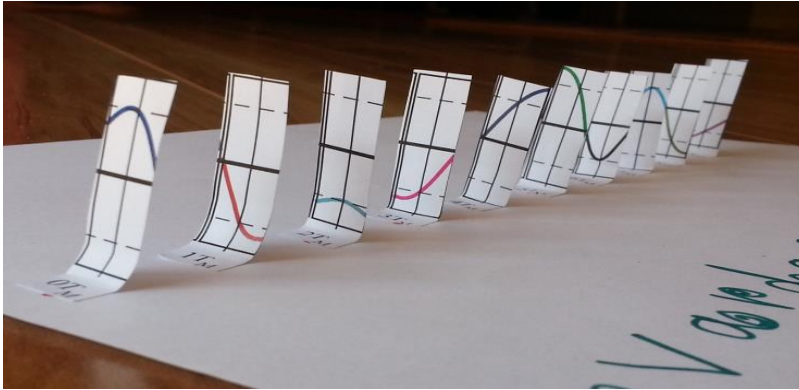
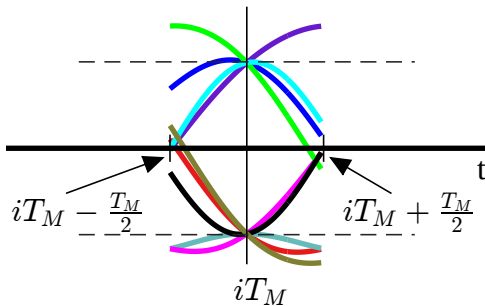
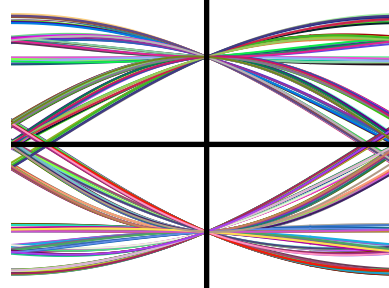


Figure 3.58: PAM symbol stream printed on a paper and cut into pieces at symbol boundaries



(a) All symbol intervals overlaid on top of one another. The time axis shifted to bring ideal sampling instant in the middle



(b) Eye diagram generated for 250 2-PAM symbols and Square-Root Raised Cosine pulse with excess bandwidth $\alpha = 0.5$

Figure 3.59: Construction of an eye diagram

Also note that the eventual pattern strongly depends on the underlying pulse auto-correlation from which the pulse shape for transmitting data is derived. Now remember that we are only using an example of a 2-PAM modulation with 10 bits, and hence 10 symbols, only. When a large number of symbols are generated for such a plot and overlaid on each other, all possible trajectories of the pulse auto-correlation dictated by the symbol sequence come into play. A diagram for Square-Root Raised Cosine pulse shape (and hence Raised Cosine pulse auto-correlation) generated for 250 symbols and excess bandwidth $\alpha = 0.5$ is shown in Figure 3.59b. *Notice that this plot resembles a human eye, hence the name eye*

diagram.

Example 3.2

Interestingly, tracing a single transition in an eye diagram gives information about 3 symbols. As an example, look at the trace around $5T_M$ in Figure 3.59a:

NOW Observe that the trace goes through $+A$ at the current sampling instant.

Past Also, it starts from a high voltage level. That is an indicator that in a clean system, its previous symbol would have been $+A$. Now compare it with Figure 3.57 and it is verified that its previous symbol is indeed $+A$.

Future Finally, its level is falling below zero and towards $-A$ which indicates that the next symbol should be $-A$. The fact that the next trace in Figure 3.57 is $-A$ verifies this observation.

This will help us when we discuss symbol timing synchronization in Chapter 7.

Eye diagram is a diagnostic tool that helps in evaluation of the effects of channel noise and Inter-Symbol Interference (ISI) on the performance of a communication system. For this purpose, an eye diagram for a real transmission system can be generated through an oscilloscope. The horizontal time base of the oscilloscope is set equal to a symbol interval T_M and the matched filtered sequence is connected to the vertical axis. This superimposes the symbol intervals into a family of traces, all displayed within the same duration. The persistence of oscilloscope display makes it look like an eye.

To see how it helps, consider Figure 3.60 that consists of two symbol durations. Relevant information that can be extracted from an eye diagram is detailed below.

Best Sampling Instant: For Nyquist pulses, no ISI occurs at the end of a symbol duration T_M – which falls in the center of the eye. This gives the maximum possible SNR on average because the sample value is farthest from the decision threshold at this point. A wrong decision will only happen if the noise is sufficient to move the sample towards the other side of this threshold.

Timing Error: A timing error occurs when the eye is not sampled at maximum average opening. For a data waveform, it means that the ideal sample is not at the peak of the auto-correlation and hence the timing instant is either early or late.

Noise Margin: If a timing error occurs, the signal is sampled closer to the decision boundary. A relatively smaller amount of noise can cause a decision error. Or in other words, noise margin gets reduced[†].

Timing Jitter: Timing jitter is a measure of average deviation around the mean zero crossings. In the old days, it was not much of a problem due to low symbol rates. Modern high speed communication systems have an increasingly shorter symbol time T_M . Since the timing jitter originates from the actual device circuitry, it becomes increasingly larger as a percentage of this symbol period.

[†]For some trajectories, a timing error actually improves the noise margin, while for many others, it reduces that margin. When the average of all trajectories is taken into account, the noise margin decreases proportionally to the timing error.

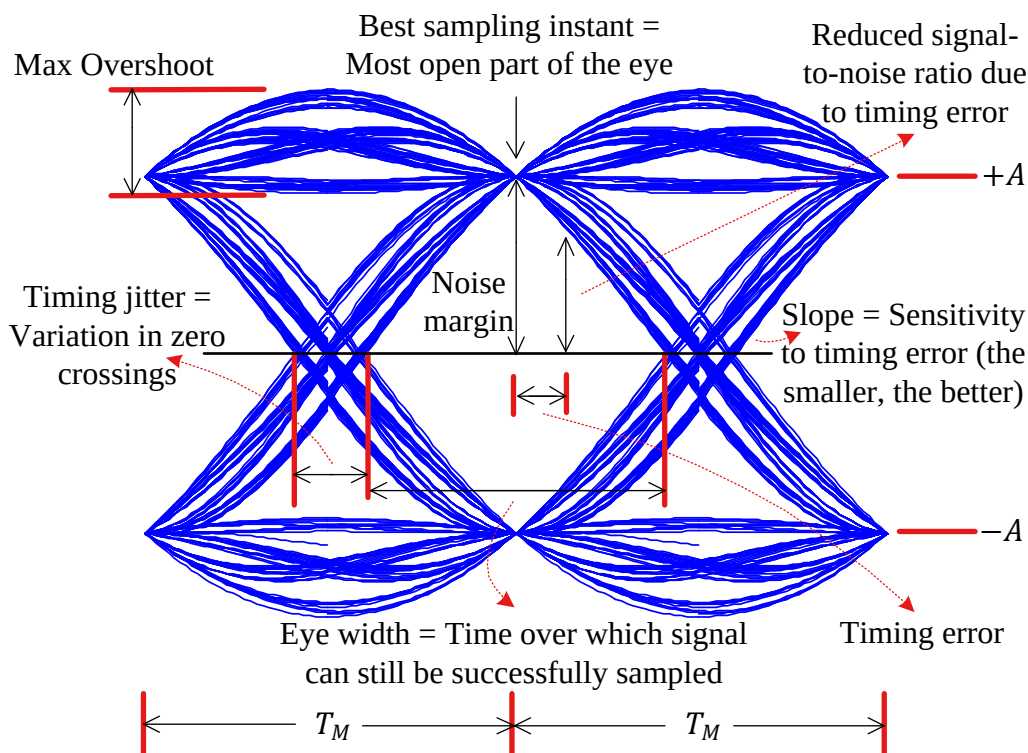


Figure 3.60: Eye diagram as a system diagnostic tool in a noiseless case

Eye Width: The wider the eye, the cleaner the channel. Channel distortion decreases the eye width and also decreases the eye opening at the best sampling instant. An open eye pattern corresponds to minimal signal distortion. Many wireless channels cause the eye to *close*: there remains no sampling instant where a best symbol estimate can be obtained.

In fact, the eye in Figure 3.60 looks so symmetric because it is drawn for a noiseless case. Even for an AWGN channel with a low SNR, the eye can close but will remain symmetric in general as illustrated in Figure 3.61. Depending on the SNR, there is distortion at the best sampling instant as opposed to the previous case. Moreover, noise margin gets reduced, as a consequence of which decision errors can occur for relatively smaller noise power.

Slope: Slope of the eye determines its sensitivity to timing errors. A large slope implies that even a little deviation in the timing instant can cause the sample value – and subsequently noise margin – to reduce significantly. Therefore, a smaller slope lessens the effect of this dependence.

Eye diagrams can also be drawn for modulation schemes packing multiple bits in one symbol. For the case $M = 4$, due to the presence of more than 2 symbols, the next symbol after every particular symbol can be $+3A$, $+A$, $-A$ or $-3A$ resulting into many different trajectories. Therefore, an eye diagram for modulation order $M > 2$ displays multiple eyes,

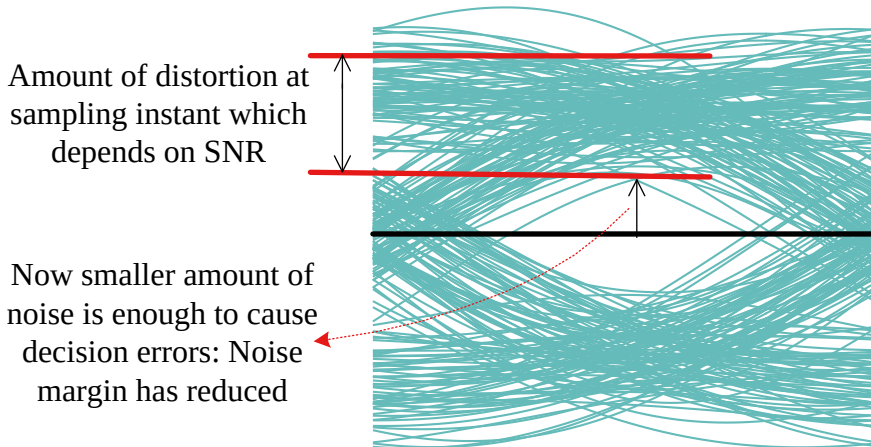


Figure 3.61: Eye diagram in an AWGN channel

an example of which is illustrated in Figure 3.62. Also notice that the larger the excess bandwidth α , the lesser the interference among adjacent symbols in time and the eye is more open.

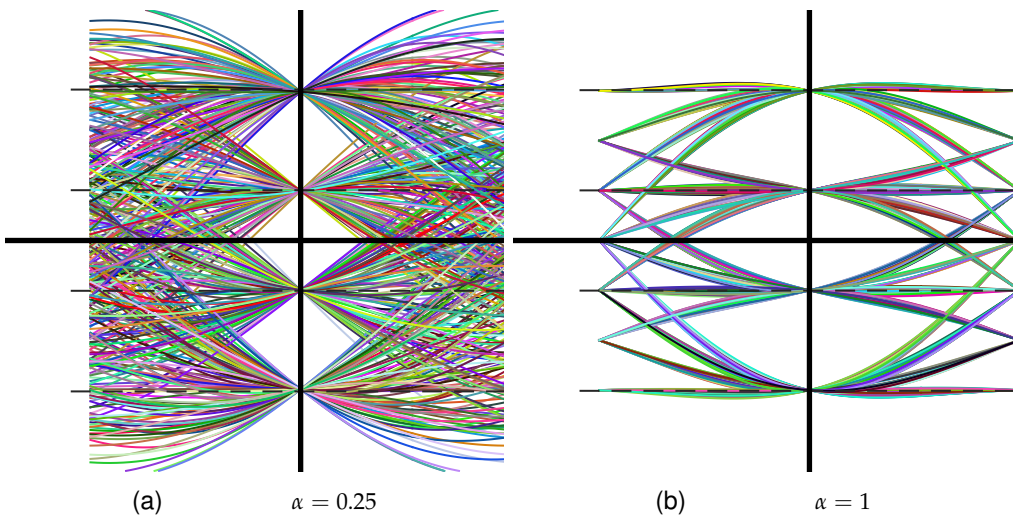


Figure 3.62: Eye diagrams generated for 400 4-PAM symbols and Square-Root Raised Cosine pulse. Higher modulations display multiple eyes. Moreover, larger excess bandwidth α results in more open eye

The eye diagrams studied before consist of a 1-dimensional modulation scheme, namely PAM. On the other hand, a QAM signal consists of two PAM signals riding on orthogonal carriers. At baseband, these two PAM signals appear as I and Q components of a complex signal at the Tx and Rx. Therefore, there are two eye diagrams for a QAM modulated signal, one for I and the other for Q and both of them are exactly the same as PAM.

Transition Diagram (Phase Trajectory)

We discussed above that in the case of QAM transmission, the matched filter output is complex valued and hence two eye diagrams are needed, one is the modulo- T_M plot of the inphase component and the other is the modulo- T_M plot of the quadrature component. Instead of drawing the I and Q parts in separate diagrams, another alternative that displays the relationship between Q versus I is a *transition diagram* or a *phase trajectory* plot. A transition diagram is just a 2D diagram depicting the continuous-time Q vs I signal in the complex plane itself *without any time axis*.

Some examples of the transition diagrams for a 4-QAM and a 16-QAM modulation schemes are drawn in Figure 3.63 for a Raised Cosine pulse with excess bandwidth $\alpha = 0.5$ and $\alpha = 1$ for zero noise. Some observations are as follows.

- The angle of the plot with respect to I -axis is the instantaneous phase of the QAM modulated signal (without any carrier).
- There is no explicit time axis in this plot but it can be traced from the plot moving from one constellation point to another.
- We can easily identify the underlying constellation points from the phase trajectory.
- A higher excess bandwidth, e.g., $\alpha = 1$, prevents any substantial overshoots from appearing when the signal transitions from one constellation point to another. This makes sense because in a time domain plot, a higher excess bandwidth gives rise to smooth transitions between symbols (very similar to an unmodulated sinusoid).

Scatter Plot

In general, *a scatter plot is a graph that shows the relationship between two sets of data*. More specifically, a scatter plot of a complex signal is a plot of its Q samples drawn versus its I samples. For the purpose of digital communications, a scatter plot can be explained as follows.

Remember that we used a continuous version of the matched filter output to understand the eye diagrams and the transition diagrams. For scatter plot, we will use the matched filter output downsampled at 1 sample/symbol as a reference. After downsampling the matched filter output to symbol rate, the samples thus obtained are mapped back to the constellation as illustrated in Figure 3.38. Before the symbol decisions are made, those samples form a cloud around the ideal constellation points as shown in Figure 3.64a. This cloud of symbol-spaced samples mapped on the original constellation diagram is called a *scatter plot*.

For a QAM modulation, this cloud of samples around the constellation points is now 2-dimensional (for PAM, there was no Q component) and can also be understood as a plot of Q versus I for each mapped value, i.e., a downsampled version of the transition diagram. This is illustrated in Figure 3.64b.

Looking at the scatter plot, one can readily deduce a lot of performance measures for the particular transmission system. One such measure widely used in practice is the *Error Vector Magnitude (EVM)* that computes the vector sum of all the Rx points from the ideal location. At this stage, however, it is enough to observe that the diameter of these clouds is a rough measure of the noise power corrupting the signal. For the ideal case of no

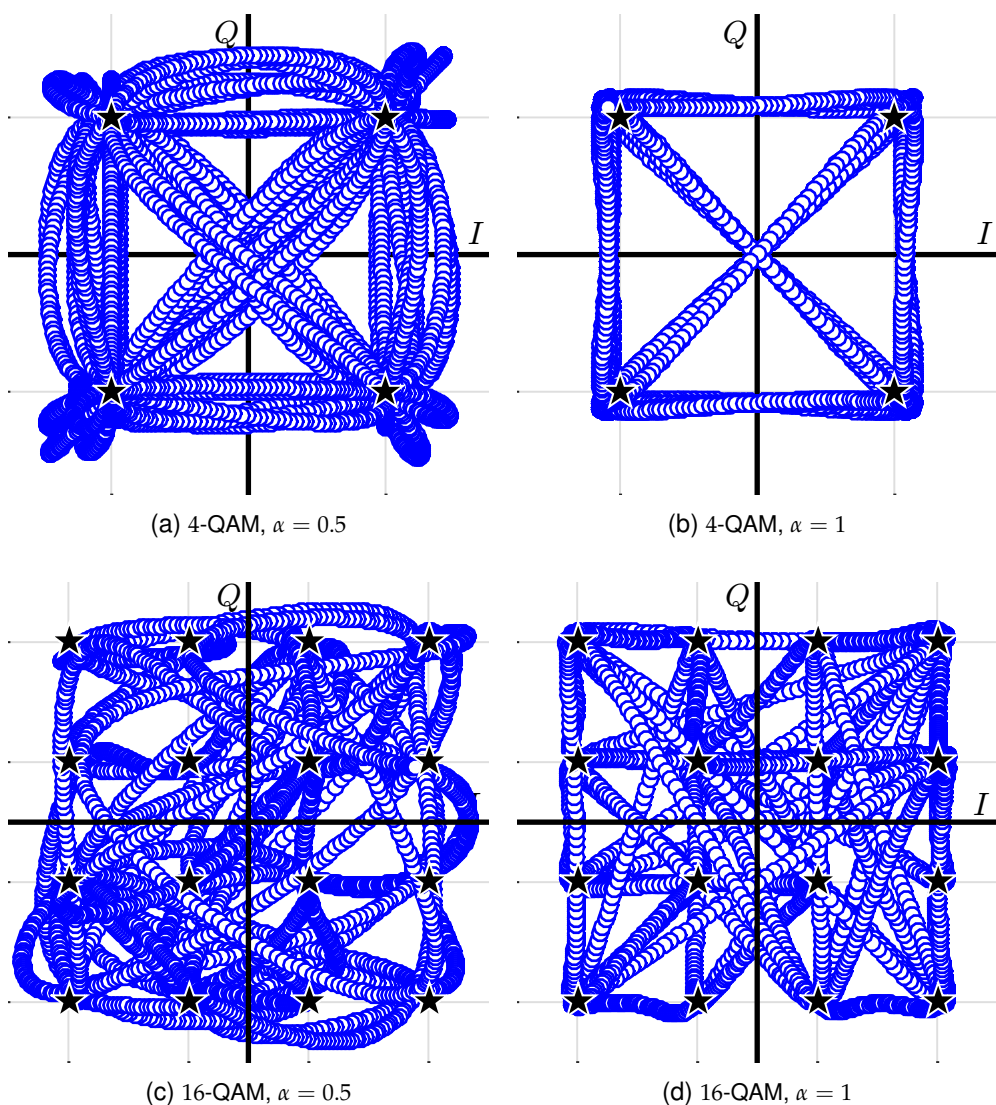
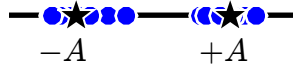


Figure 3.63: Transition diagram or phase trajectory for 4-QAM and 16-QAM for a Raised Cosine pulse with excess bandwidth $\alpha = 0.5$ and $\alpha = 1$ and zero noise

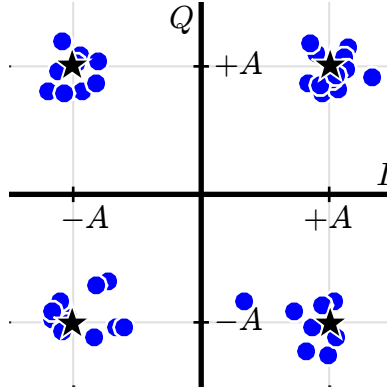
noise, this diameter is zero and all the optimally timed samples coincide with their respective constellation points.

3.9 Modulation Bandwidths

After covering linear modulations and pulse shaping, we can correctly determine the occupied bandwidth for each modulation scheme. Figure 3.31 shows the bandwidth of a Square-Root Raised Cosine pulse shape as $(1 + \alpha)/2T_M$. We have also seen that the spectrum ap-



(a) 2-PAM modulation in the presence of AWGN



(b) 4-QAM modulation in the presence of AWGN

Figure 3.64: Scatter plots for PAM and QAM

proximately remains the same, provided that there is enough randomness in the bit stream and the resulting symbols are equally likely and independent from each other. Therefore, the bandwidth for a PAM modulated signal can be given as

$$BW_{\text{PAM}} = \frac{1 + \alpha}{2T_M} \quad (3.44)$$

QAM is basically a similar modulation scheme except that it is modulated on a carrier. After the spectral upconversion, both positive and negative portions of the baseband spectrum appear at $+F_C$ and $-F_C$. The bandwidth – positive portion of the spectrum – hence becomes double.

$$BW_{\text{QAM}} = 2 \frac{1 + \alpha}{2T_M} = \frac{1 + \alpha}{T_M} \quad (3.45)$$

The same equation holds for PSK modulation. In case some other pulse shape is used with a bandwidth B , the factor $0.5(1 + \alpha)$ can be replaced with an appropriate factor in the above equations.

3.10 Computing Error Rates

Having built a simple digital communication system, it is necessary to know how to measure its performance. As the names say, *Symbol Error Rate (SER)* and *Bit Error Rate (BER)* are the probabilities of receiving a symbol and bit in error, respectively. SER and BER can be approximated through simulating a complete digital communication system involving a

large number of bits and comparing the ratio of symbols or bits received in error to the total number of bits. Hence,

$$SER = \frac{\text{No. of symbols in error}}{\text{Total no. of transmitted symbols}} \quad (3.46)$$

and

$$BER = \frac{\text{No. of bits in error}}{\text{Total no. of transmitted bits}} \quad (3.47)$$

Till now, the only imperfection between Tx and Rx entities we have covered is AWGN. It then makes sense that more distortions should corrupt the signal further and result in more errors. Hence, SER or BER plotted against some measure of signal-to-noise ratio (SNR) is the benchmark against which the performance of a system can be measured. Next, we discuss how we choose a specific measure of SNR against which SER and BER should be plotted.

Figure of Merit: E_b/N_0

The **SNR** in a digital communication system is defined as a ratio of signal power in a symbol P_M to noise power P_w as

$$SNR = \frac{P_M}{P_w} \quad (3.48)$$

where the signal power P_M is the average symbol energy of the modulation per unit symbol time.

$$P_M = \frac{E_M}{T_M} \quad (3.49)$$

For PAM and QAM modulation schemes, average E_M was defined in Eq (3.18) and Eq (3.42). Moreover, how to determine the noise power P_w was detailed in Section 2.8, Eq (2.47).

When specified in dB units and using a factor of 10 for power conversions, it is

$$\begin{aligned} SNR_{dB} &= P_{M,dB} - P_{w,dB} \\ &= 10 \log_{10} P_M - 10 \log_{10} P_w \end{aligned} \quad (3.50)$$

A visual measure of finding SNR_{dB} is shown in Figure 3.65.

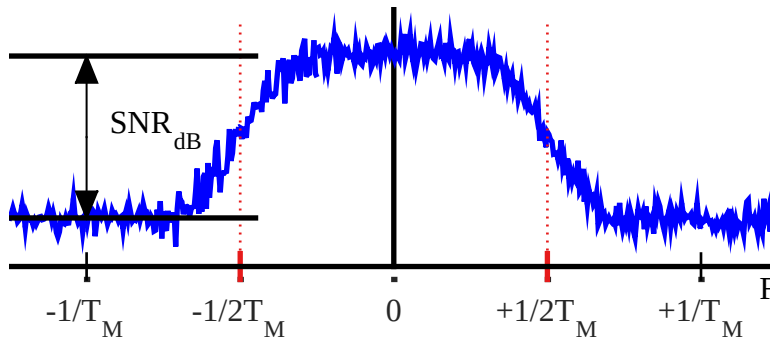


Figure 3.65: A visual measure of SNR_{dB}